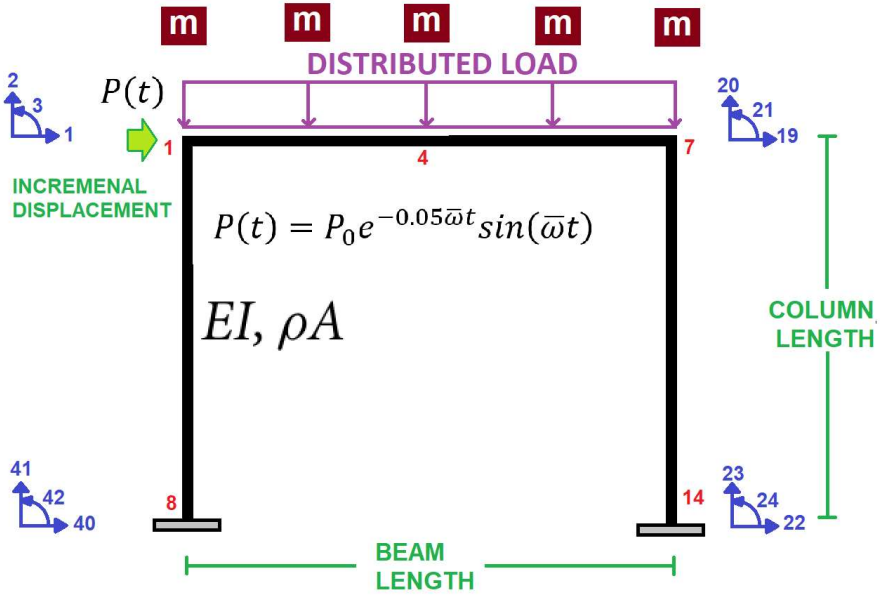
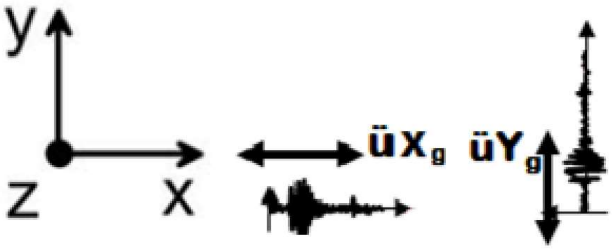
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A COMPOSITE SECTION FRAME: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

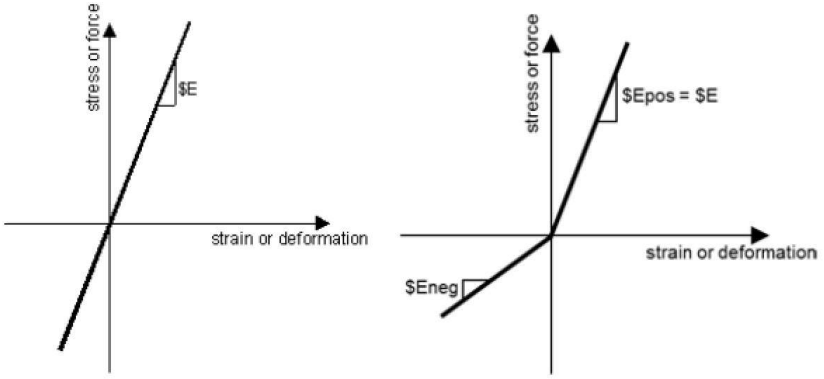




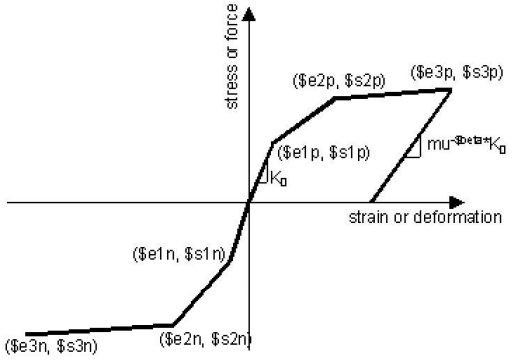
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

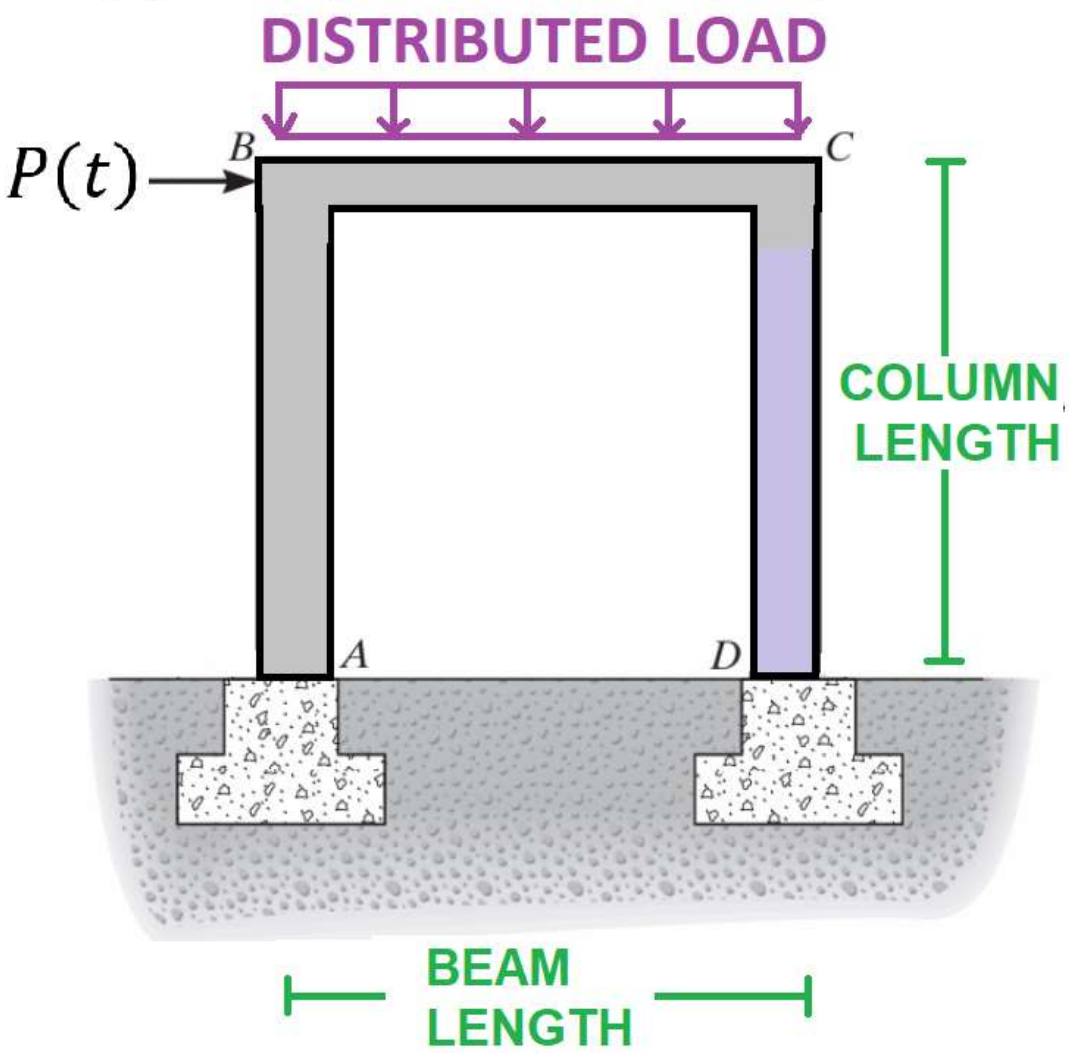


ELEMENT ELASTIC MATERIAL



ELEMENT INELASTIC MATERIAL

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$



$$\rho A \frac{\partial^2 v(x,t)}{\partial t^2} + EI \frac{\partial^4 v(x,t)}{\partial x^4} = P(t)$$

Spyder (Python 3.12)

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...ENERGY_DISSIPATION_CAPACITY_INDEX\10_COMPOSITE_FRAME_8_ANA

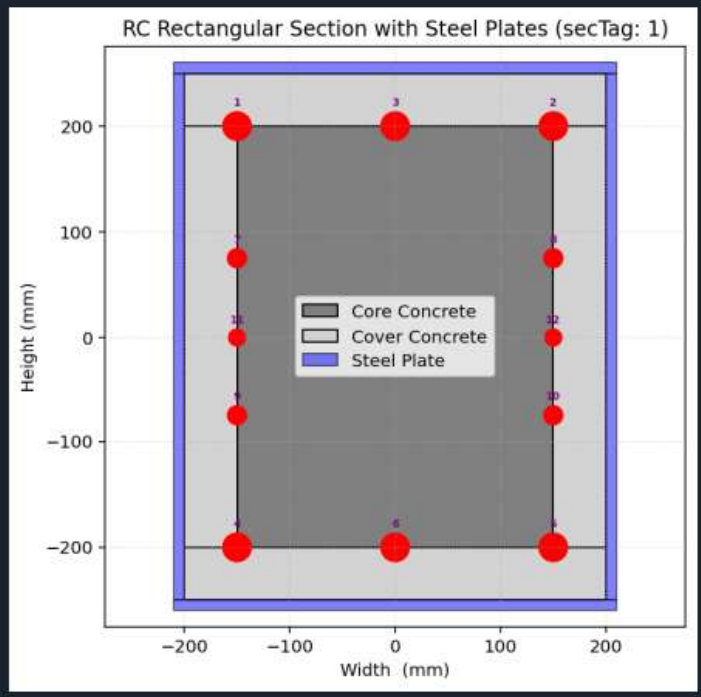
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...AME_8_ANA\COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py | COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

```python
#----------------------------------------------------------------------
# CONFINED CONCRETE FIBER RECTANGULAR WITH PLATE SECTION - QUAD FIBERS
#----------------------------------------------------------------------
def CONCRETE_CONFINED_REC_PLATE_SECTION_QUAD_EXTRA(secTag, MAT_TYPE, HSec, BSec, coverH, co
                                    plateThickness=10.0, PLOT=True, CONCRETE_DENSITY=2500
    """
    Creates a rectangular reinforced concrete fiber section with surrounding steel plates i
    Uses plt.Rectangle() for concrete and steel plates, and plt.scatter() for reinforcement

    Parameters:
    secTag (int): Section tag identifier
    HSec (float): Height of the section
    BSec (float): Width of the section
    coverH (float): Cover thickness in the height direction
    coverB (float): Cover thickness in the width direction
    coreID (int): Material tag for confined core concrete
    coverID (int): Material tag for unconfined cover concrete
    steelID (int): Material tag for reinforcing steel
    steelPlateID (int): Material tag for the steel plates
    numBarsTop (int): Number of top reinforcing bars
    barAreaTop (float): Cross-sectional area of each top reinforcing bar
    numBarsBot (int): Number of bottom reinforcing bars
    barAreaBot (float): Cross-sectional area of each bottom reinforcing bar
    numBarsIntTot (int): Total number of intermediate reinforcing bars
    barAreaInt (float): Cross-sectional area of each intermediate reinforcing bar
    plateThickness (float): Thickness of the steel plates (default=10.0)
    PLOT (bool): Whether to plot the section (default=True)
    CONCRETE_DENSITY (float): Density of concrete (default=2500 kg/m³ converted to consiste
    STEEL_DENSITY (float): Density of steel (default=7850 kg/m³ converted to consistent uni

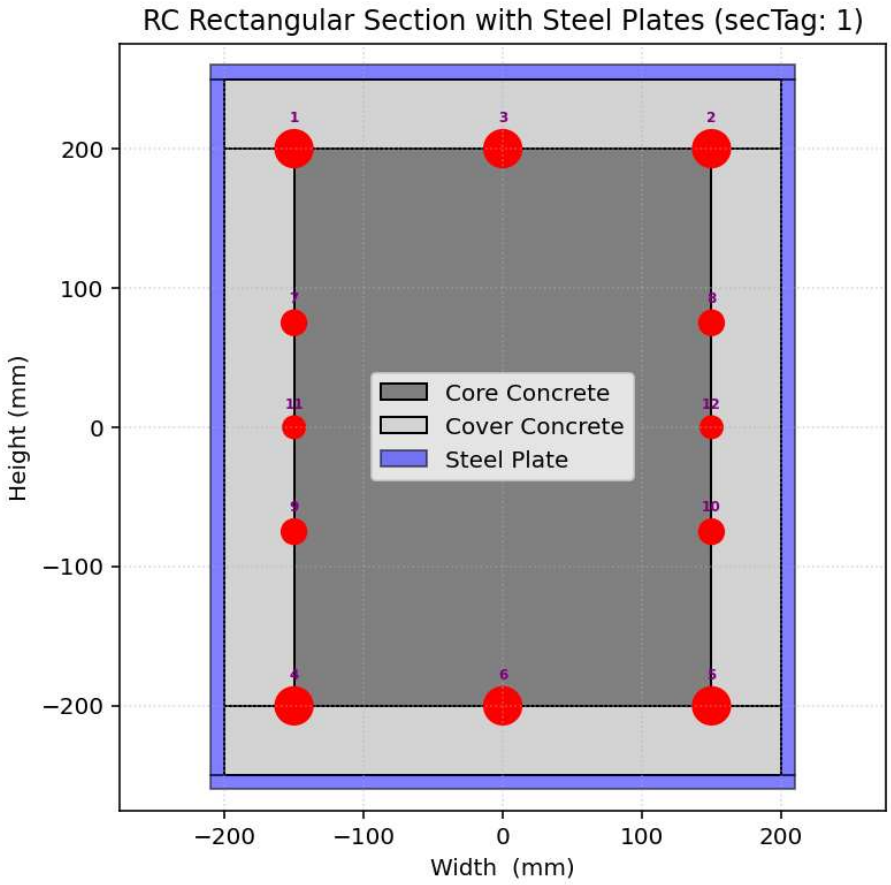
    THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
    """
    #import openseespy.opensees as ops
    import matplotlib.pyplot as plt
```

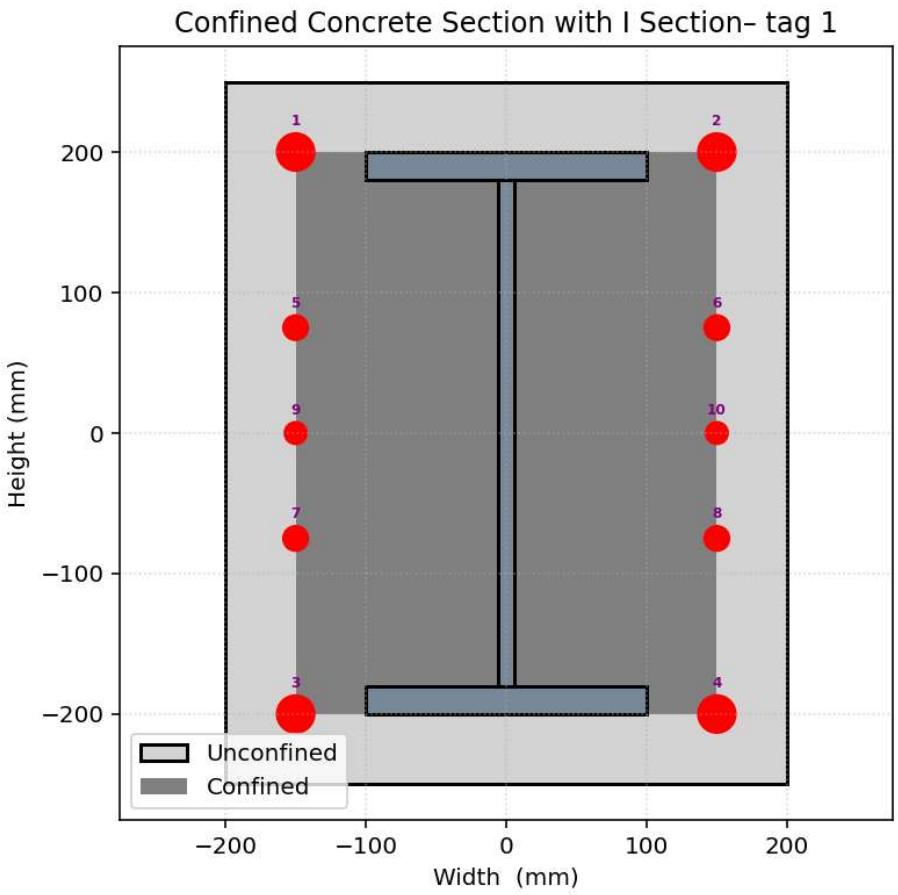


IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 39%

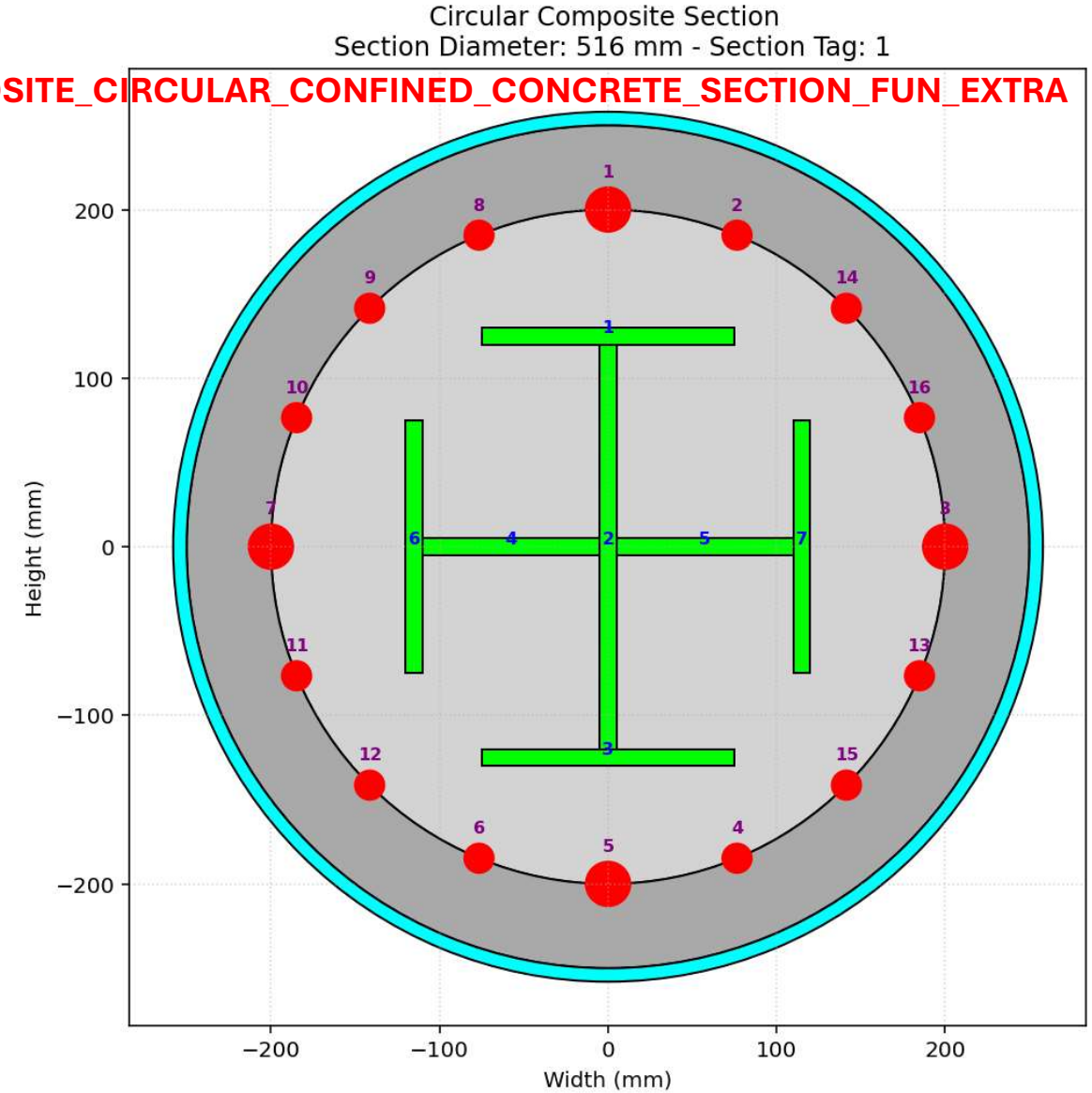
COMPOSITE_CONCRETE_SECTION_FUN_EXTRA



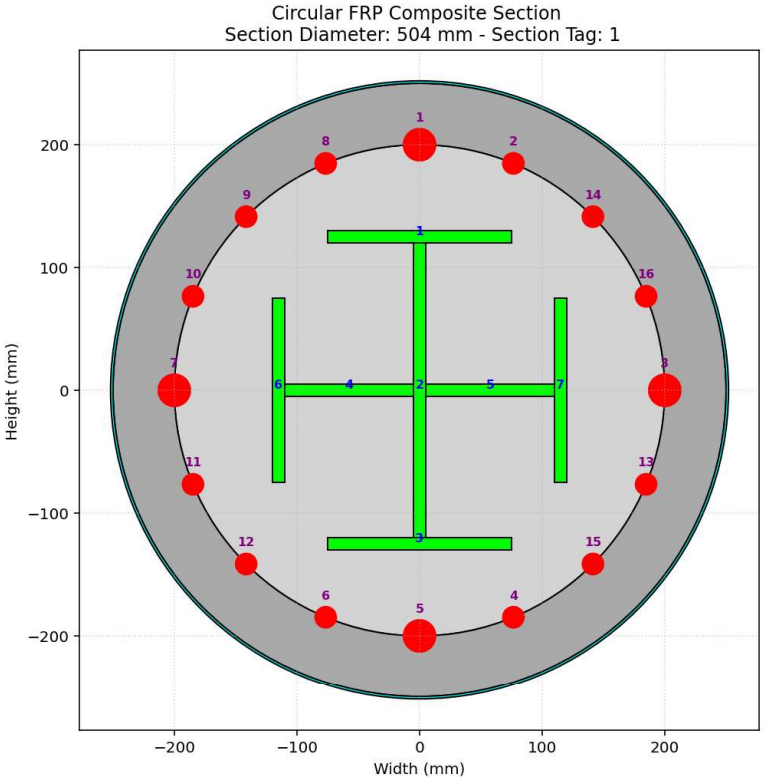


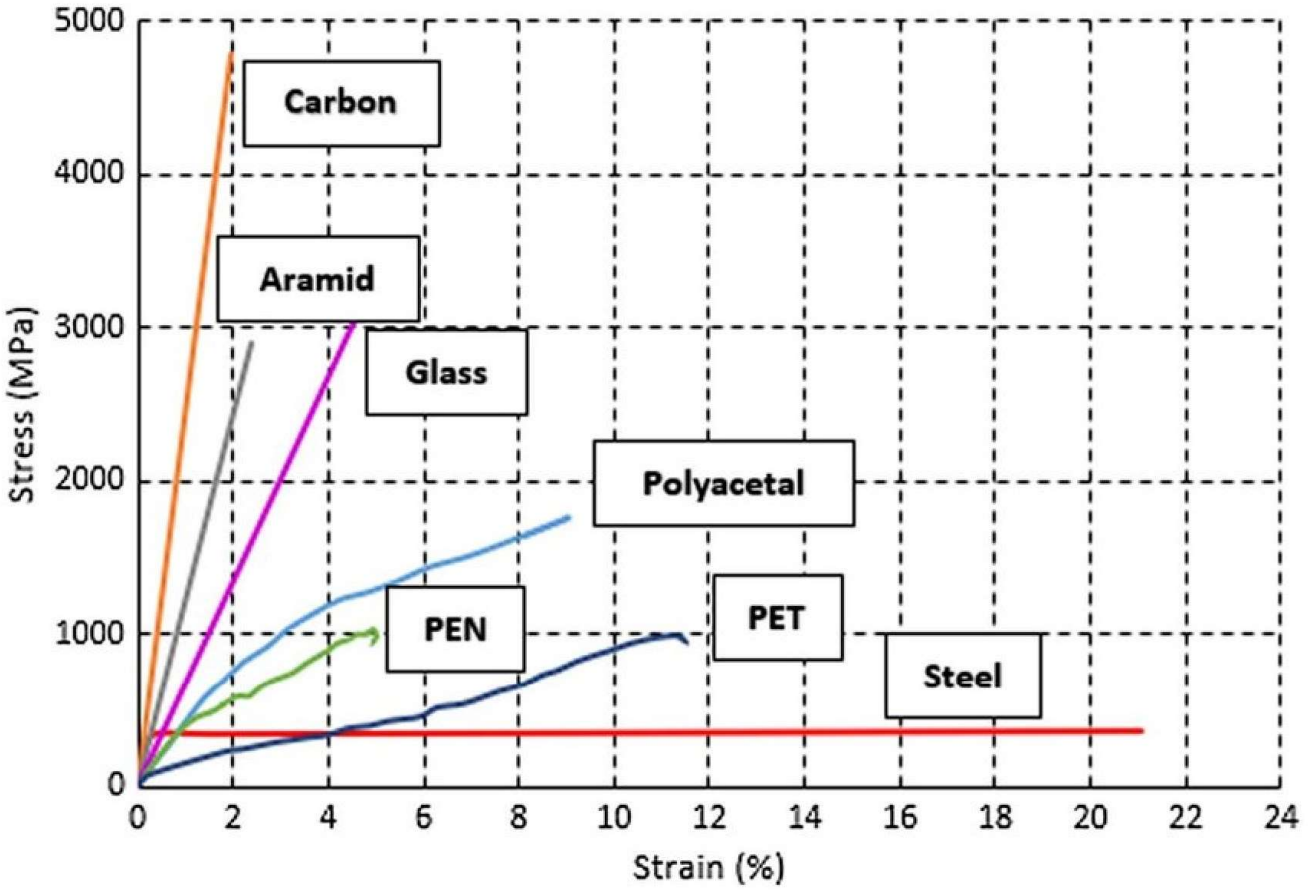
COMPOSITE_REC_I_SECTION_FUN

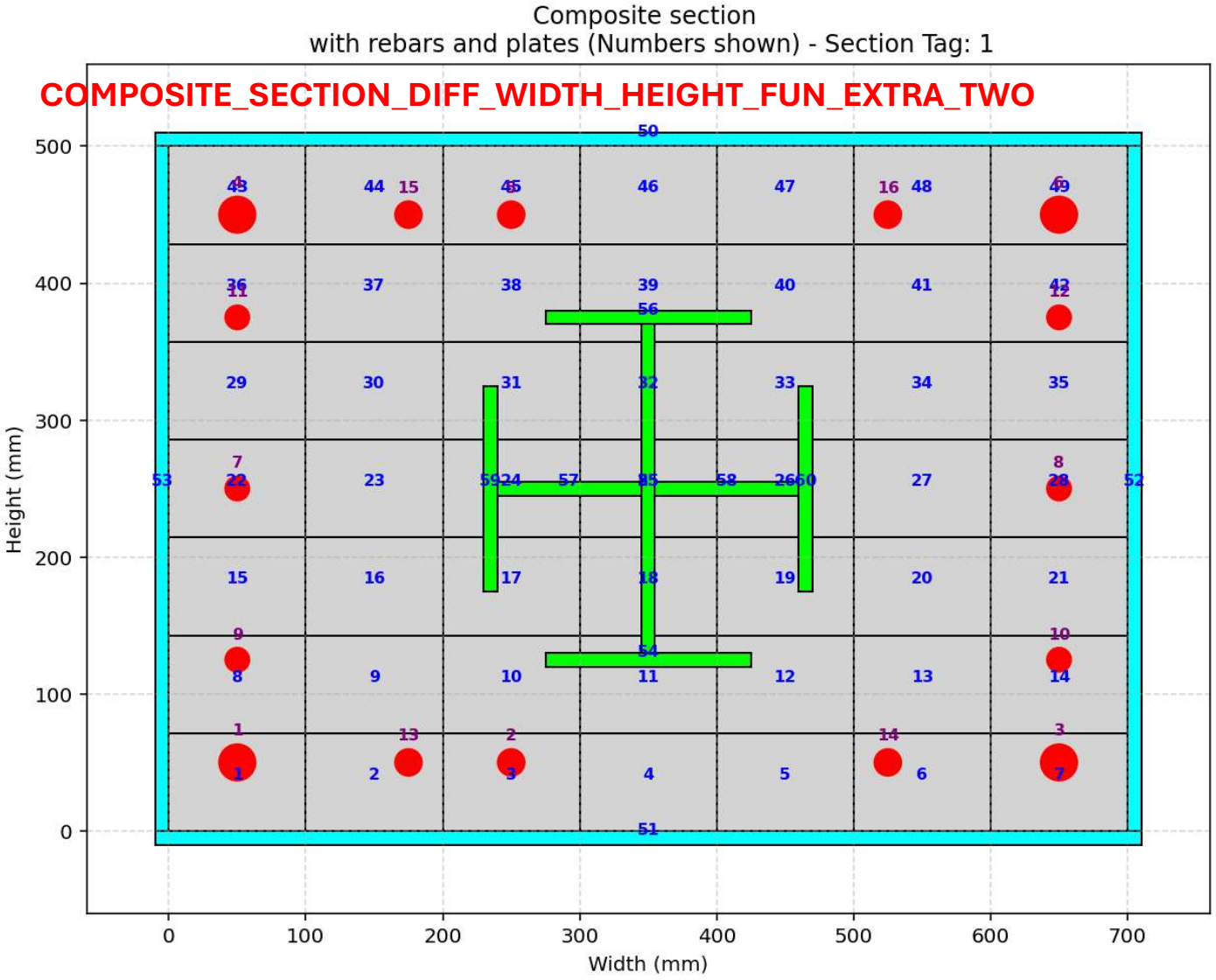
COMPOSITE_CIRCULAR_CONFINED_CONCRETE_SECTION_FUN_EXTRA

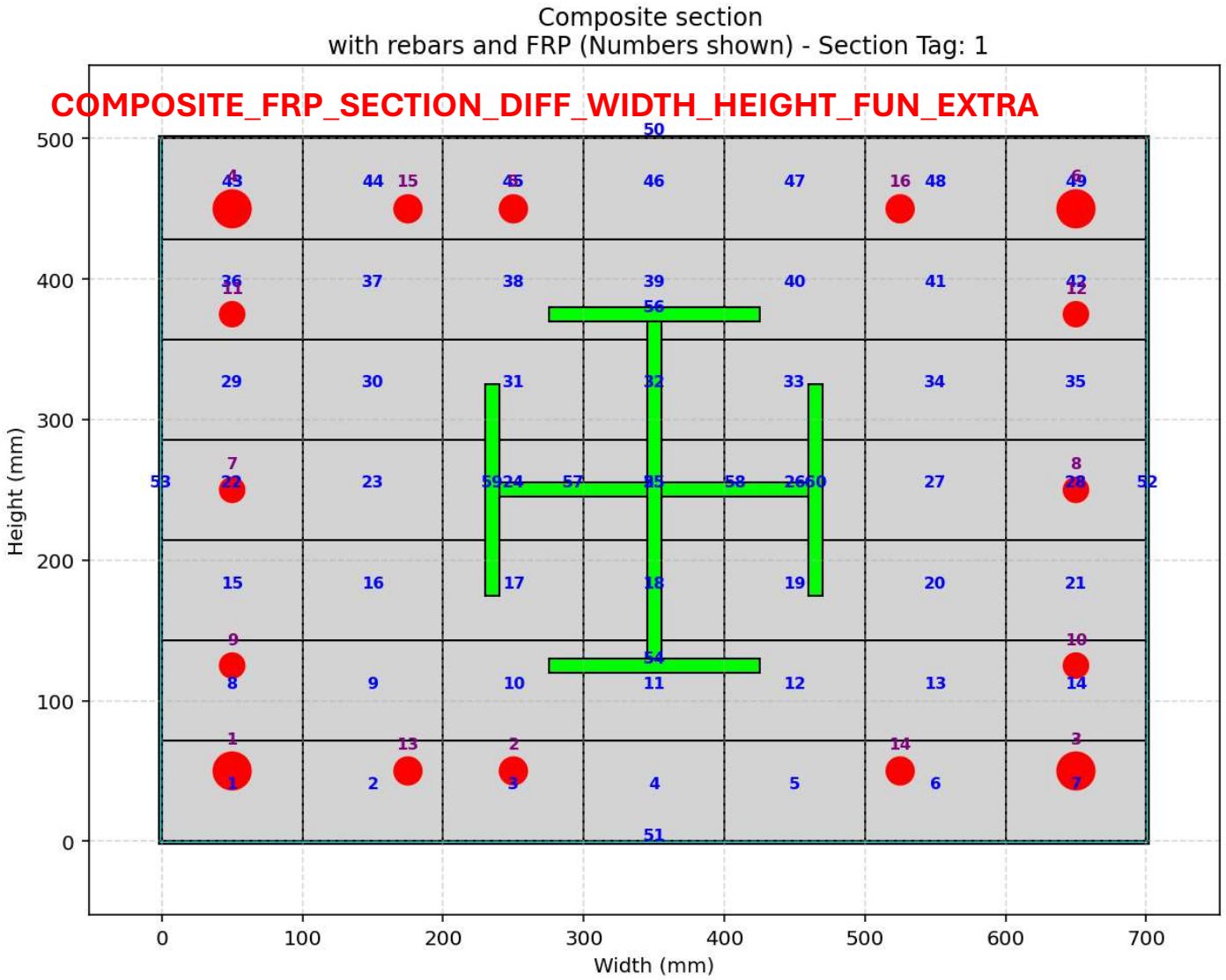


COMPOSITE_FRP_CIRCULAR_CONFINED_CONCRETE_SECTION_FUN_EXTRA






Composite section
with rebars and plates (Numbers shown) - Section Tag: 1
COMPOSITE_SECTION_DIFF_WIDTH_HEIGHT_FUN_EXTRA_TWO
Height (mm)
Width (mm)
0
100
200
300
400
500
600
700


Composite section
with rebars and FRP (Numbers shown) - Section Tag: 1
COMPOSITE_FRP_SECTION_DIFF_WIDTH_HEIGHT_FUN_EXTRA
Height (mm)
Width (mm)
0
100
200
300
400
500
600
700

STATIC ANALYSIS

Spyder (Python 3.12)

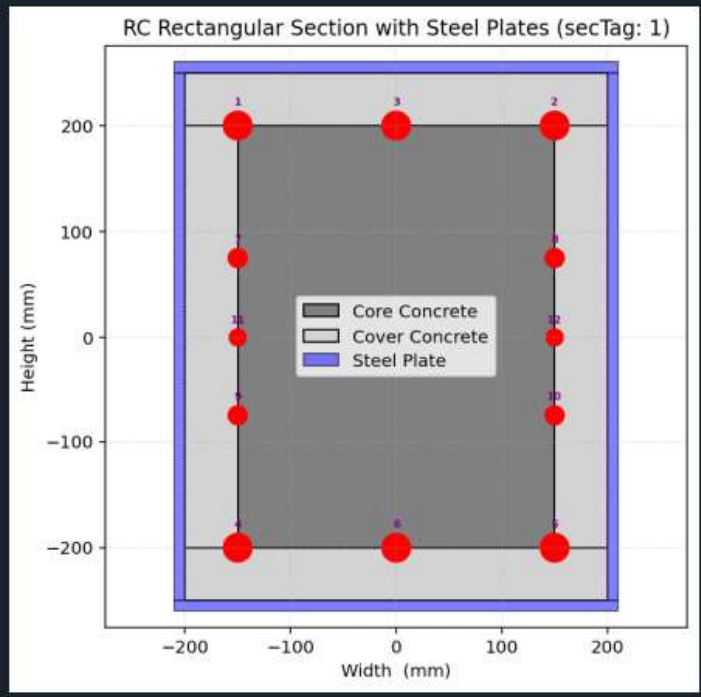
File Edit Search Source Run Debug Consoles Projects Tools View Help

...ENERGY_DISSIPATION_CAPACITY_INDEX\10_COMPOSITE_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...FRAME_8_ANA\P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py

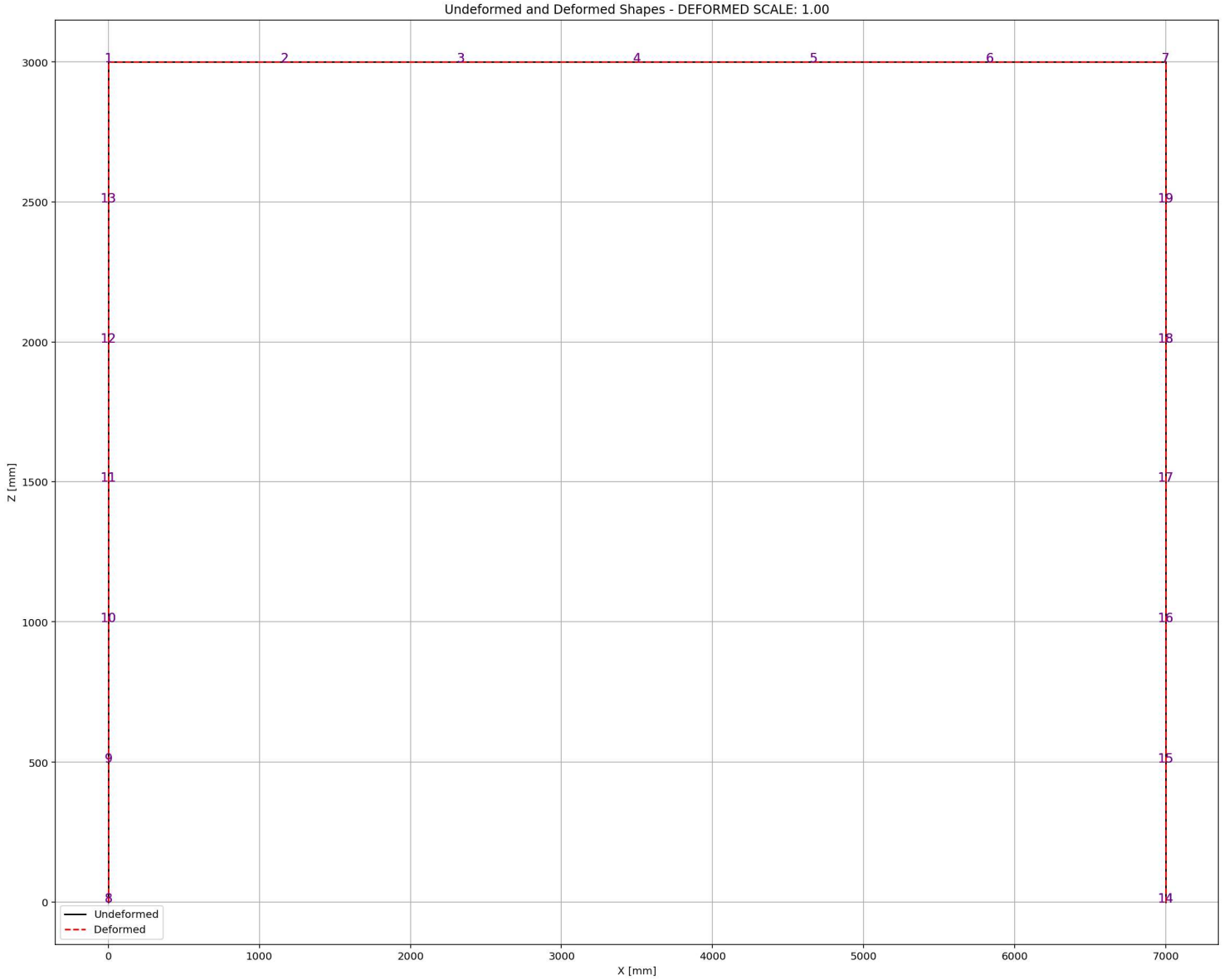
P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py | COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

```
396         return PERIODmin, PERIODmax
397
398     if ANAL_TYPE == 'STATIC':
399         ops.integrator('LoadControl', 1.0)
400         ops.analysis('Static')
401         OK = ops.analyze(1)
402         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
403         ops.reactions()
404         reactionX.append(ops.nodeReaction(8, 1)+ops.nodeReaction(14, 1))     # BASE REACTI
405         reactionY.append(ops.nodeReaction(8, 2)+ops.nodeReaction(14, 2))     # BASE REACTI
406         disp_mid.append(ops.nodeDisp(center_node, 1))                        # DISPLACEMEN
407         dispX.append(ops.nodeDisp(center_node, 1))                           # DISPLACEMEN
408         dispY.append(ops.nodeDisp(center_node, 2))                           # DISPLACEMEN
409         # Store axial forces, strain and stress
410         for ele_id in ele_axialforce.keys():
411             ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'force')[0])       # [N
412             ele_shearforce[ele_id].append(ops.eleResponse(ele_id, 'force')[1])       # [N
413             ele_momentforce[ele_id].append(ops.eleResponse(ele_id, 'force')[2])      # [N
414         # Store displacements
415         for node_id in node_displacementsX.keys():
416             node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
417             node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
418         else:
419             print('\n\nSTATIC ANALYSIS DONE.\n\n')
420
421         DATA = (reactionX, reactionY, disp_mid,
422                 ele_axialforce, ele_shearforce, ele_momentforce,
423                 node_displacementsX, node_displacementsY,
424                 dispX, dispY)
425
426         return  DATA
427
428     if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
429         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 84, Col 34 UTF-8 CRLF RW Mem 40%


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed

PUSHOVER ANALYSIS

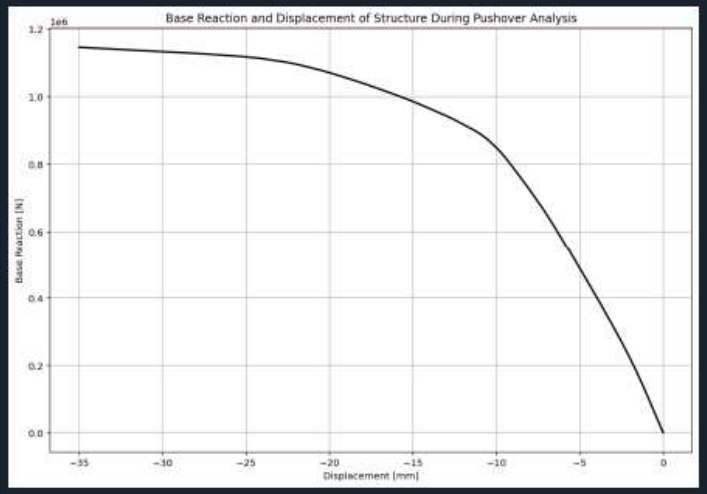
Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...FRAME_8_ANA\P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py

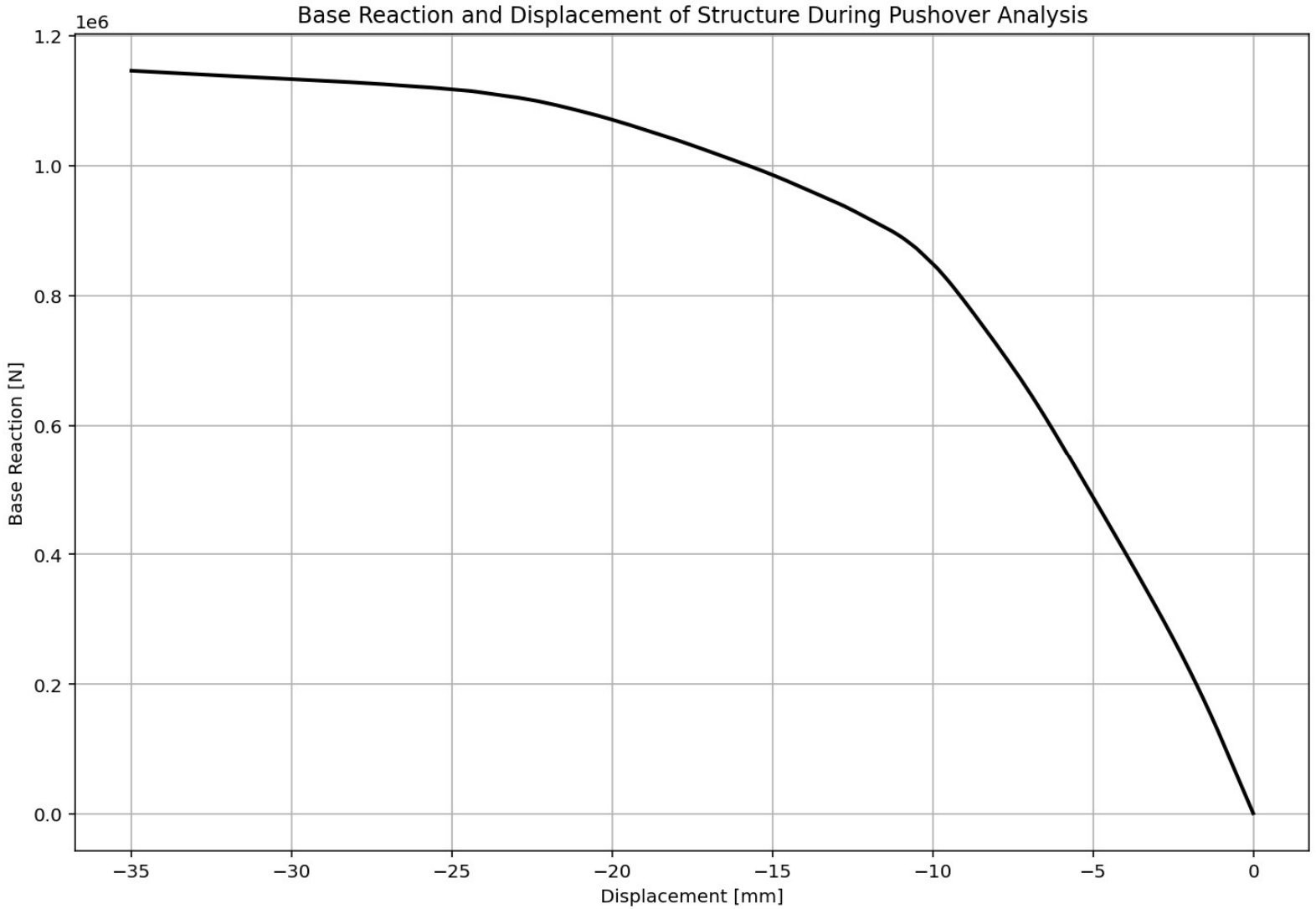
P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py | COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

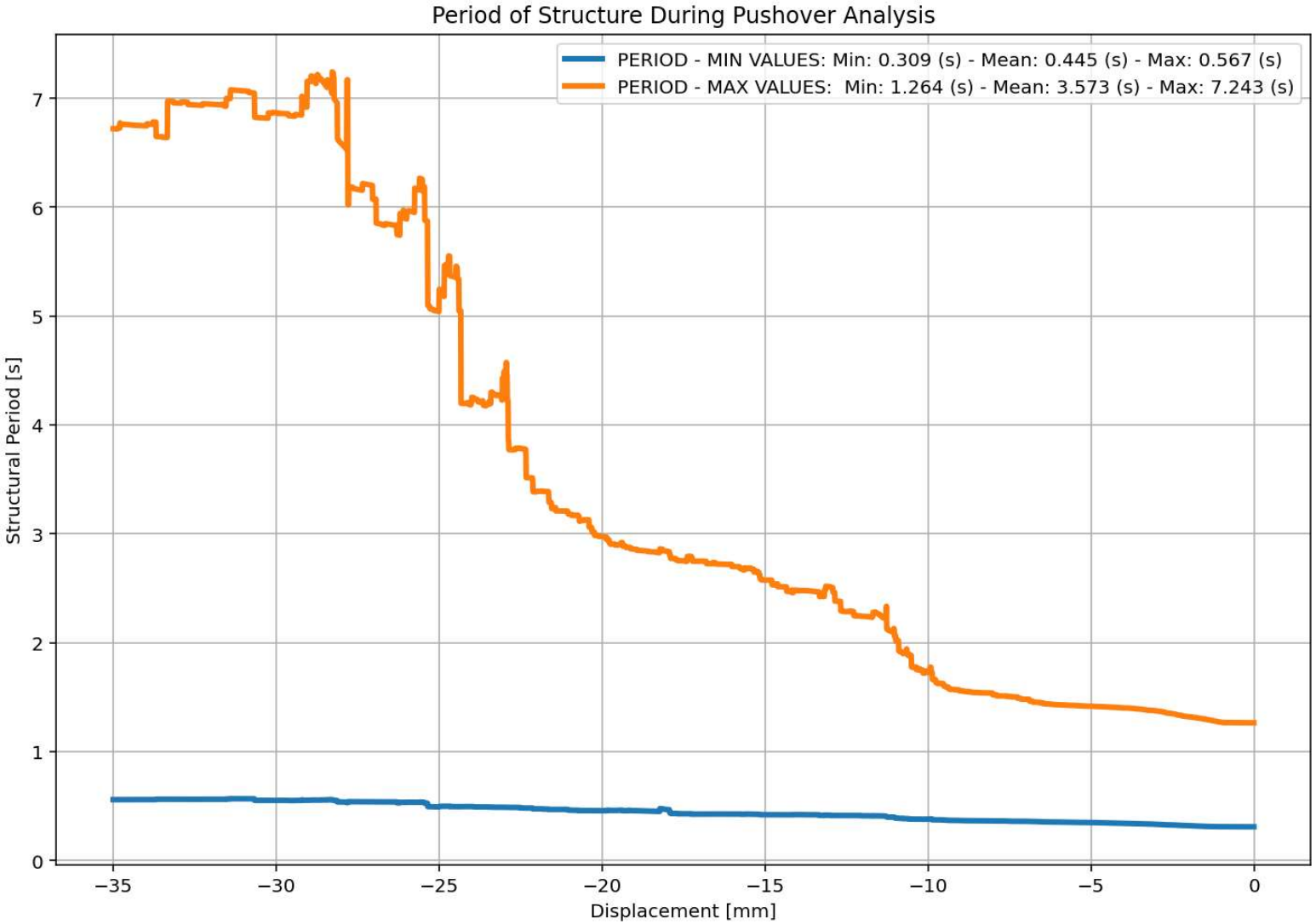
```
427
428     if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
429         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
430         DINCR = -0.01   # [mm] Incremental Displacement
431         DMAX = -35.0    # [mm] Max. Displacement
432         ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
433         ops.analysis('Static')
434         Nsteps =  int(np.abs(DMAX/ DINCR))
435         STEP = 0.0
436         for step in range(Nsteps):
437             OK = ops.analyze(1)
438             S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
439             ops.reactions()
440             reactionX.append(ops.nodeReaction(8, 1)+ops.nodeReaction(14, 1))     # BASE RE
441             reactionY.append(ops.nodeReaction(8, 2)+ops.nodeReaction(14, 2))     # BASE RE
442             disp_mid.append(ops.nodeDisp(center_node, 1))                        # DISPLAC
443             dispX.append(ops.nodeDisp(center_node, 1))                           # DISPLAC
444             dispY.append(ops.nodeDisp(center_node, 2))                           # DISPLAC
445             # Store elements force, strain and stress
446             for ele_id in ele_axialforce.keys():
447                 ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'force')[0])
448                 ele_shearforce[ele_id].append(ops.eleResponse(ele_id, 'force')[1])
449                 ele_momentforce[ele_id].append(ops.eleResponse(ele_id, 'force')[2])
450             # Store displacements
451             for node_id in node_displacementsX.keys():
452                 node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
453                 node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
454             # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
455             #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(3, 0.5*DR, DR, 0, 1)
456             PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(3, PLOT=True)
457             PERIOD_MIN.append(PERIODmin)
458             PERIOD_MAX.append(PERIODmax)
459             # SECTION STRESS-STRAIN FOR bOTTOM FIBER
460             stressB, strainB = S10.SECTION_STRESS_STRAIN(center_node, SEC_TAG, -Depth*0.5,
```

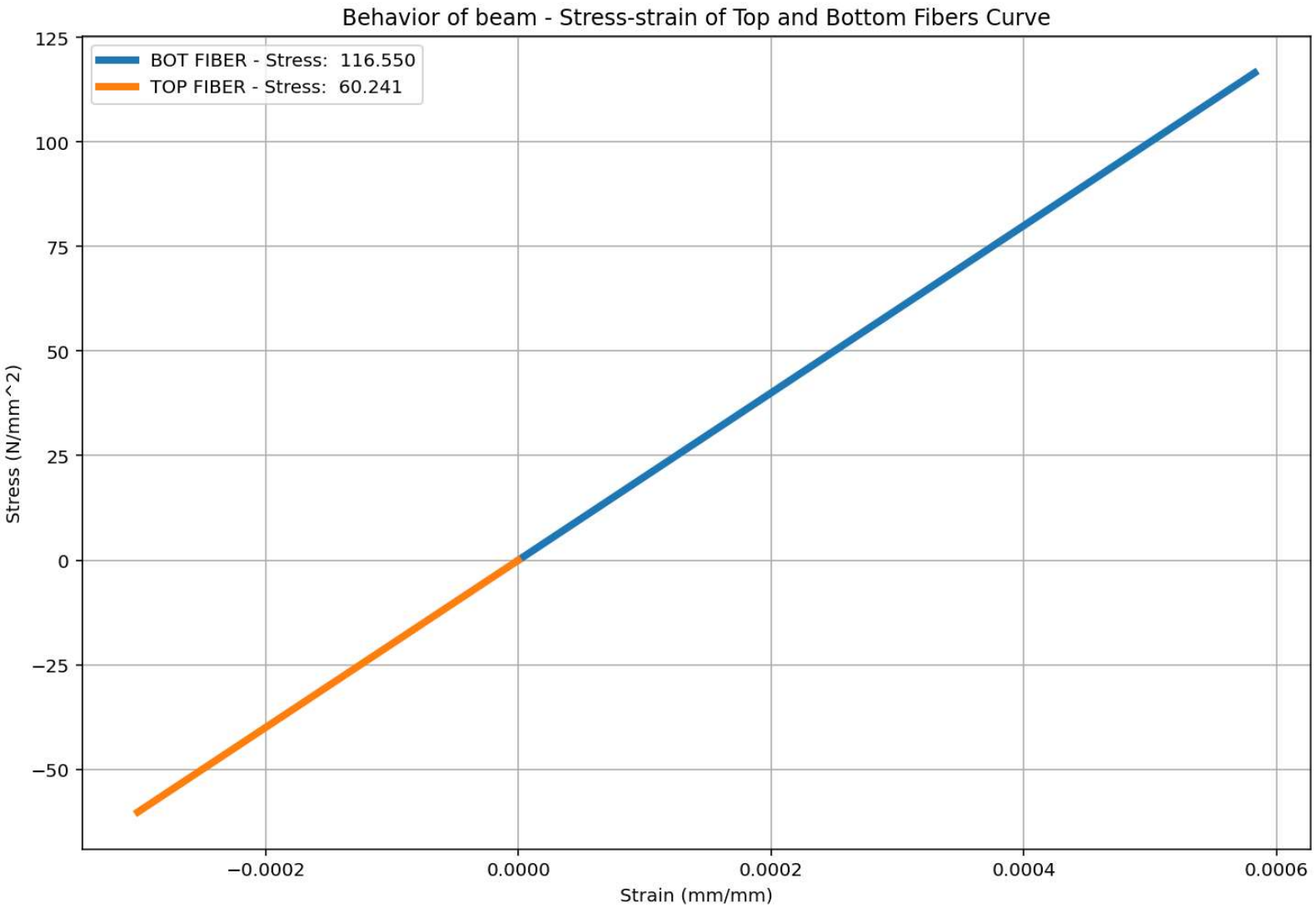


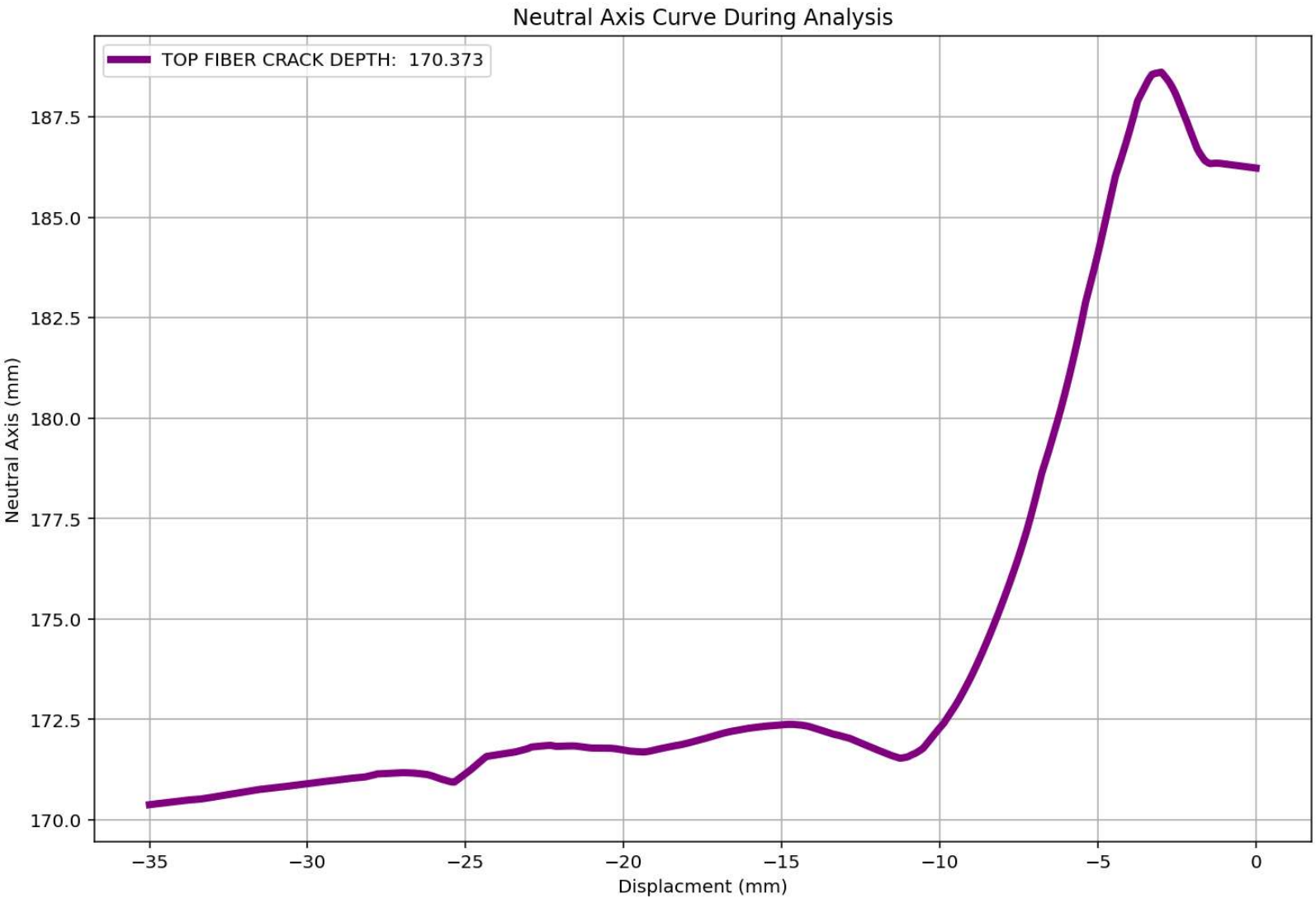
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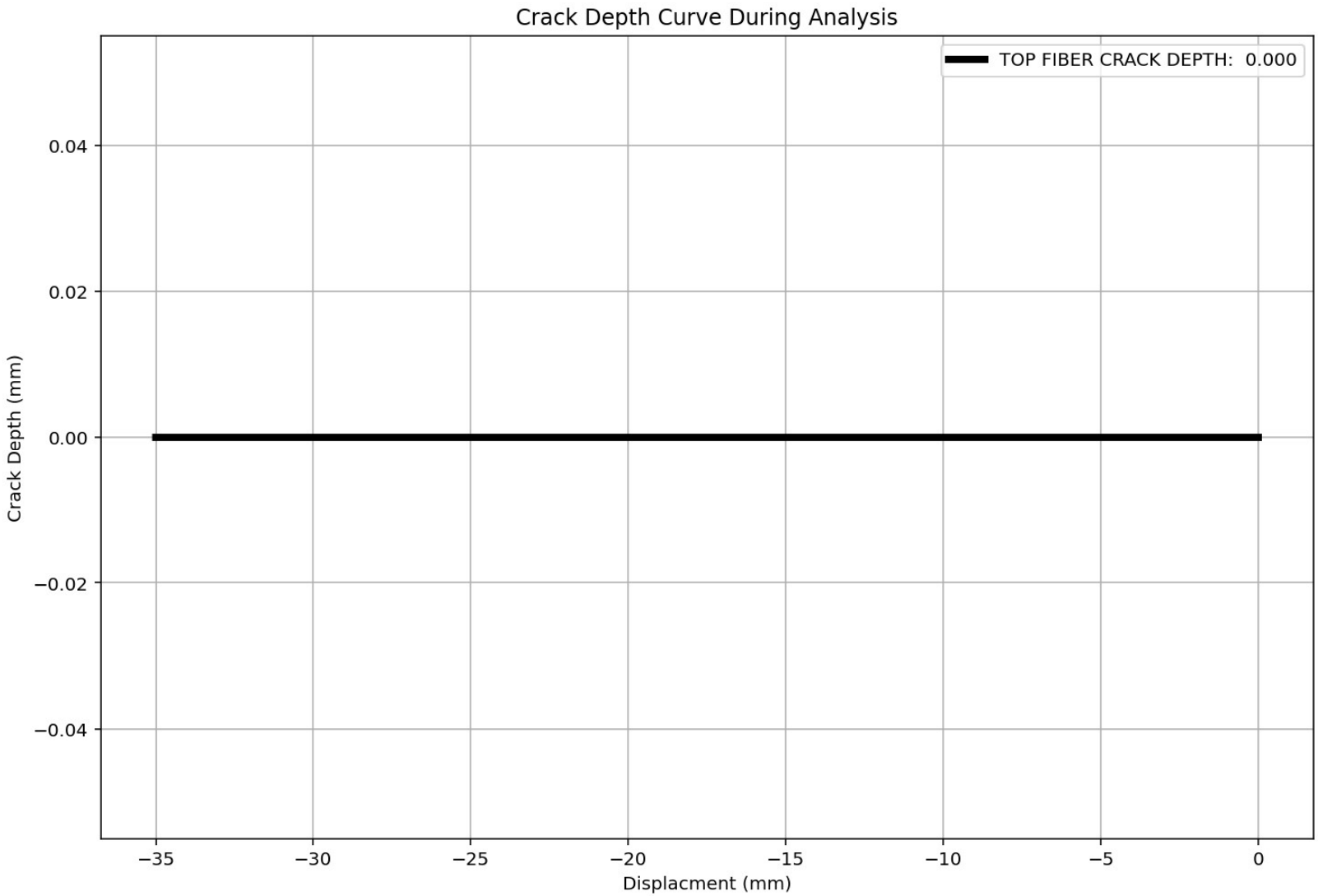
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 84, Col 34 | UTF-8 | CRLF | RW | Mem 40%

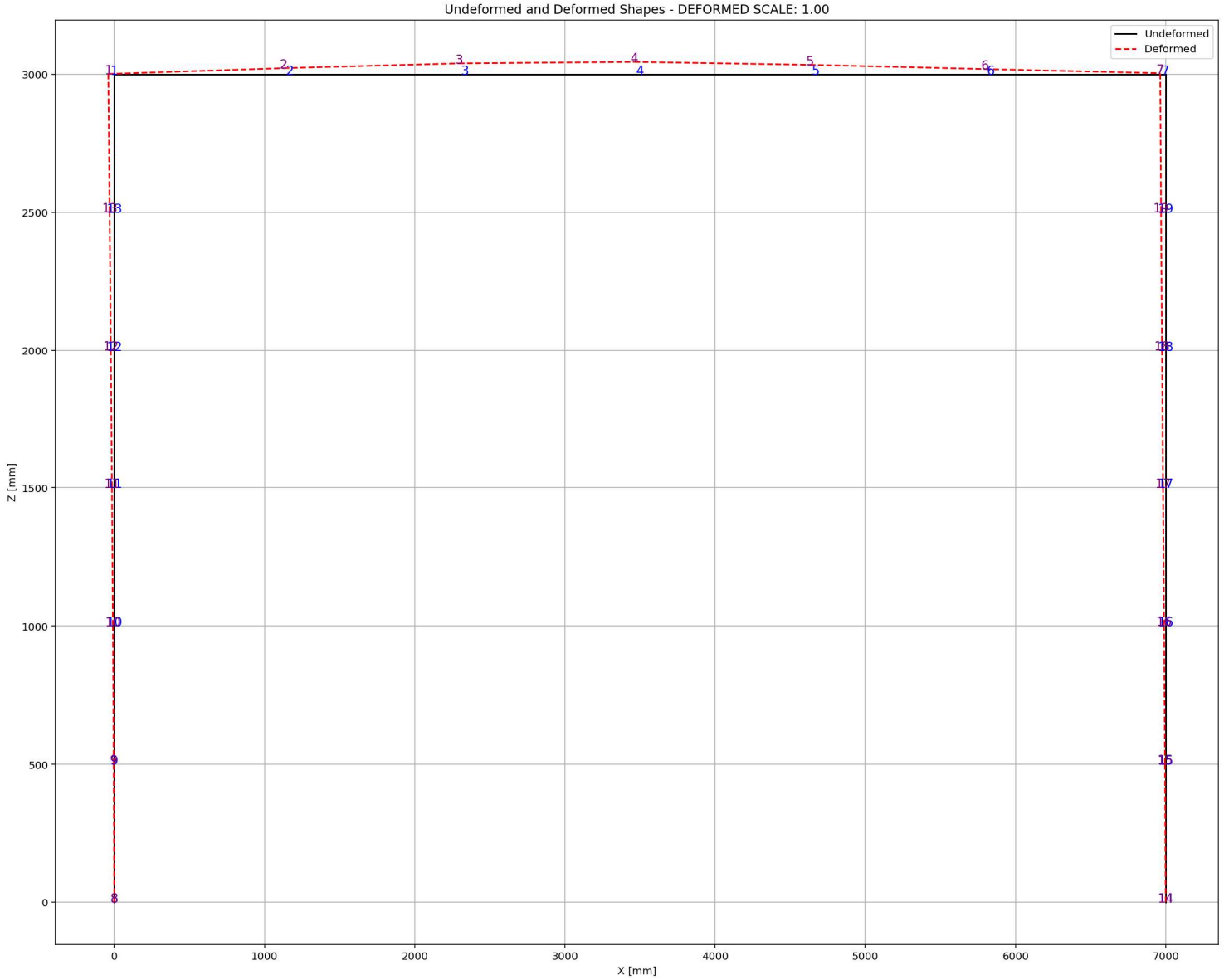

Base Reaction and Displacement of Structure During Pushover Analysis
1e6
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Base Reaction [N]
−35
−30
−25
−20
−15
−10
−5
0
Displacement [mm]


Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: 0.309 (s) - Mean: 0.445 (s) - Max: 0.567 (s)
PERIOD - MAX VALUES: Min: 1.264 (s) - Mean: 3.573 (s) - Max: 7.243 (s)
Structural Period [s]
0
1
2
3
4
5
6
7
−35
−30
−25
−20
−15
−10
−5
0
Displacement [mm]


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 116.550
TOP FIBER - Stress: 60.241
125
100
75
50
25
0
−25
−50
Stress (N/mm^2)
−0.0002
0.0000
0.0002
0.0004
0.0006
Strain (mm/mm)


Neutral Axis Curve During Analysis
TOP FIBER CRACK DEPTH: 170.373
Neutral Axis (mm)
187.5
185.0
182.5
180.0
177.5
175.0
172.5
170.0
−35
−30
−25
−20
−15
−10
−5
0
Displacment (mm)


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−35
−30
−25
−20
−15
−10
−5
0
Displacment (mm)


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

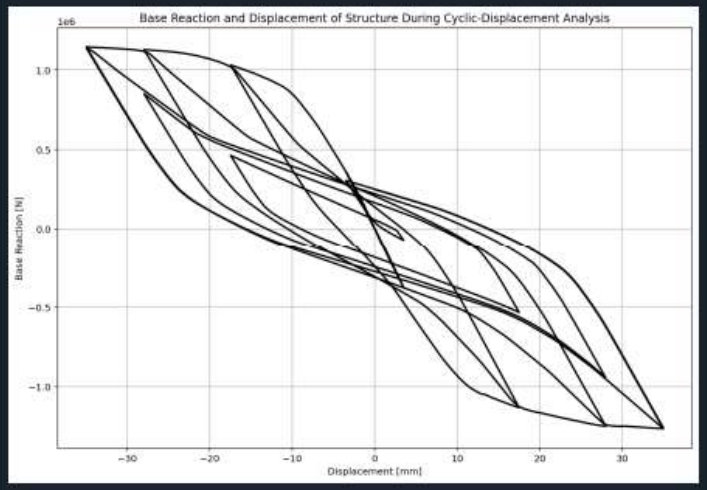
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...ENERGY_DISSIPATION_CAPACITY_INDEX\10_COMPOSITE_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...FRAME_8_ANA\P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py

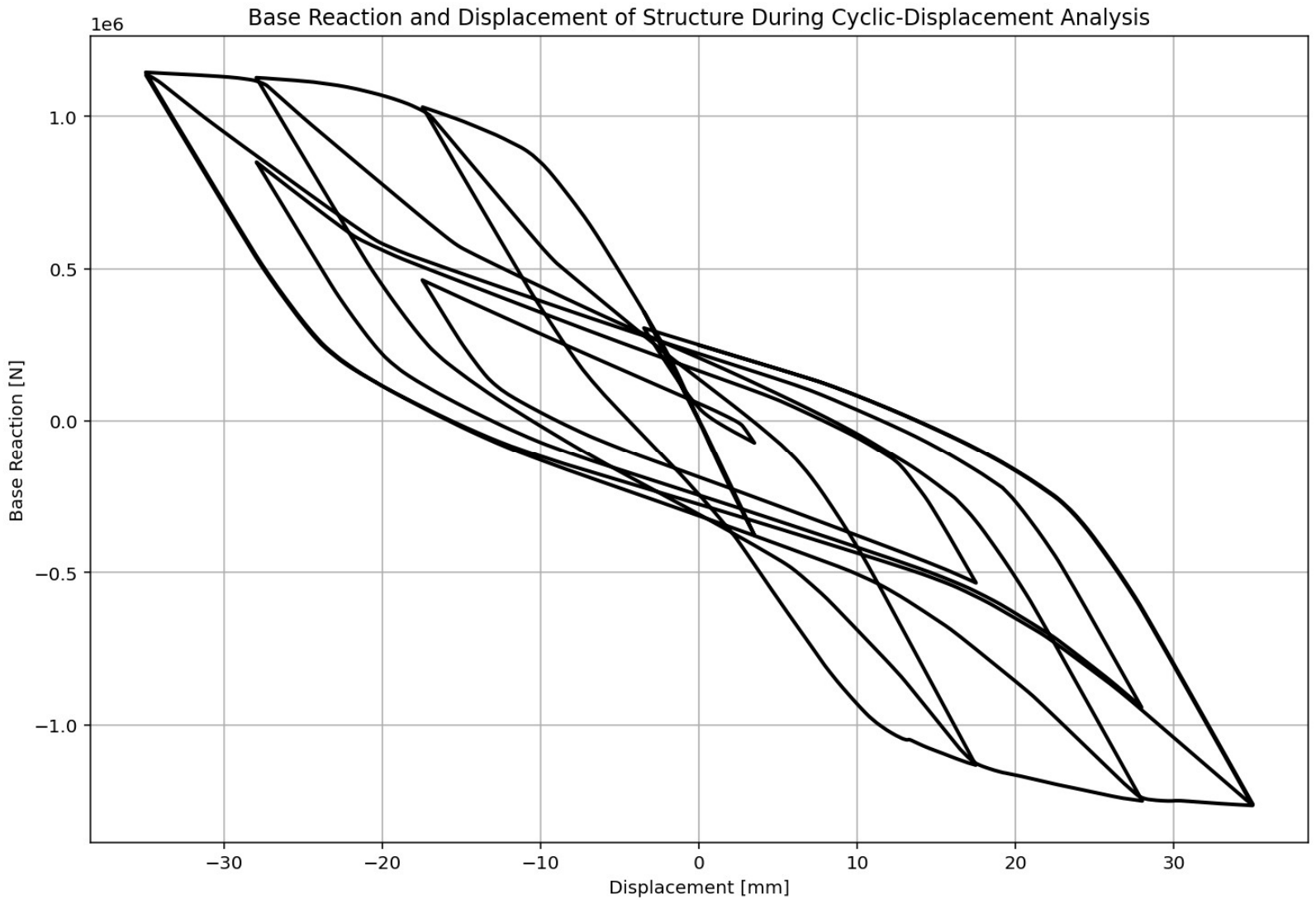
P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py | COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

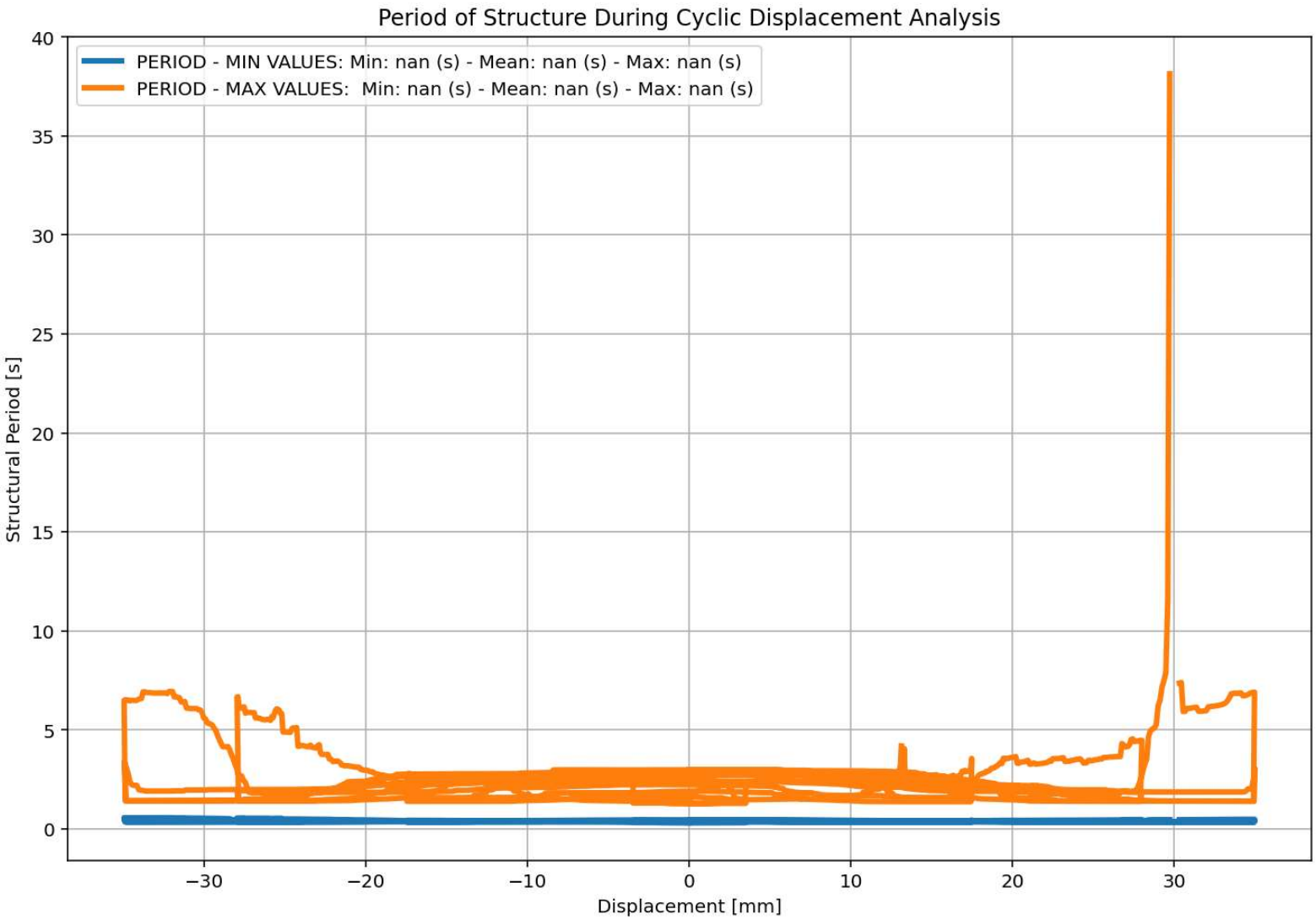
```
492
493     if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
494         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
495         DMAX = -35.0    # [mm] Max. Displacement
496         n_points = 10000
497         # 4. CYCLIC DISPALCEMENT PROTOCOL
498         # Key strain points (same logic as your protocol)
499         key_disp = np.array([
500              0.0,
501              0.1*DMAX,   -0.1*DMAX,
502              0.5*DMAX,   -0.5*DMAX,
503              0.8*DMAX,   -0.8*DMAX,
504              DMAX,       -DMAX,
505              0.1*DMAX,   -0.1*DMAX,
506              0.5*DMAX,   -0.5*DMAX,
507              0.8*DMAX,   -0.8*DMAX,
508              DMAX,       -DMAX,
509              0.0
510         ])
511
512         # Generate 10000-point displacement protocol
513         disp_protocol = np.interp(
514             np.linspace(0, len(key_disp) - 1, n_points),
515             np.arange(len(key_disp)),
516             key_disp
517         )
518
519         ops.analysis('Static')
520         STEP = 0.0
521         for target_disp in disp_protocol:
522             current_disp = ops.nodeDisp(center_node, 1) # DISPALCEMENT APPLIED IN NODE IN
523             dU = target_disp - current_disp
524             ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
525             OK = ops.analyze(1)
```

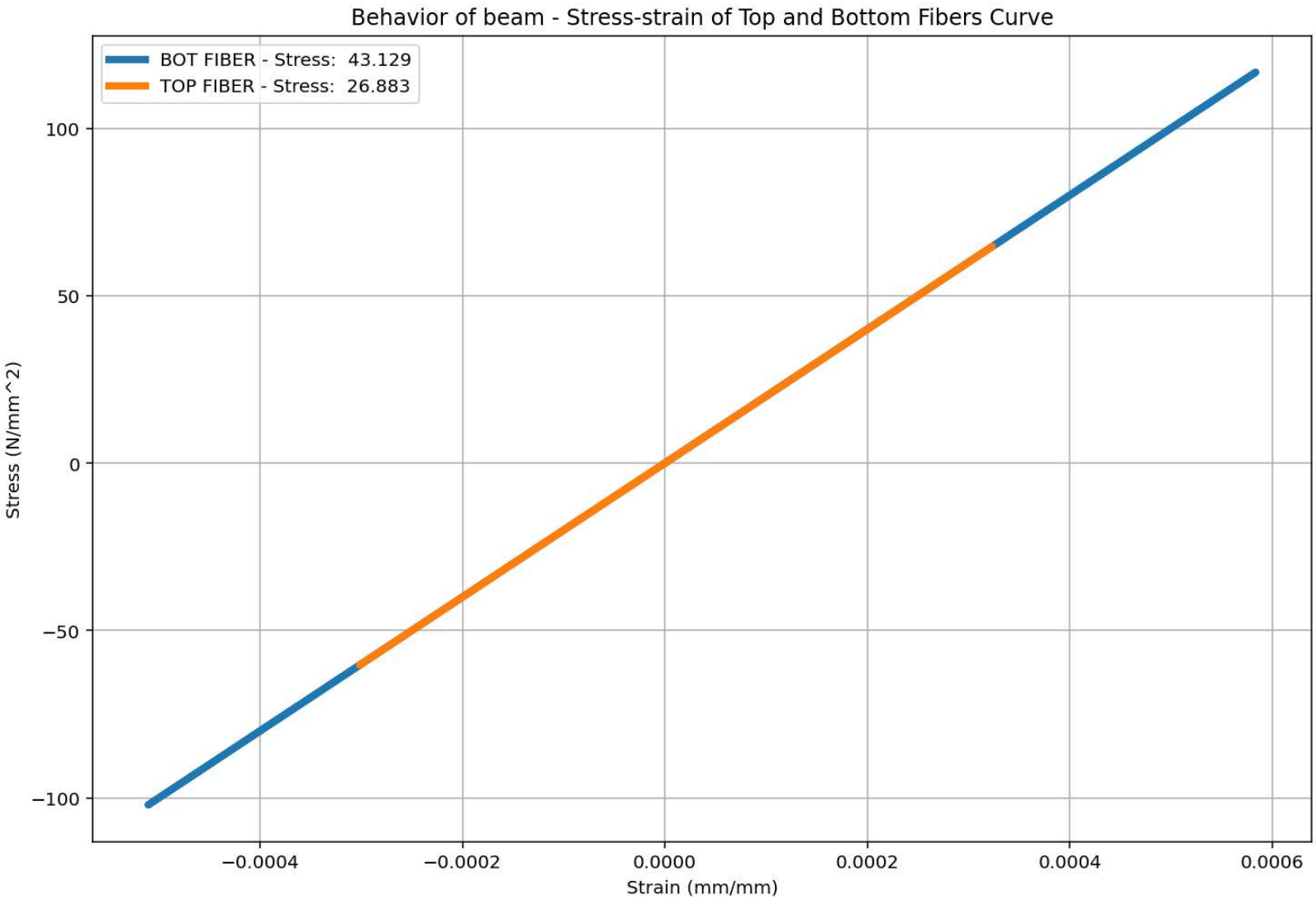


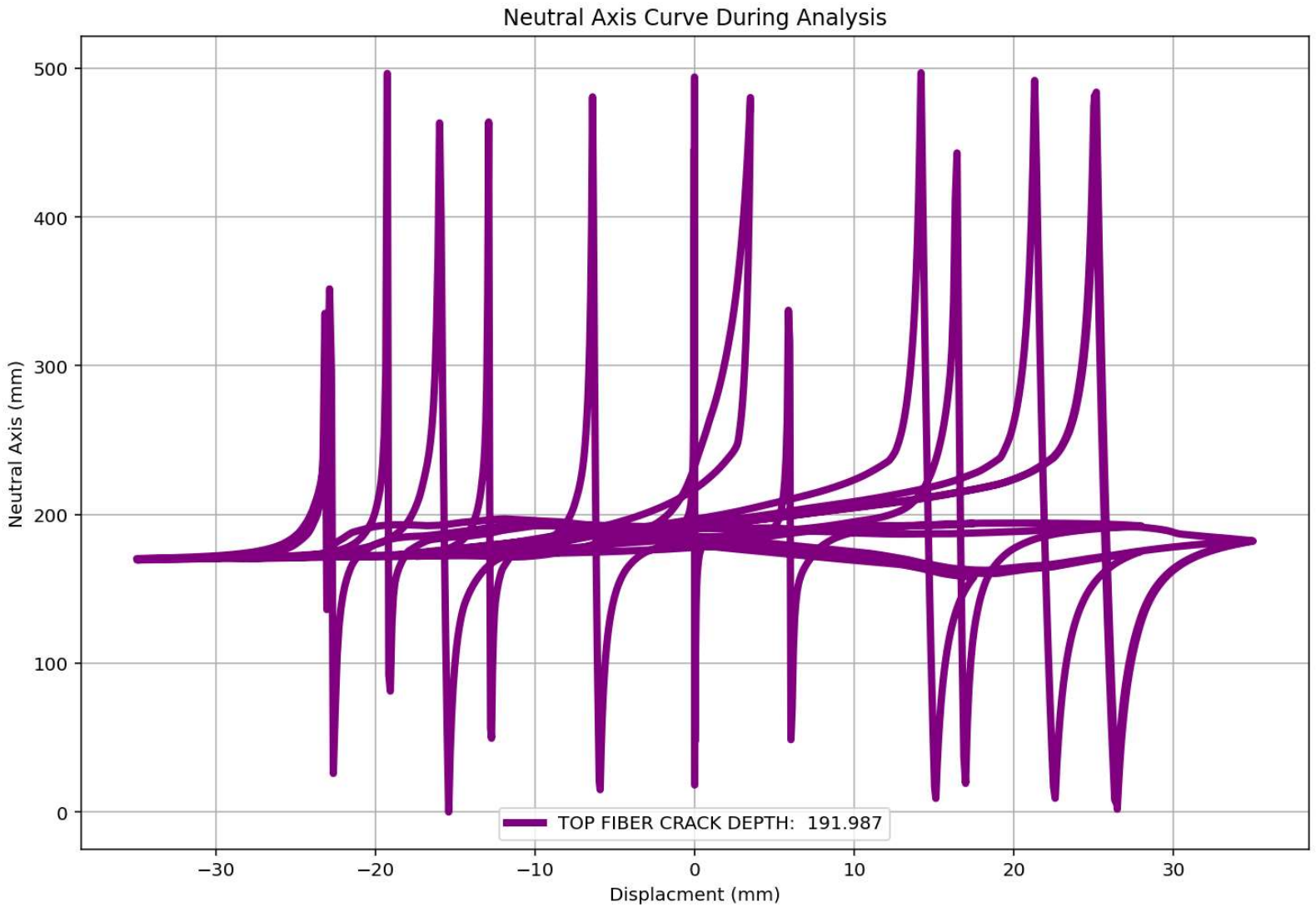
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

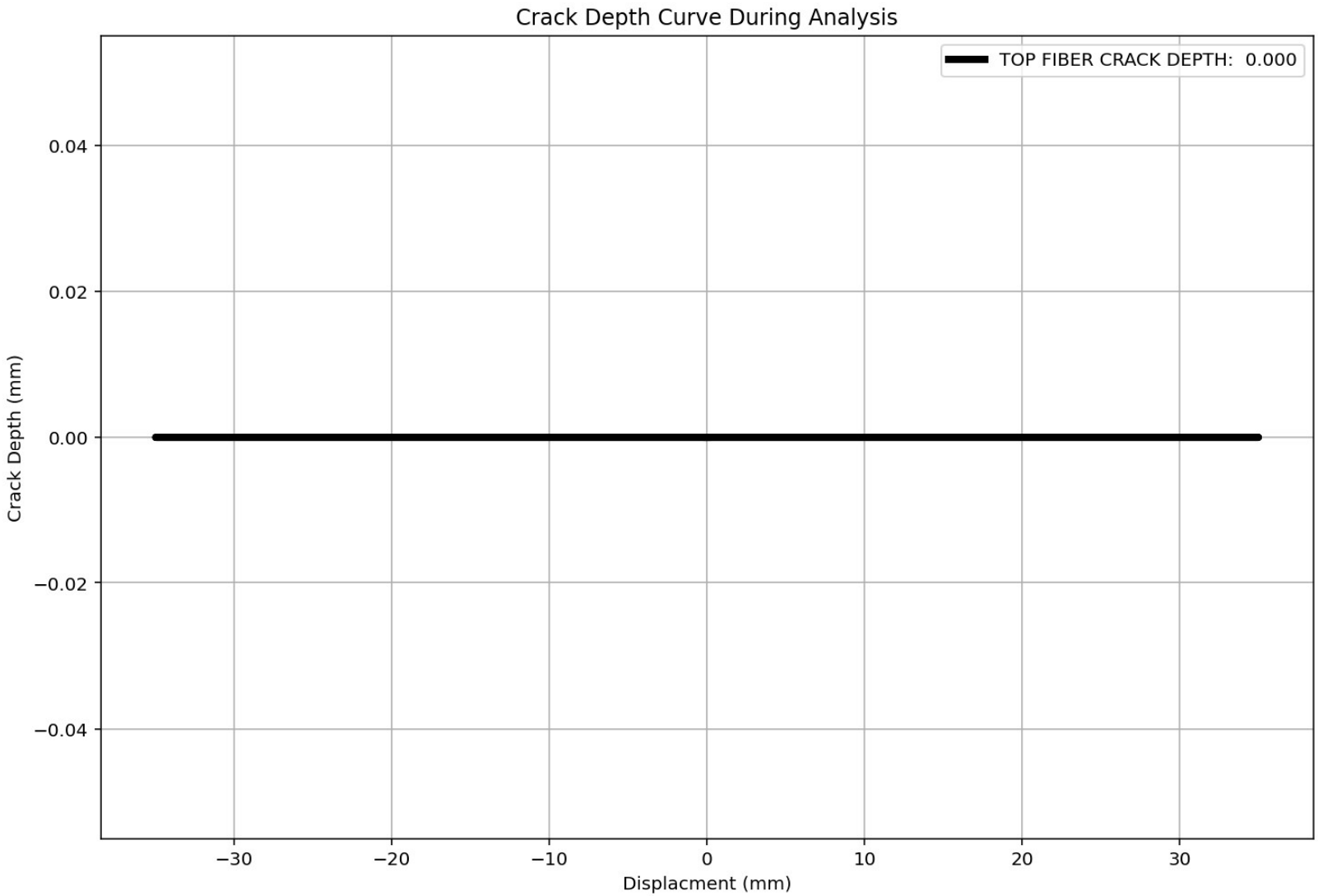
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 84, Col 34 | UTF-8 | CRLF | RW | Mem 41%

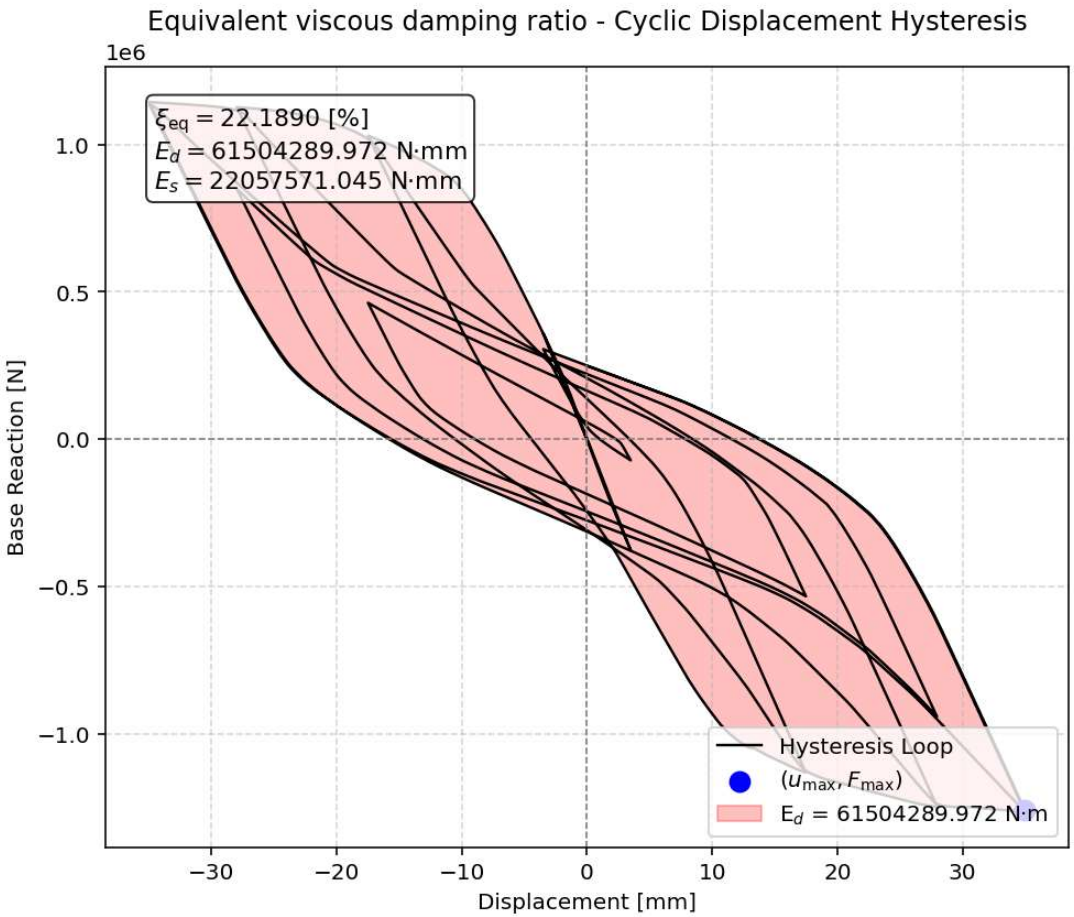

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
1e6
1.0
0.5
0.0
−0.5
−1.0
Base Reaction [N]
−30
−20
−10
0
10
20
30
Displacement [mm]

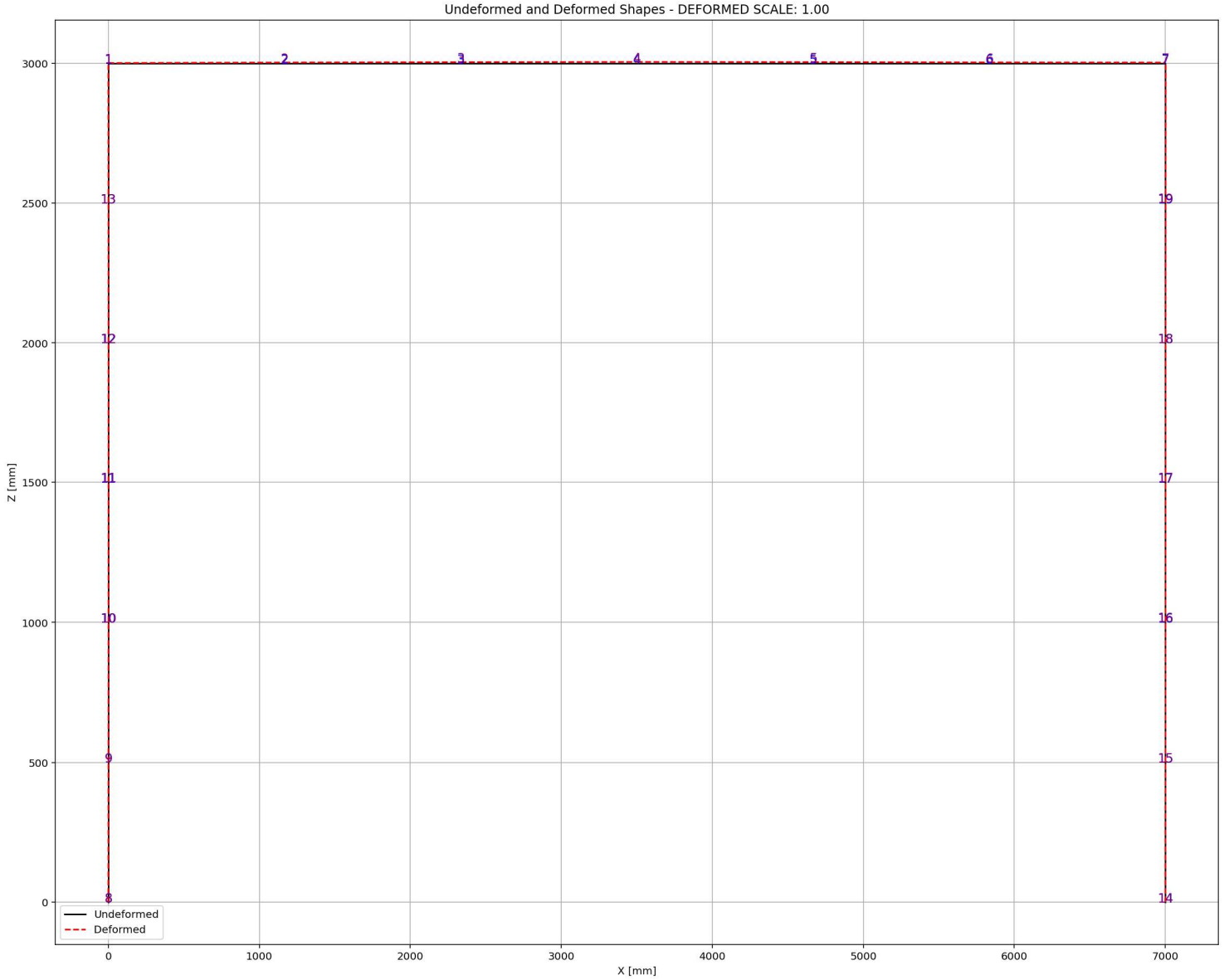

Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Displacement [mm]
40
35
30
25
20
15
10
5
0
−30
−20
−10
0
10
20
30


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 43.129
TOP FIBER - Stress: 26.883
100
50
0
−50
−100
Stress (N/mm^2)
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
Strain (mm/mm)


Neutral Axis Curve During Analysis
Neutral Axis (mm)
500
400
300
200
100
0
−30
−20
−10
0
10
20
30
Displacment (mm)
TOP FIBER CRACK DEPTH: 191.987


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−30
−20
−10
0
10
20
30
Displacment (mm)


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
1e6
$\xi_{eq} = 22.1890$ [%]
$E_d = 61504289.972$ N·mm
$E_s = 22057571.045$ N·mm
Base Reaction [N]
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 61504289.972 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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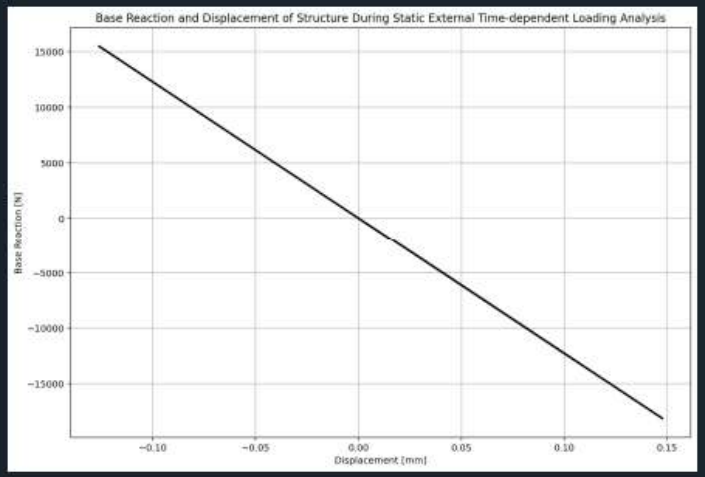
...ENERGY_DISSIPATION_CAPACITY_INDEX\10_COMPOSITE_FRAME_8_ANA

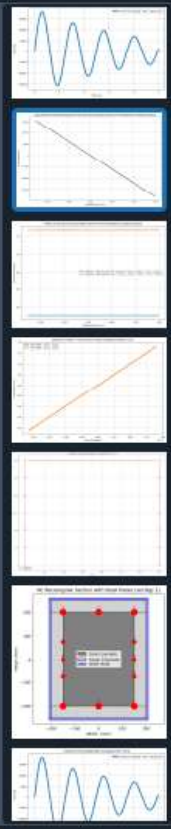
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...FRAME_8_ANA\P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py

P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py | COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

```
581     if ANAL_TYPE == 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING': # STATIC TIME-HISTORY ANALYS
582         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
583         # IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
584         duration = 20.0             # [s] Analysis duration
585         dt = 0.01                   # [s] Time step
586         DT = dt                     # [s] Time step
587         DT_time = 5.0               # [s] Total external Load Analysis Durations [*******]
588         force_amplitude = 19600.0   # [N] Amplitude Force
589         omega_DT = 5.0715           # [rad/s] Natural angular frequency
590         time_steps = int(duration/dt)
591
592         # Check function
593         def CHECK_FUN(DT_time ,duration):
594             if DT_time > duration:
595                 print('\n\nAnalysis Duration Must be greater than External Load Duration\n
596                 exit()
597             return -1
598
599         CHECK_FUN(DT_time ,duration)
600         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 s
601             import numpy as np
602             import matplotlib.pyplot as plt
603             # External Load Durations
604             num_steps = int(DT_time / DT)
605             load_time = np.linspace(0, DT_time, num_steps)
606             target_frequency = 1.0 * omega_DT  # Target excitation frequency
607             DT_load = force_amplitude * np.sin(target_frequency * load_time)
608             # Plot External Time-dependent Loading
609             plt.figure(figsize=(10, 6))
610             plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load):
611             plt.title('External Time-dependent Loading Over Time')
612             plt.xlabel('Time (s)')
613             plt.ylabel('Force (N)')
614             plt.grid(True)
```

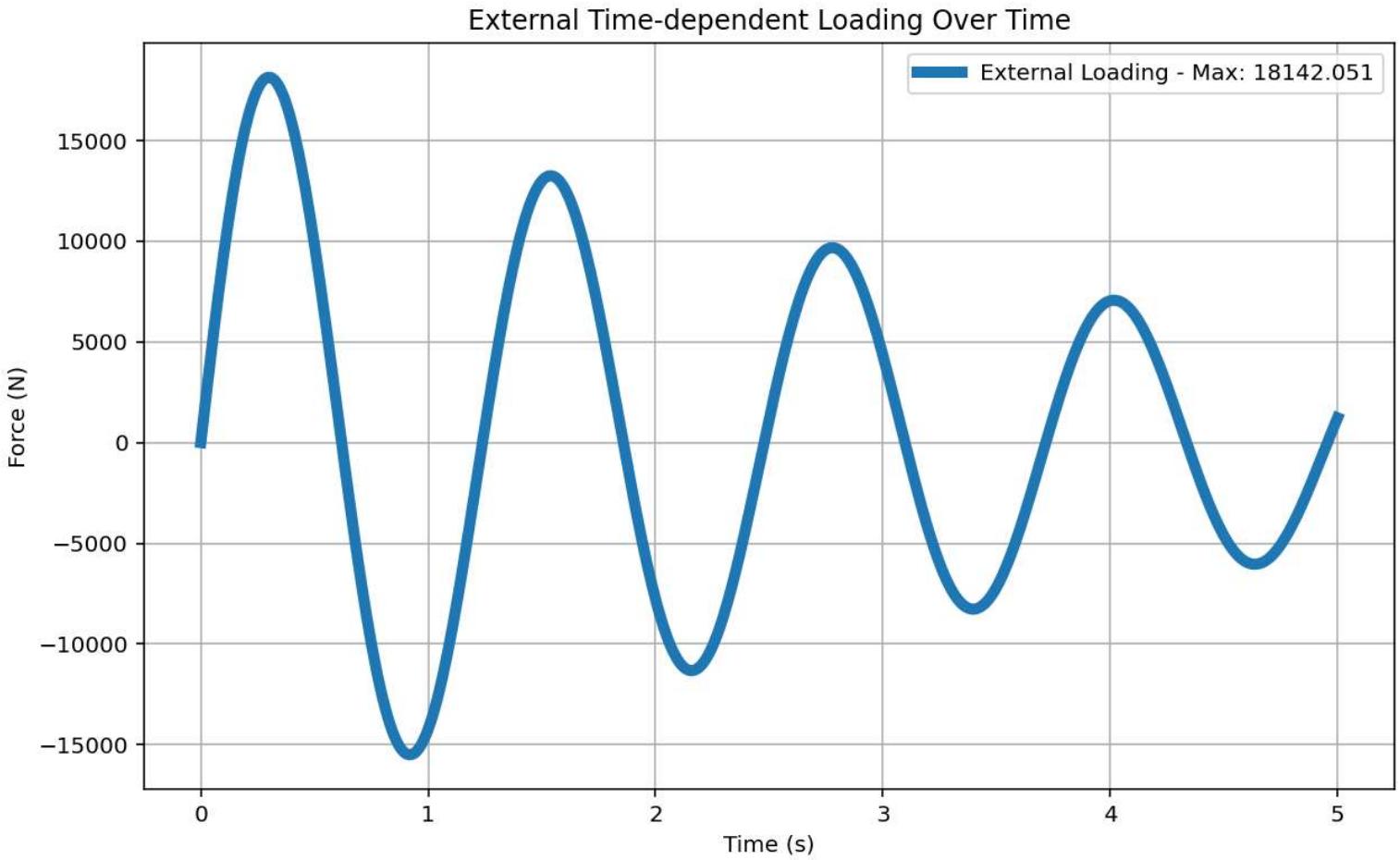
30 %

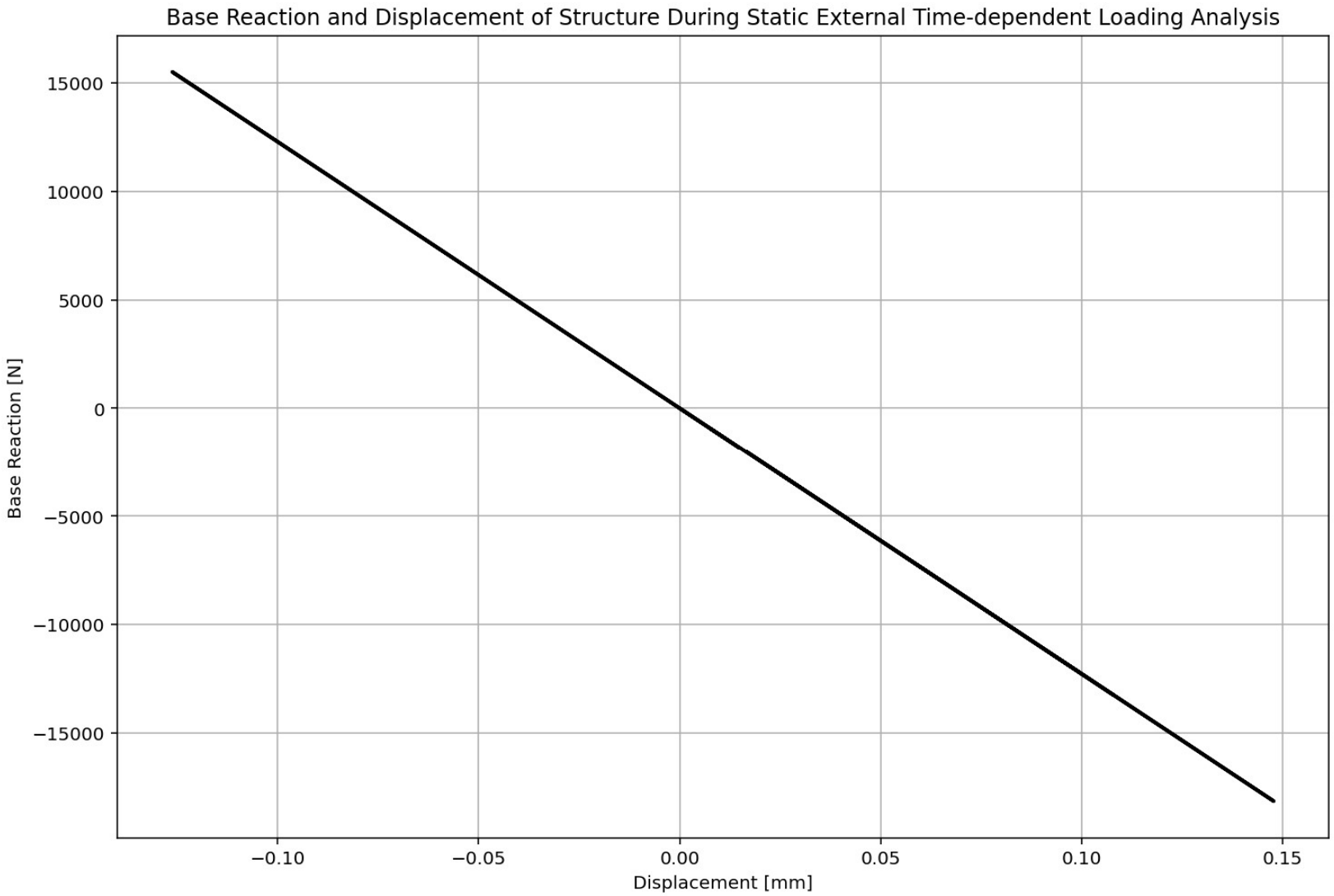


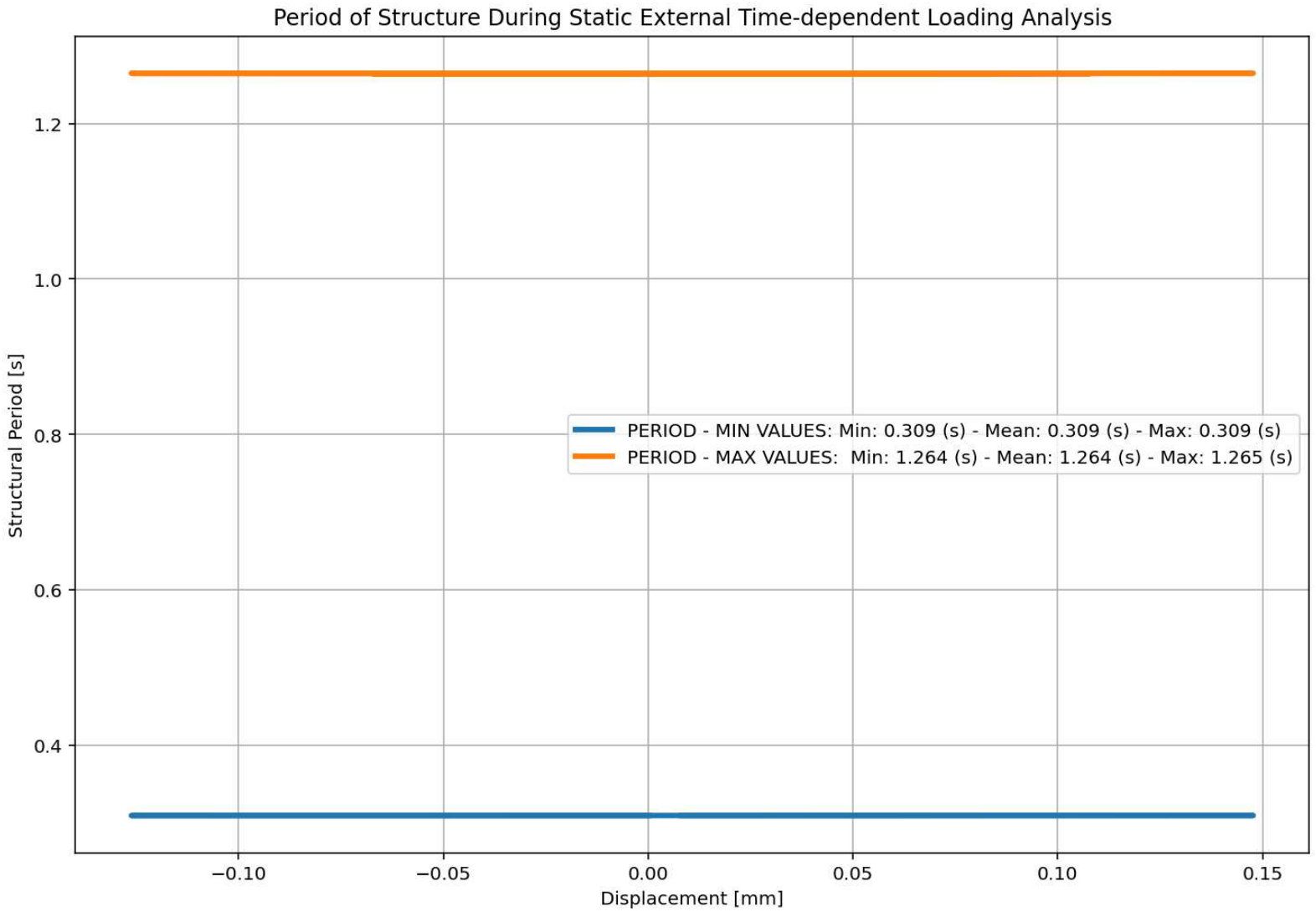


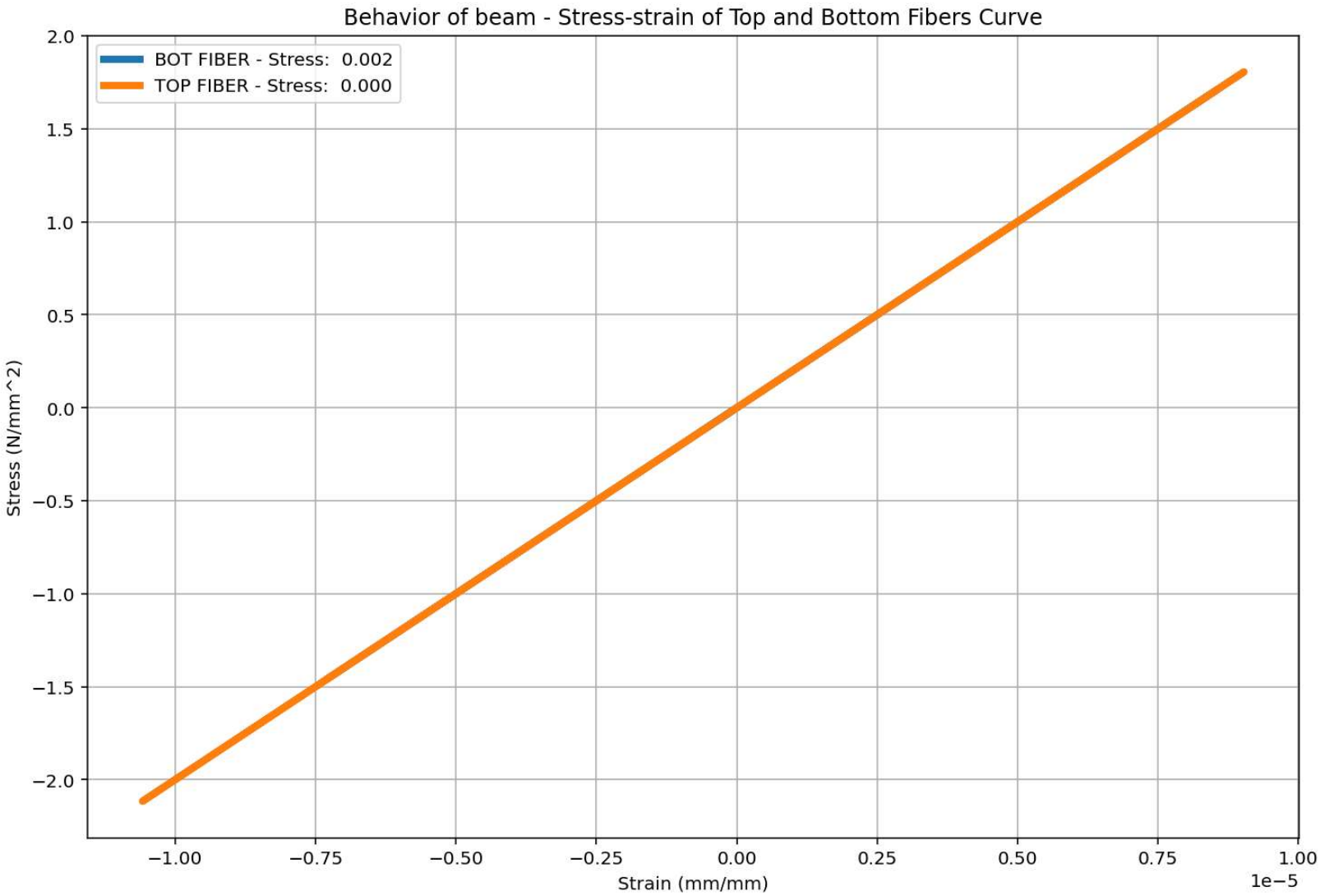
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

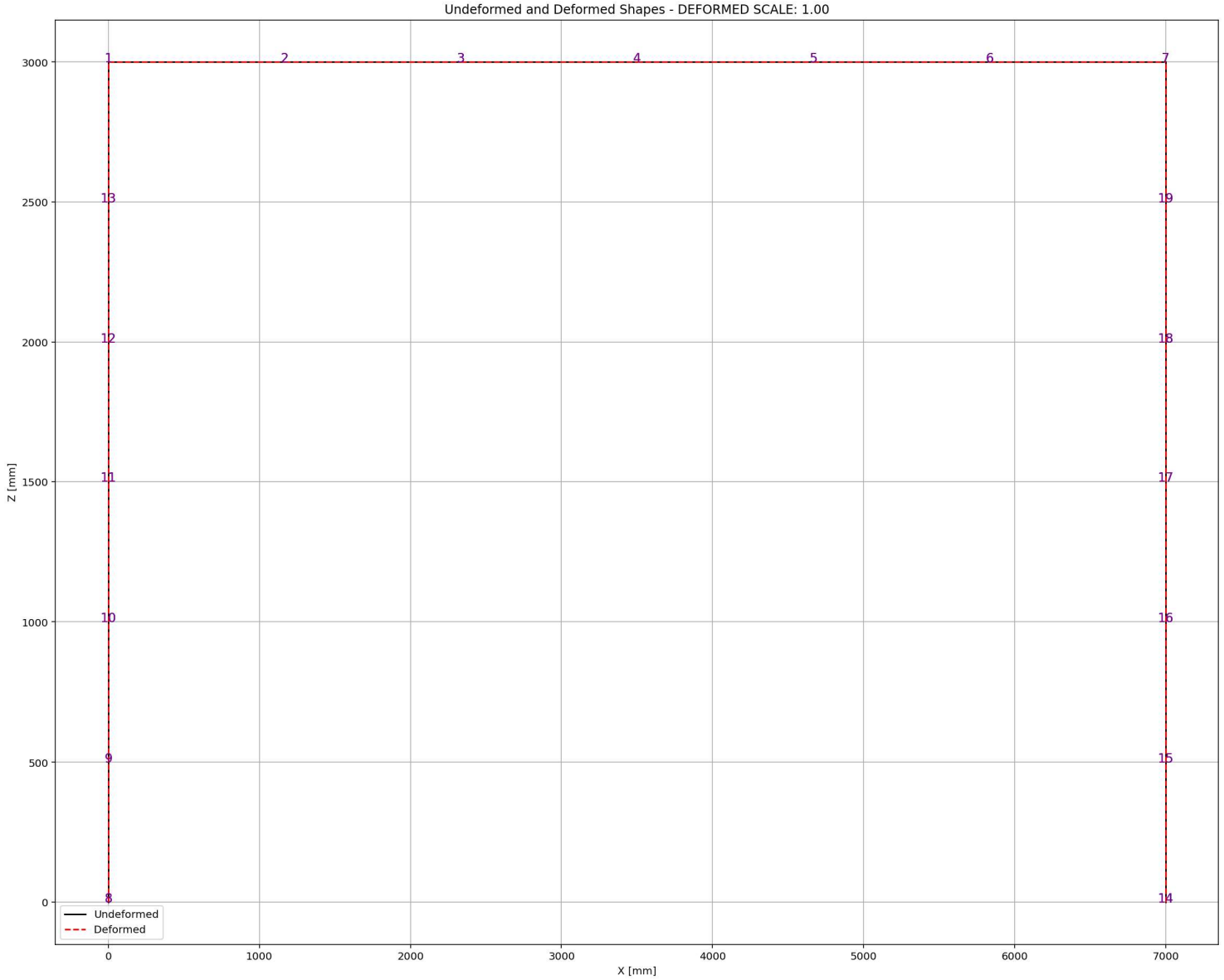
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 84, Col 34 UTF-8 CRLF RW Mem 41%


External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
15000
10000
5000
0
−5000
−10000
−15000
−0.10
−0.05
0.00
0.05
0.10
0.15
Displacement [mm]


Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
1.2
1.0
0.8
0.6
0.4
−0.10
−0.05
0.00
0.05
0.10
0.15
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.309 (s) - Mean: 0.309 (s) - Max: 0.309 (s)
PERIOD - MAX VALUES: Min: 1.264 (s) - Mean: 1.264 (s) - Max: 1.265 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.002
TOP FIBER - Stress: 0.000
Stress (N/mm^2)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Strain (mm/mm)
1e−5


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

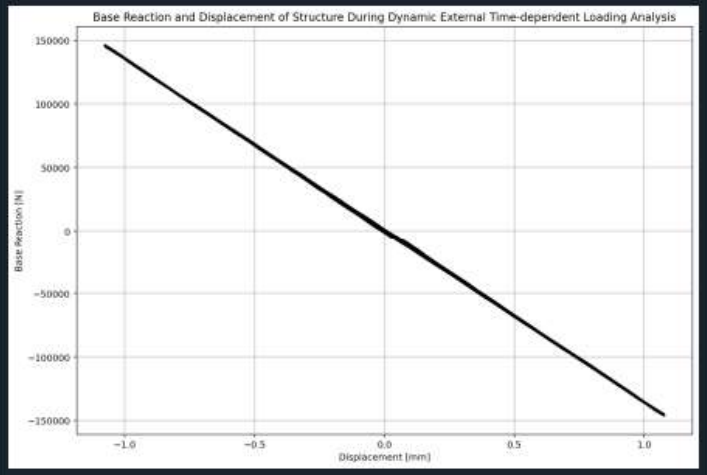
File Edit Search Source Run Debug Consoles Projects Tools View Help

...ENERGY_DISSIPATION_CAPACITY_INDEX\10_COMPOSITE_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...FRAME_8_ANA\P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py

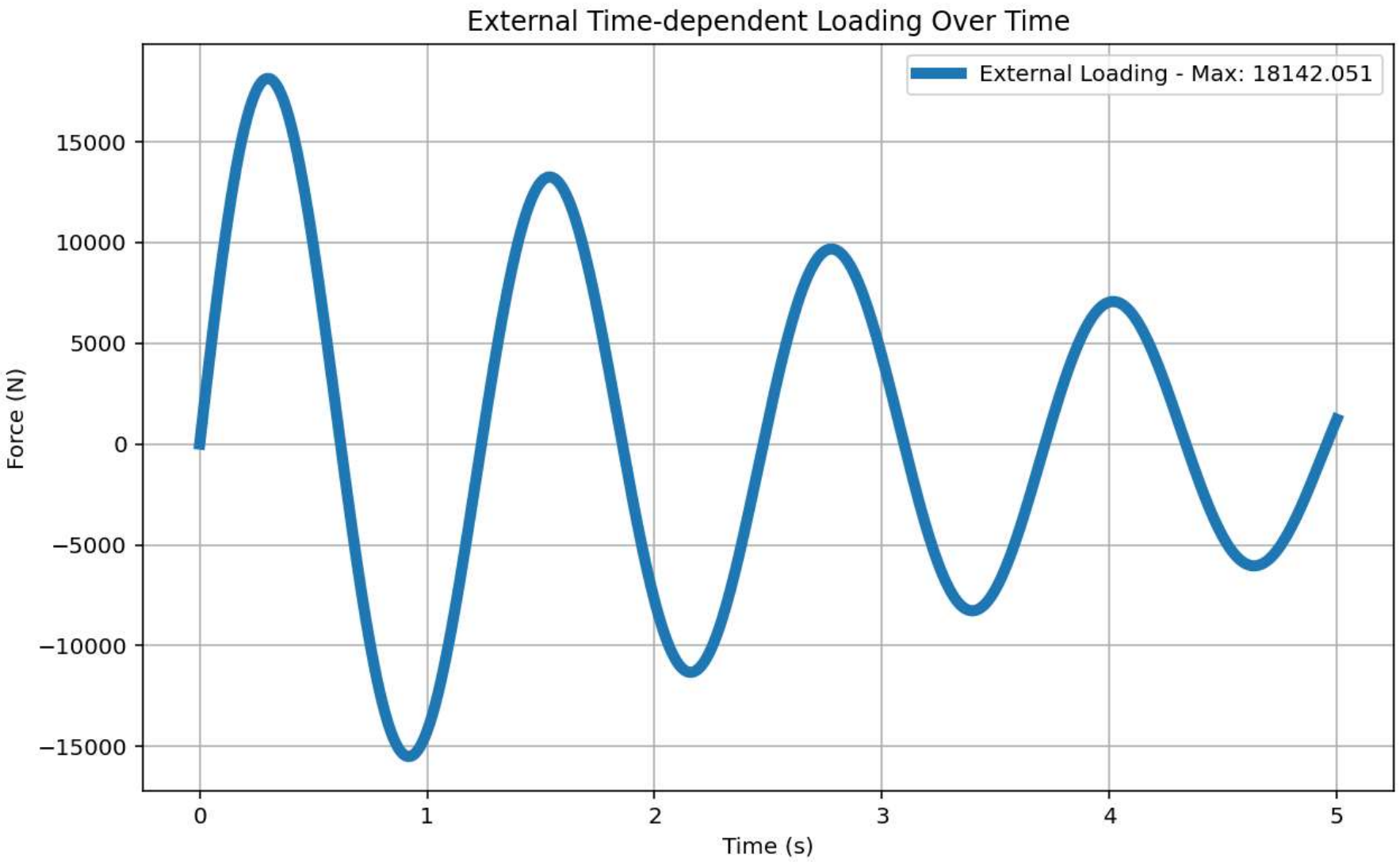
P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py | COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

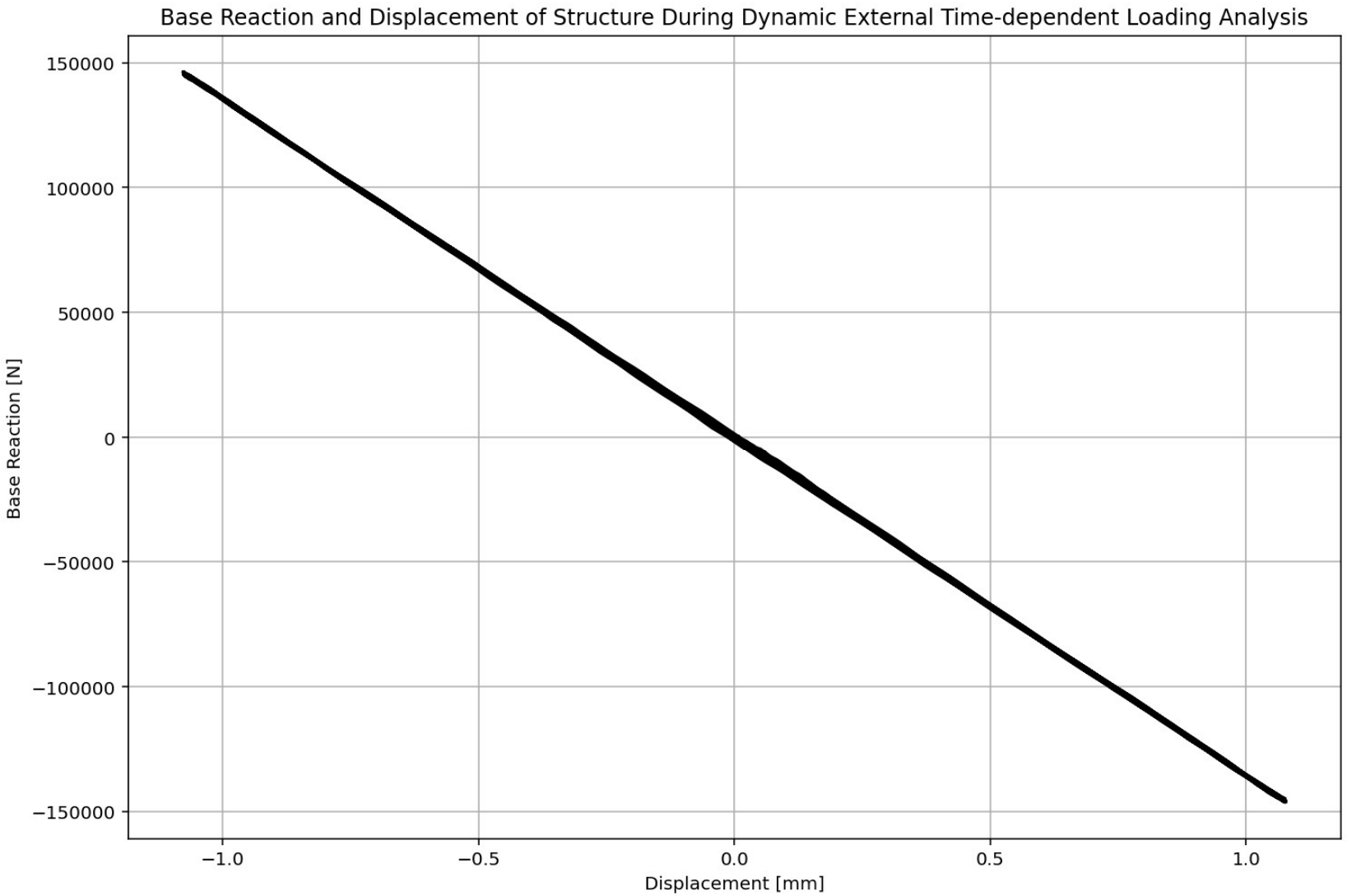
```
716
717     if ANAL_TYPE == 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING': # DYNAMIC TIME-HISTORY ANAL
718         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
719         duration = 20.0             # [s] Analysis duration
720         dt = 0.01                   # [s] Time step
721         DT = dt                     # [s] Time step
722         DT_time = 5.0               # [s] Total external Load Analysis Durations [*******]
723         force_amplitude = 19600.0   # [N] Amplitude Force
724         omega_DT = 5.0715           # [rad/s] Natural angular frequency
725         DR = 0.03                   # Damping Ratio
726
727         # Check function
728         def CHECK_FUN(DT_time ,duration):
729             if DT_time > duration:
730                 print('\n\nAnalysis Duration Must be greater than External Load Duration\n
731                 exit()
732             return -1
733
734         CHECK_FUN(DT_time ,duration)
735
736         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 s
737             import numpy as np
738             import matplotlib.pyplot as plt
739             # External Load Durations
740             num_steps = int(DT_time / DT)
741             load_time = np.linspace(0, DT_time, num_steps)
742             target_frequency = 1.0 * omega_DT  # Target excitation frequency
743             DT_load = force_amplitude * np.sin(target_frequency * load_time)
744             # Plot External Time-dependent Loading
745             plt.figure(figsize=(10, 6))
746             plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load):
747             plt.title('External Time-dependent Loading Over Time')
748             plt.xlabel('Time (s)')
749             plt.ylabel('Force (N)')
```

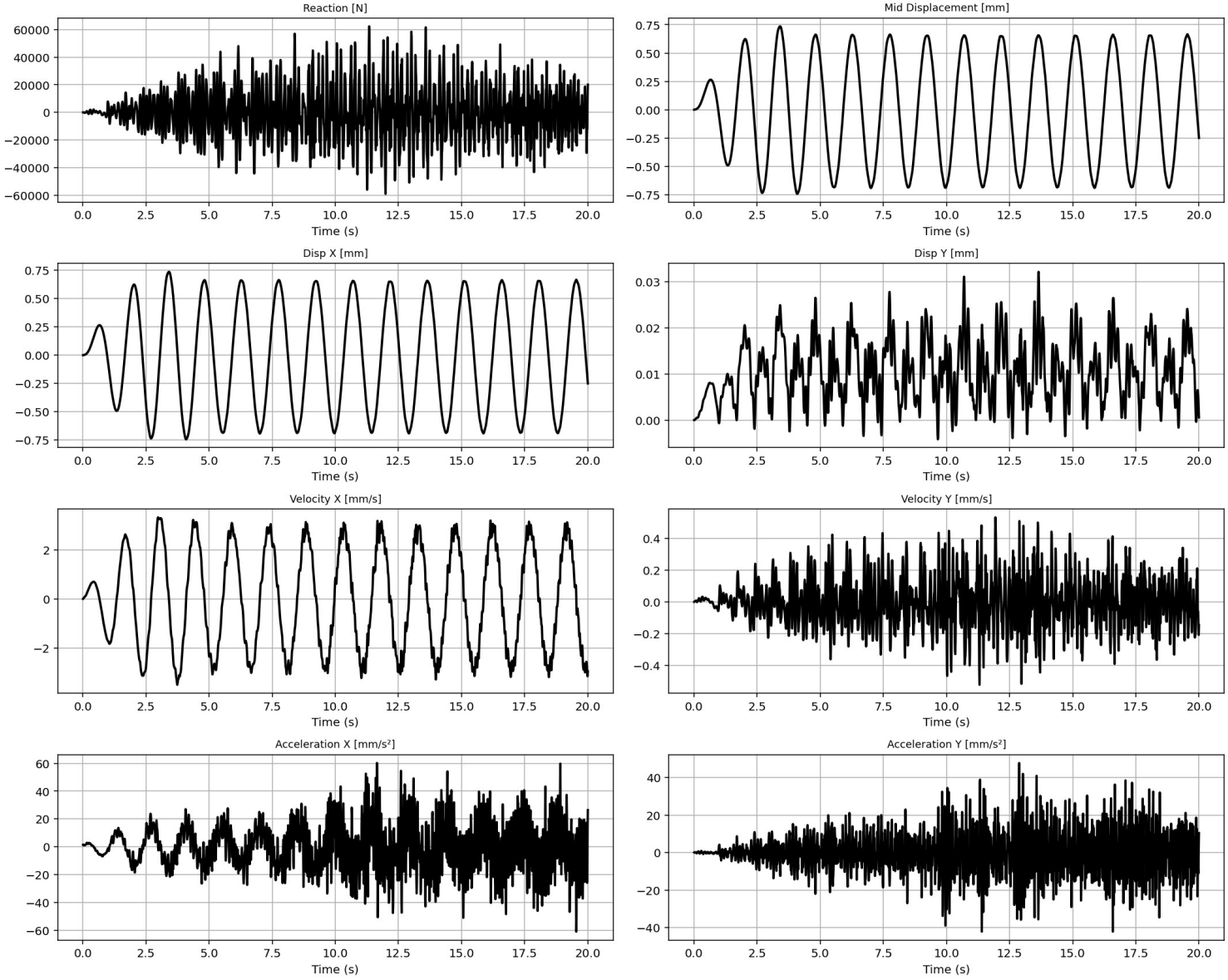


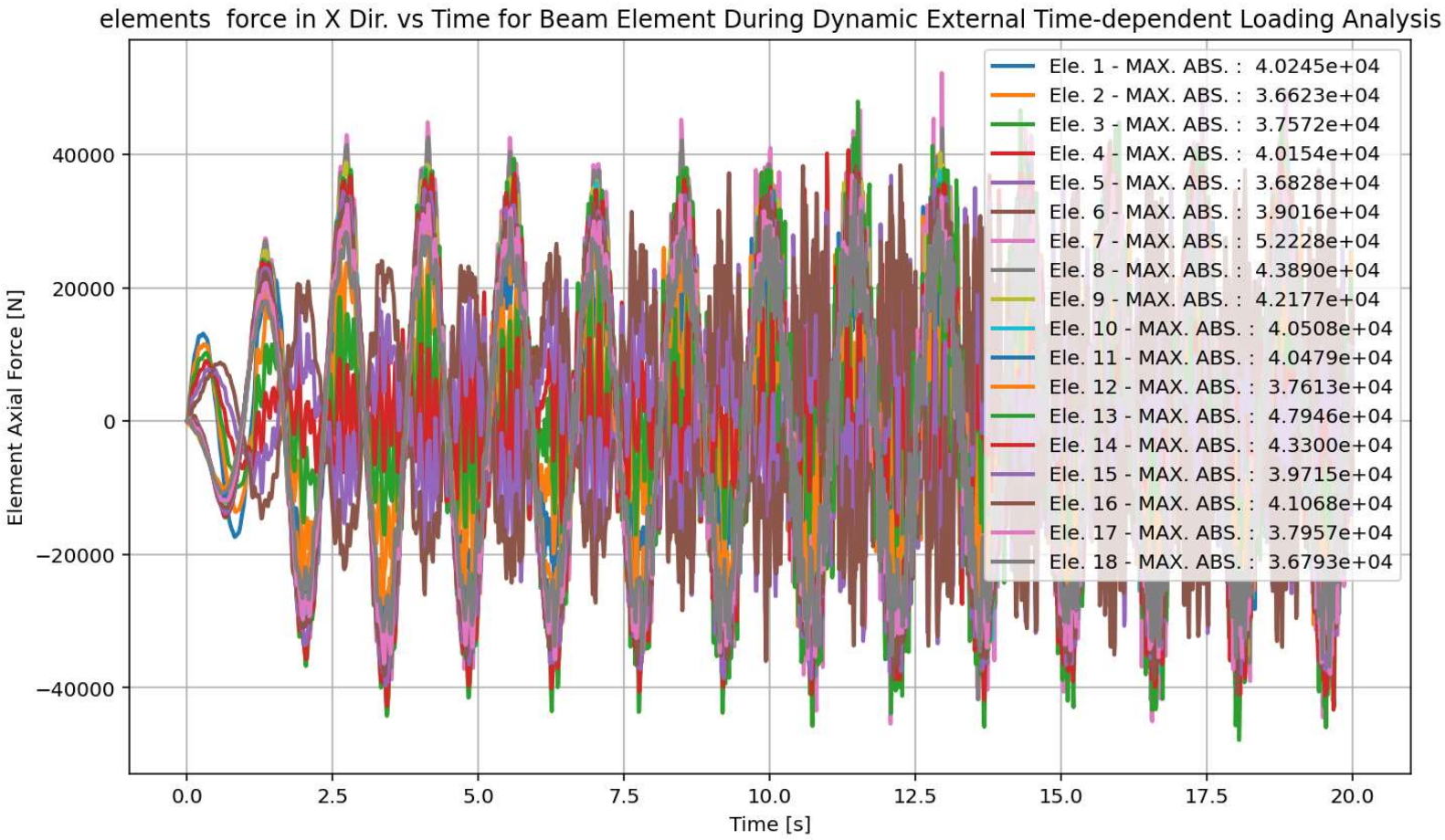
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

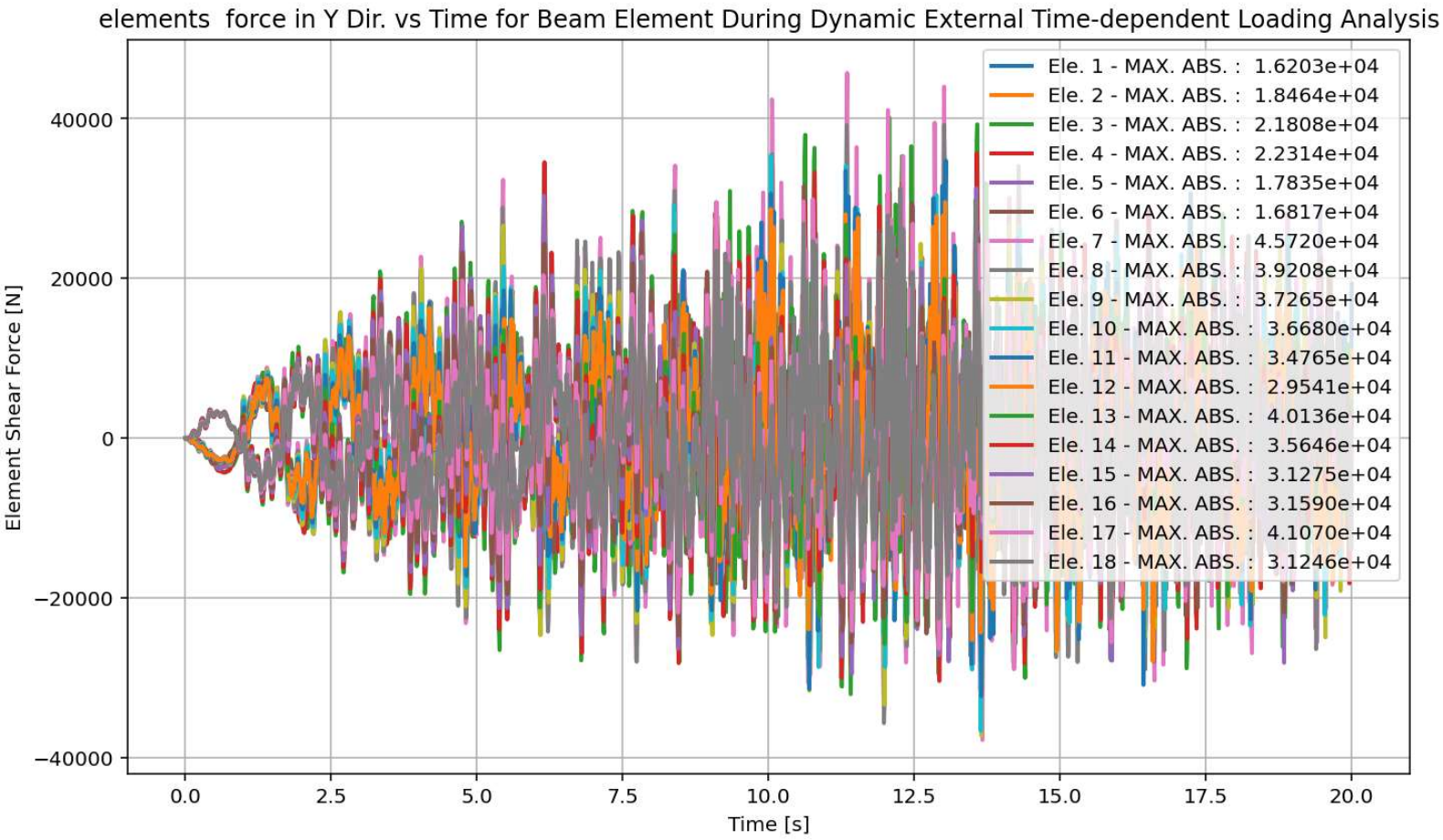
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 84, Col 34 UTF-8 CRLF RW Mem 41%

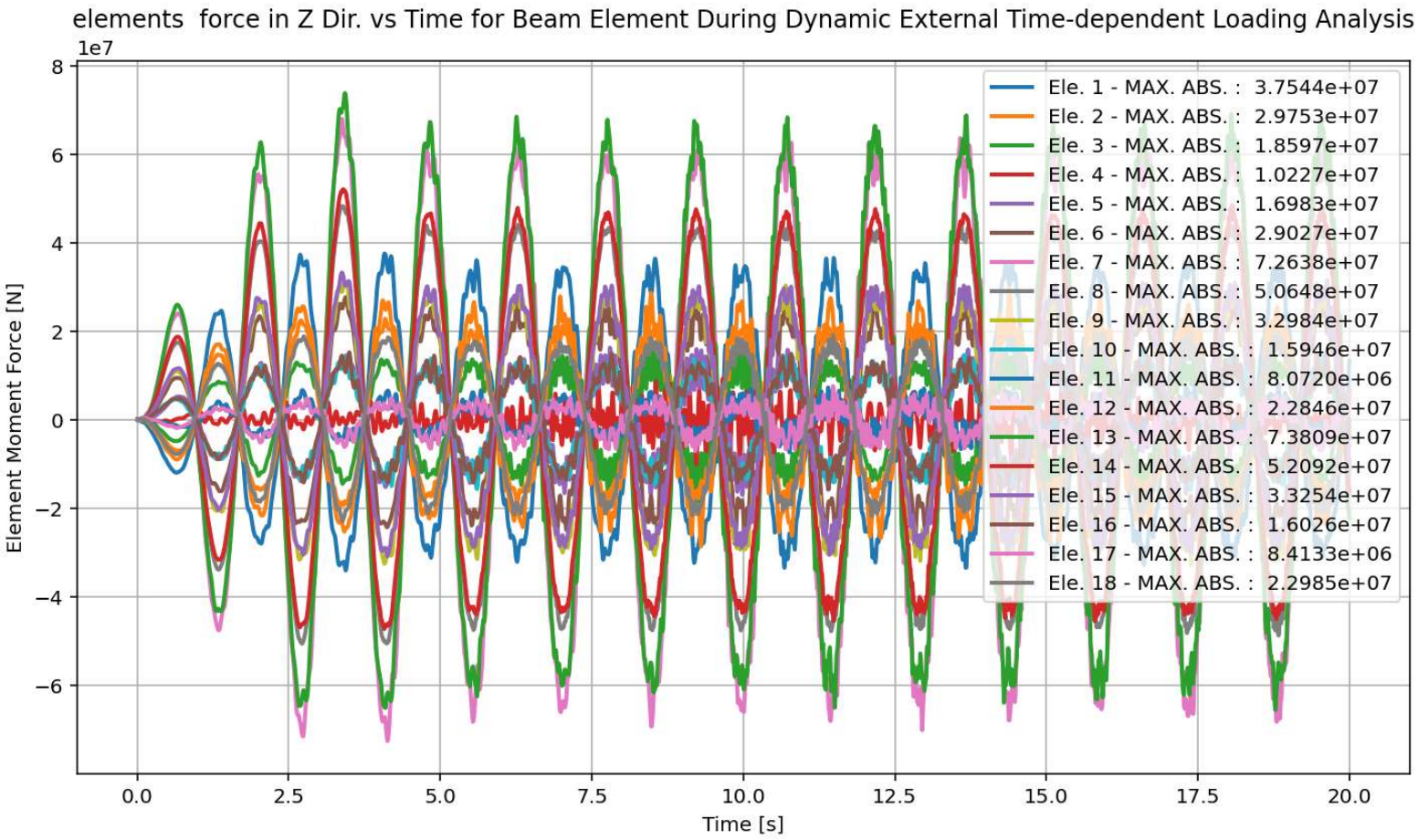

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)

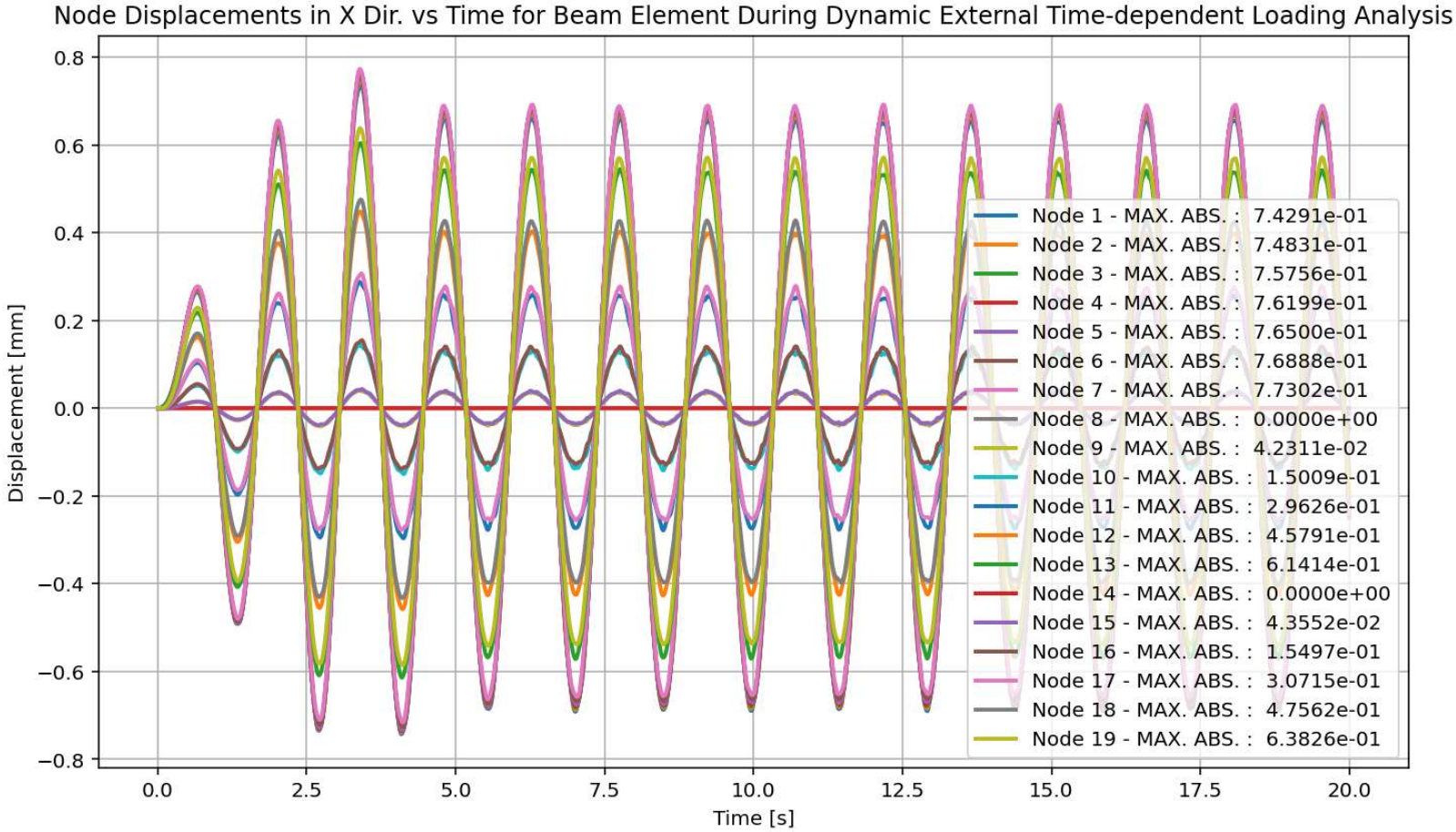

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
150000
100000
50000
0
−50000
−100000
−150000
Base Reaction [N]
−1.0
−0.5
0.0
0.5
1.0
Displacement [mm]

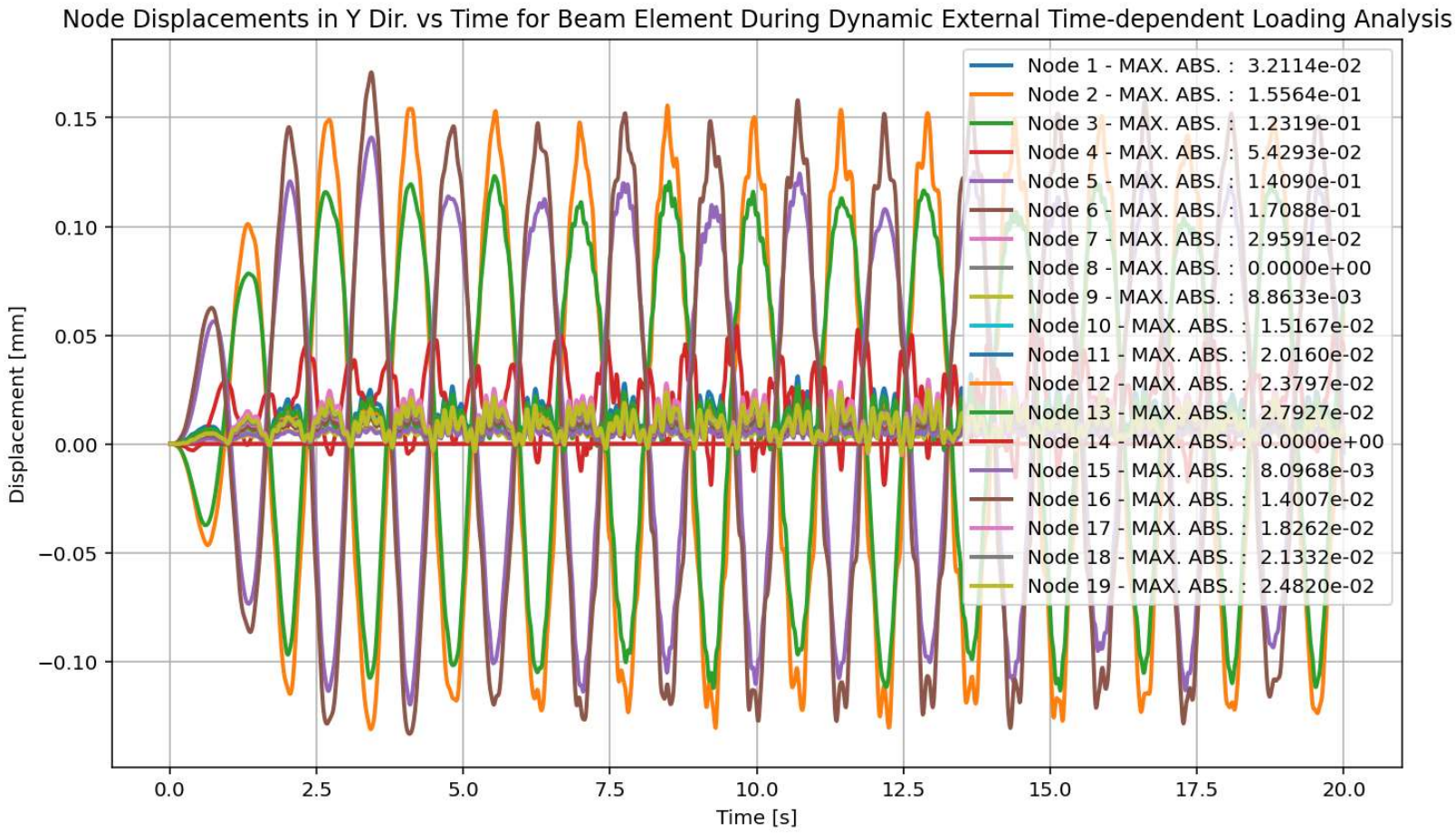

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

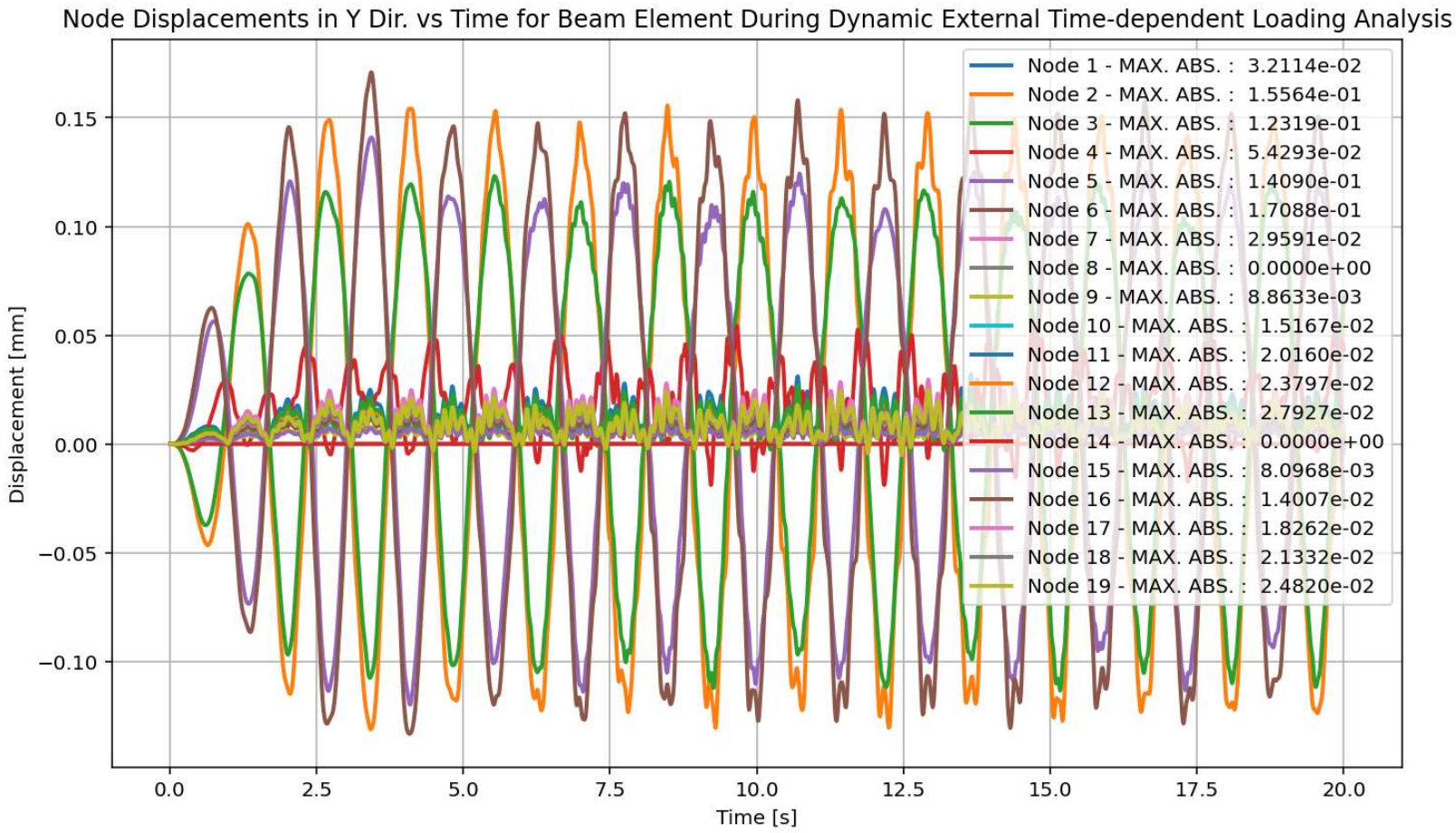

elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Axial Force [N]
Time [s]
40000
20000
0
−20000
−40000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 4.0245e+04
Ele. 2 - MAX. ABS. : 3.6623e+04
Ele. 3 - MAX. ABS. : 3.7572e+04
Ele. 4 - MAX. ABS. : 4.0154e+04
Ele. 5 - MAX. ABS. : 3.6828e+04
Ele. 6 - MAX. ABS. : 3.9016e+04
Ele. 7 - MAX. ABS. : 5.2228e+04
Ele. 8 - MAX. ABS. : 4.3890e+04
Ele. 9 - MAX. ABS. : 4.2177e+04
Ele. 10 - MAX. ABS. : 4.0508e+04
Ele. 11 - MAX. ABS. : 4.0479e+04
Ele. 12 - MAX. ABS. : 3.7613e+04
Ele. 13 - MAX. ABS. : 4.7946e+04
Ele. 14 - MAX. ABS. : 4.3300e+04
Ele. 15 - MAX. ABS. : 3.9715e+04
Ele. 16 - MAX. ABS. : 4.1068e+04
Ele. 17 - MAX. ABS. : 3.7957e+04
Ele. 18 - MAX. ABS. : 3.6793e+04

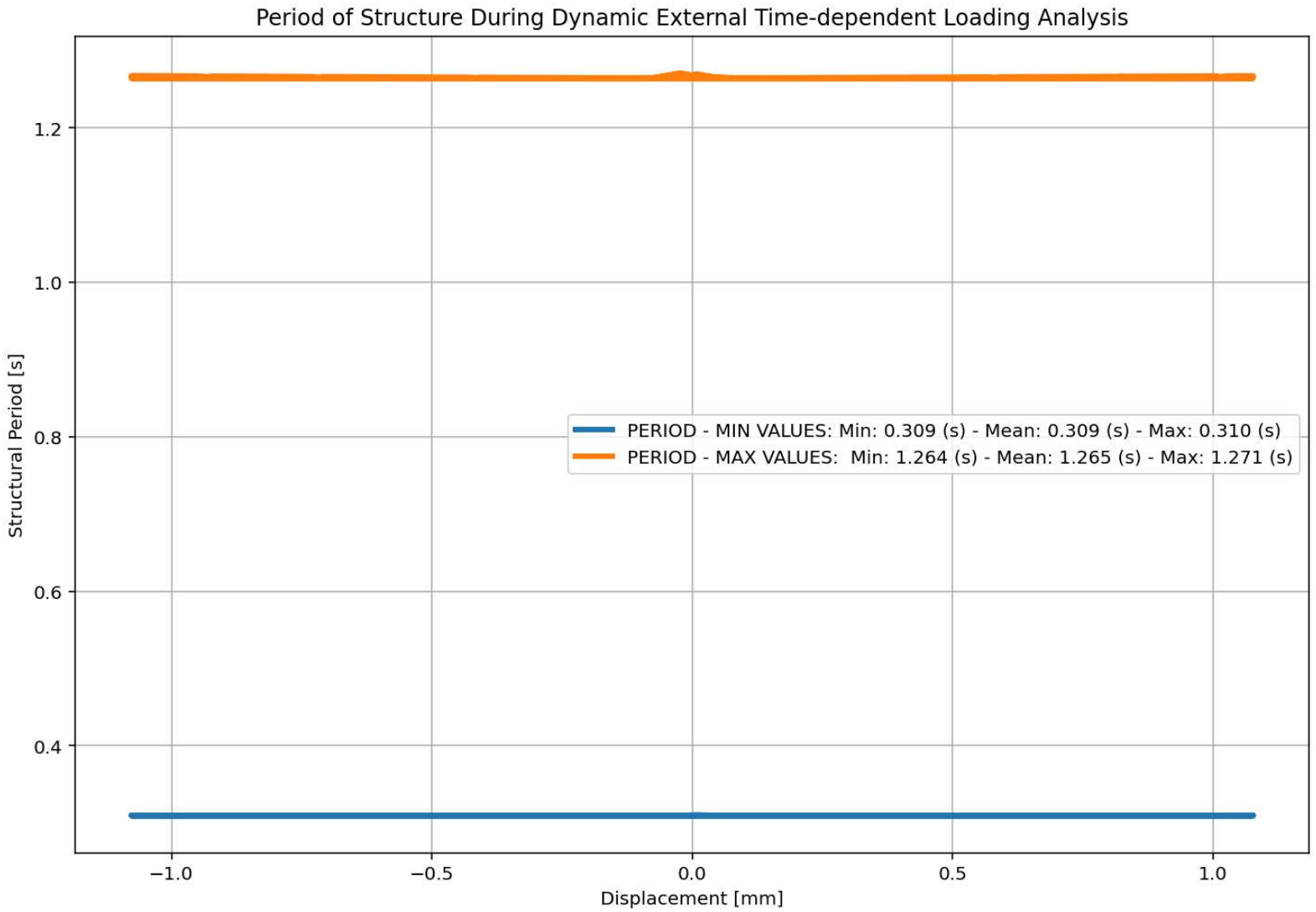

elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Shear Force [N]
Time [s]
40000
20000
0
−20000
−40000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 1.6203e+04
Ele. 2 - MAX. ABS. : 1.8464e+04
Ele. 3 - MAX. ABS. : 2.1808e+04
Ele. 4 - MAX. ABS. : 2.2314e+04
Ele. 5 - MAX. ABS. : 1.7835e+04
Ele. 6 - MAX. ABS. : 1.6817e+04
Ele. 7 - MAX. ABS. : 4.5720e+04
Ele. 8 - MAX. ABS. : 3.9208e+04
Ele. 9 - MAX. ABS. : 3.7265e+04
Ele. 10 - MAX. ABS. : 3.6680e+04
Ele. 11 - MAX. ABS. : 3.4765e+04
Ele. 12 - MAX. ABS. : 2.9541e+04
Ele. 13 - MAX. ABS. : 4.0136e+04
Ele. 14 - MAX. ABS. : 3.5646e+04
Ele. 15 - MAX. ABS. : 3.1275e+04
Ele. 16 - MAX. ABS. : 3.1590e+04
Ele. 17 - MAX. ABS. : 4.1070e+04
Ele. 18 - MAX. ABS. : 3.1246e+04

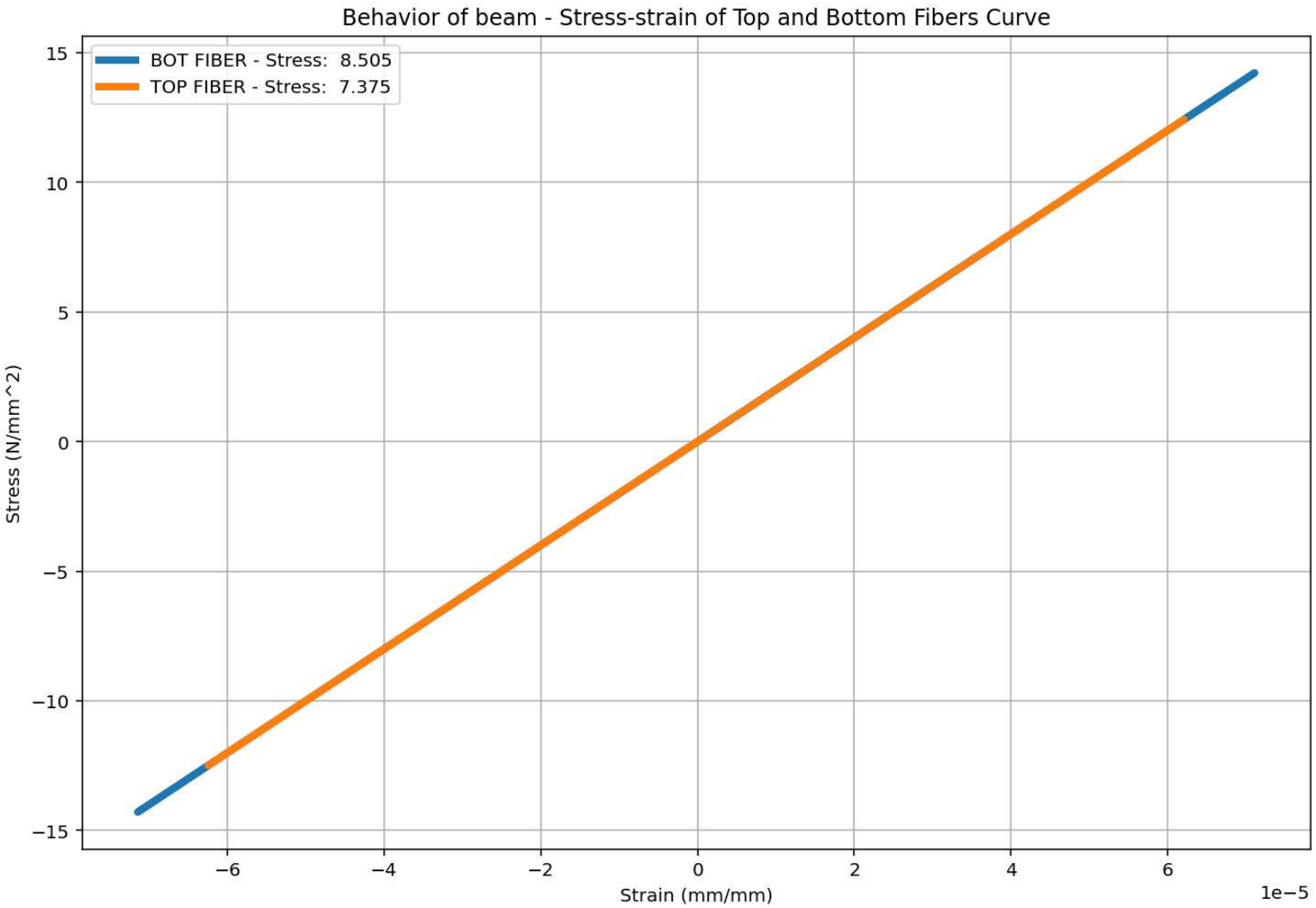

elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
1e7
8
6
4
2
0
−2
−4
−6
Element Moment Force [N]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 3.7544e+07
Ele. 2 - MAX. ABS. : 2.9753e+07
Ele. 3 - MAX. ABS. : 1.8597e+07
Ele. 4 - MAX. ABS. : 1.0227e+07
Ele. 5 - MAX. ABS. : 1.6983e+07
Ele. 6 - MAX. ABS. : 2.9027e+07
Ele. 7 - MAX. ABS. : 7.2638e+07
Ele. 8 - MAX. ABS. : 5.0648e+07
Ele. 9 - MAX. ABS. : 3.2984e+07
Ele. 10 - MAX. ABS. : 1.5946e+07
Ele. 11 - MAX. ABS. : 8.0720e+06
Ele. 12 - MAX. ABS. : 2.2846e+07
Ele. 13 - MAX. ABS. : 7.3809e+07
Ele. 14 - MAX. ABS. : 5.2092e+07
Ele. 15 - MAX. ABS. : 3.3254e+07
Ele. 16 - MAX. ABS. : 1.6026e+07
Ele. 17 - MAX. ABS. : 8.4133e+06
Ele. 18 - MAX. ABS. : 2.2985e+07

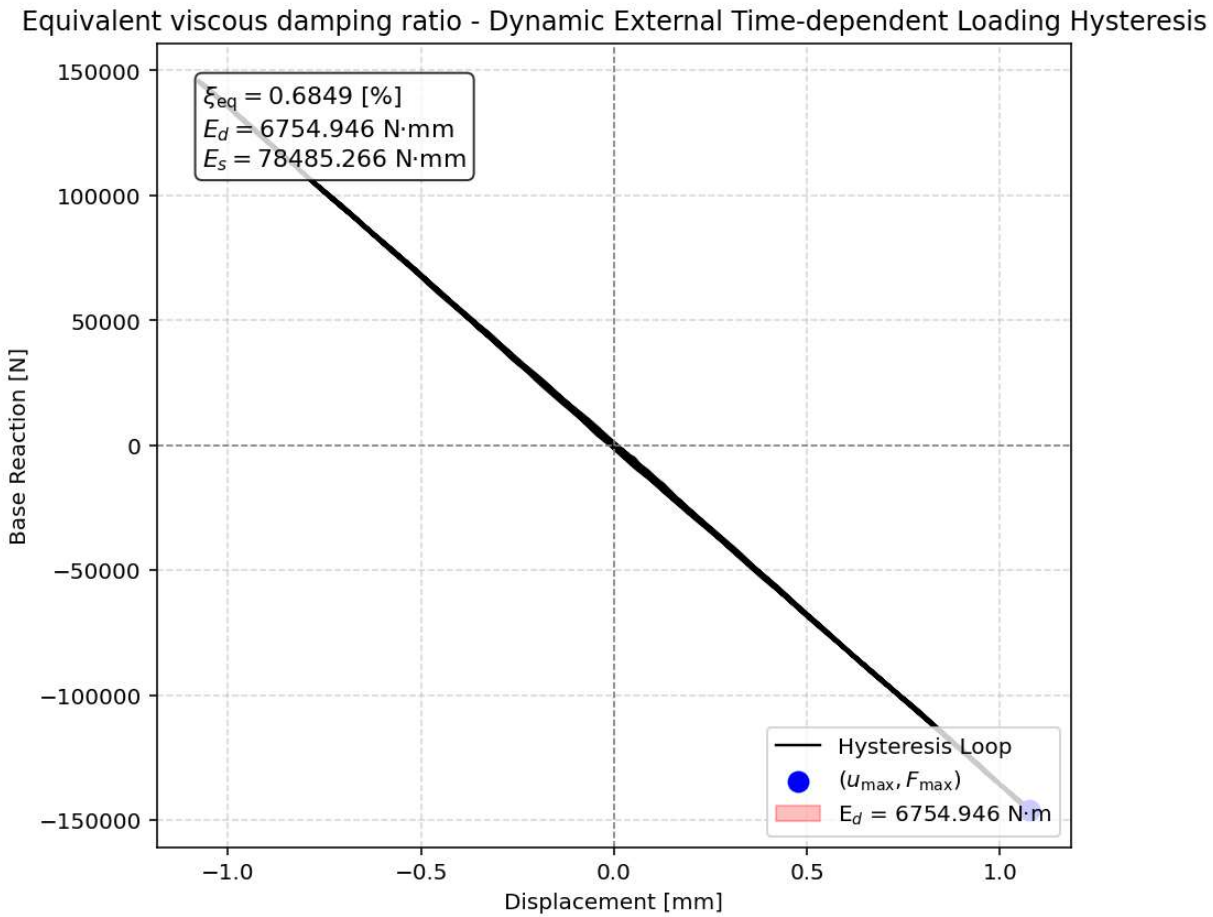

Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 7.4291e-01
Node 2 - MAX. ABS. : 7.4831e-01
Node 3 - MAX. ABS. : 7.5756e-01
Node 4 - MAX. ABS. : 7.6199e-01
Node 5 - MAX. ABS. : 7.6500e-01
Node 6 - MAX. ABS. : 7.6888e-01
Node 7 - MAX. ABS. : 7.7302e-01
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 4.2311e-02
Node 10 - MAX. ABS. : 1.5009e-01
Node 11 - MAX. ABS. : 2.9626e-01
Node 12 - MAX. ABS. : 4.5791e-01
Node 13 - MAX. ABS. : 6.1414e-01
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 4.3552e-02
Node 16 - MAX. ABS. : 1.5497e-01
Node 17 - MAX. ABS. : 3.0715e-01
Node 18 - MAX. ABS. : 4.7562e-01
Node 19 - MAX. ABS. : 6.3826e-01

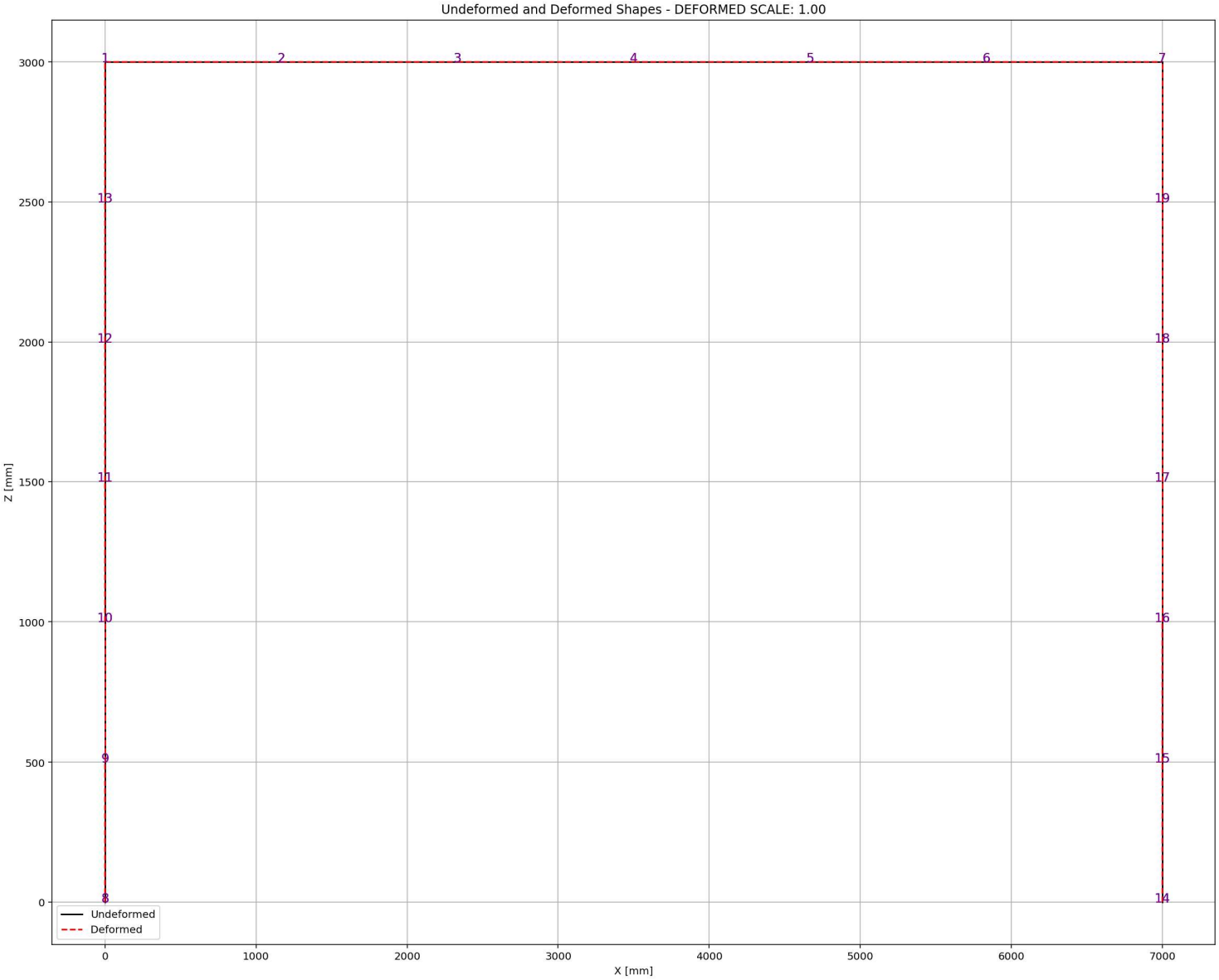

Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.15
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 3.2114e-02
Node 2 - MAX. ABS. : 1.5564e-01
Node 3 - MAX. ABS. : 1.2319e-01
Node 4 - MAX. ABS. : 5.4293e-02
Node 5 - MAX. ABS. : 1.4090e-01
Node 6 - MAX. ABS. : 1.7088e-01
Node 7 - MAX. ABS. : 2.9591e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 8.8633e-03
Node 10 - MAX. ABS. : 1.5167e-02
Node 11 - MAX. ABS. : 2.0160e-02
Node 12 - MAX. ABS. : 2.3797e-02
Node 13 - MAX. ABS. : 2.7927e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 8.0968e-03
Node 16 - MAX. ABS. : 1.4007e-02
Node 17 - MAX. ABS. : 1.8262e-02
Node 18 - MAX. ABS. : 2.1332e-02
Node 19 - MAX. ABS. : 2.4820e-02


Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.15
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 3.2114e-02
Node 2 - MAX. ABS. : 1.5564e-01
Node 3 - MAX. ABS. : 1.2319e-01
Node 4 - MAX. ABS. : 5.4293e-02
Node 5 - MAX. ABS. : 1.4090e-01
Node 6 - MAX. ABS. : 1.7088e-01
Node 7 - MAX. ABS. : 2.9591e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 8.8633e-03
Node 10 - MAX. ABS. : 1.5167e-02
Node 11 - MAX. ABS. : 2.0160e-02
Node 12 - MAX. ABS. : 2.3797e-02
Node 13 - MAX. ABS. : 2.7927e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 8.0968e-03
Node 16 - MAX. ABS. : 1.4007e-02
Node 17 - MAX. ABS. : 1.8262e-02
Node 18 - MAX. ABS. : 2.1332e-02
Node 19 - MAX. ABS. : 2.4820e-02


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
1.2
1.0
0.8
0.6
0.4
−1.0
−0.5
0.0
0.5
1.0
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.309 (s) - Mean: 0.309 (s) - Max: 0.310 (s)
PERIOD - MAX VALUES: Min: 1.264 (s) - Mean: 1.265 (s) - Max: 1.271 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 8.505
TOP FIBER - Stress: 7.375
Stress (N/mm^2)
Strain (mm/mm)
1e−5
15
10
5
0
−5
−10
−15
−6
−4
−2
0
2
4
6


Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis
$\xi_{eq} = 0.6849$ [%]
$E_d = 6754.946$ N·mm
$E_s = 78485.266$ N·mm
Base Reaction [N]
Displacement [mm]
150000
100000
50000
0
−50000
−100000
−150000
−1.0
−0.5
0.0
0.5
1.0
Hysteresis Loop
(u_{max}, F_{max})
E_d = 6754.946 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

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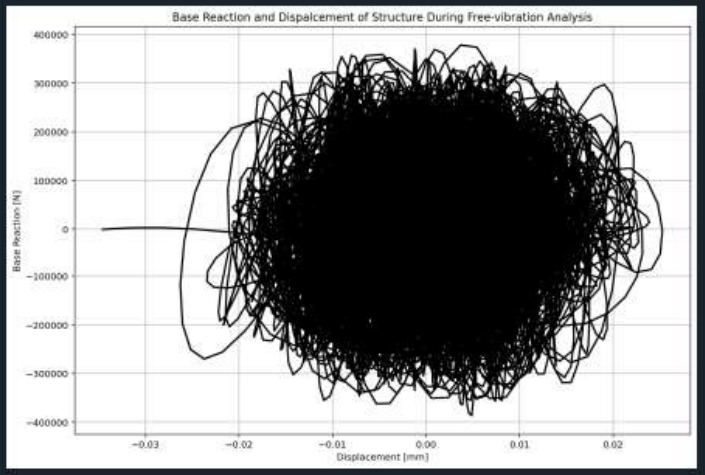
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...FRAME_8_ANA\P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py

P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py | COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

```
873
874     if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
875         #%% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
876         u0 = -0.035                         # [mm] Initial displacement
877         v0 = 0.15                           # [mm/s] Initial velocity
878         a0 = 0.065                          # [mm/s^2] Initial acceleration
879         IU = True                           # Free Vibration with Initial Displacement
880         IV = True                           # Free Vibration with Initial Velocity
881         IA = True                           # Free Vibration with Initial Acceleration
882         duration = 20.0                     # [s] Analysis duration
883         dt = 0.001                          # [s] Time step
884         DR = 0.03                           # Damping Ratio
885
886         if IU == True:
887             # Define initial displacment
888             ops.setNodeDisp(center_node, 1, u0, '-commit') # INFO LINK: https://openseespy
889         if IV == True:
890             # Define initial velocity
891             ops.setNodeVel(center_node, 1, v0, '-commit')  # INFO LINK: https://openseespy
892         if IA == True:
893             # Define initial  acceleration
894             ops.setNodeAccel(center_node, 1, a0, '-commit') # INFO LINK: https://openseesp
895
896         ops.constraints('Plain')
897         ops.numberer('Plain')
898         ops.system('BandGeneral')
899         ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK: https://opens
900         #ops.integrator('CentralDifference')  # JUST FOR LINEAR ANALYSIS - INFO LINK: http
901         alpha=0.5; beta=0.25;
902         ops.integrator('Newmark', alpha, beta) # INFO LINK: https://openseespydoc.readthed
903         #alpha=2/3;gamma=1.5-alpha; gamma=1.5-alpha;beta=(2-alpha)**2/4;
904         #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://openseespydoc.read
905         ops.algorithm('Newton')  # INFO LINK: https://openseespydoc.readthedocs.io/en/late
906         ops.analysis('Transient') # INFO LINK: https://openseespydoc.readthedocs.io/en/lat
```

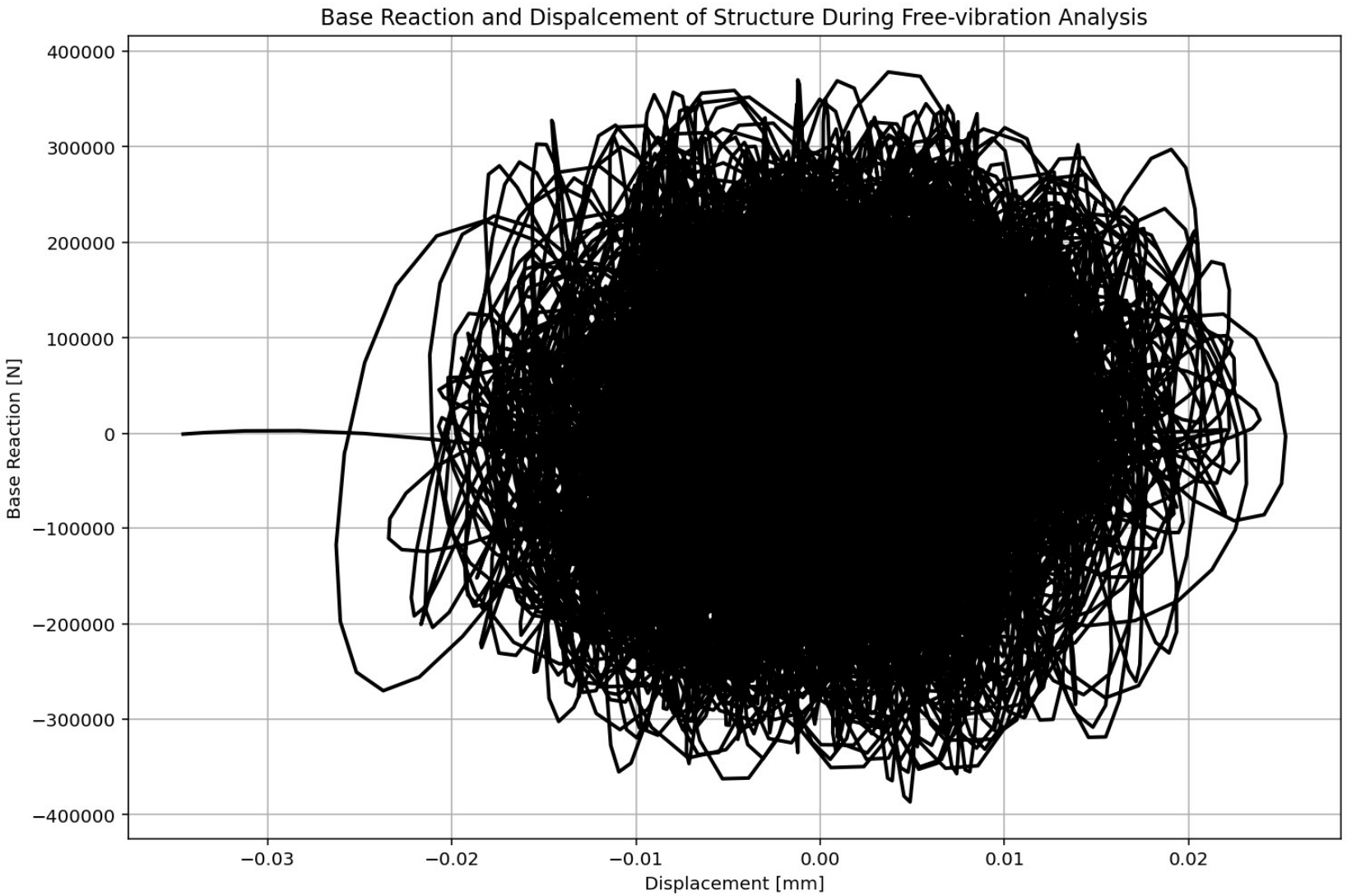
..ENERGY_DISSIPATION_CAPACITY_INDEX\10_COMPOSITE_FRAME_8_ANA

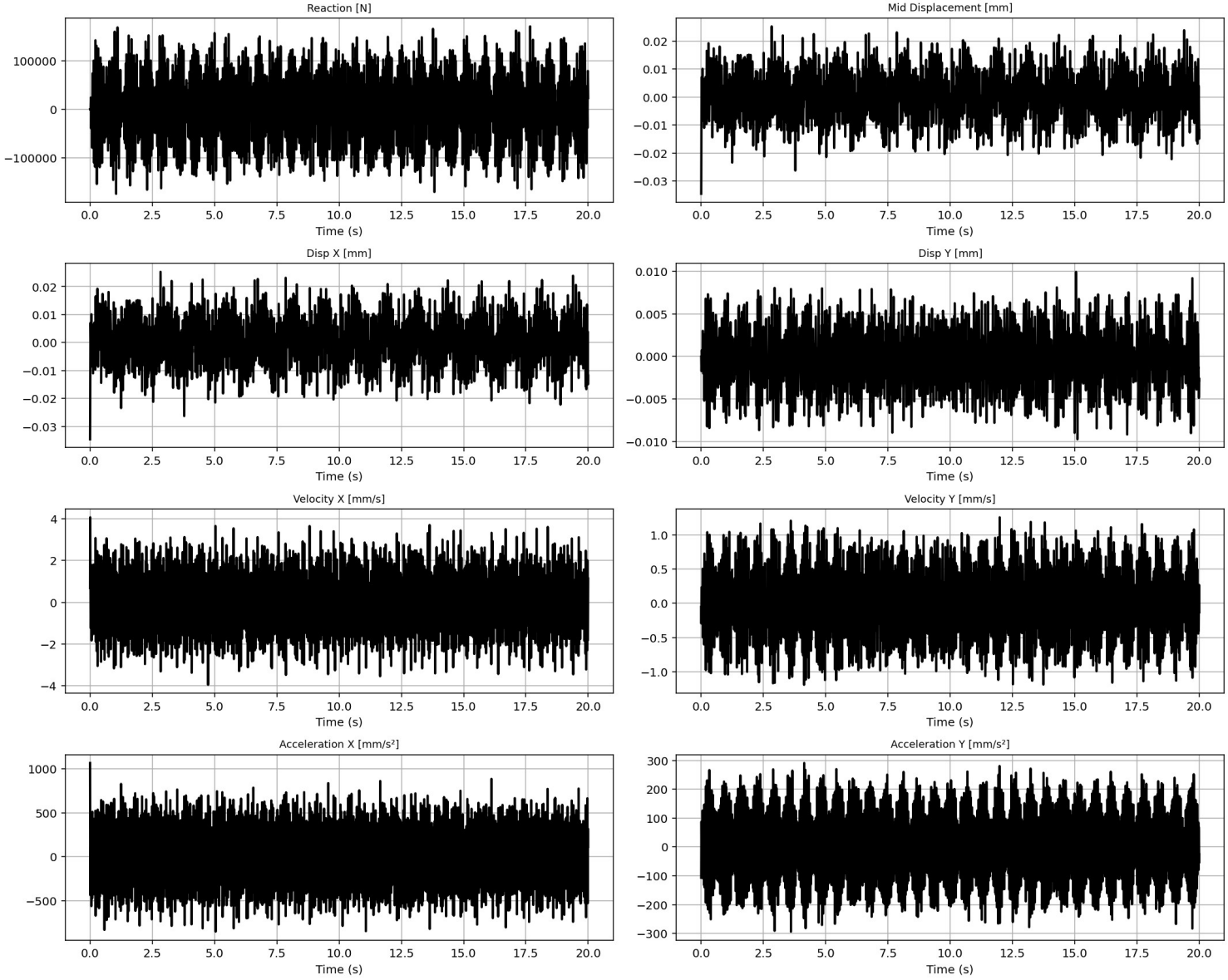
30 %

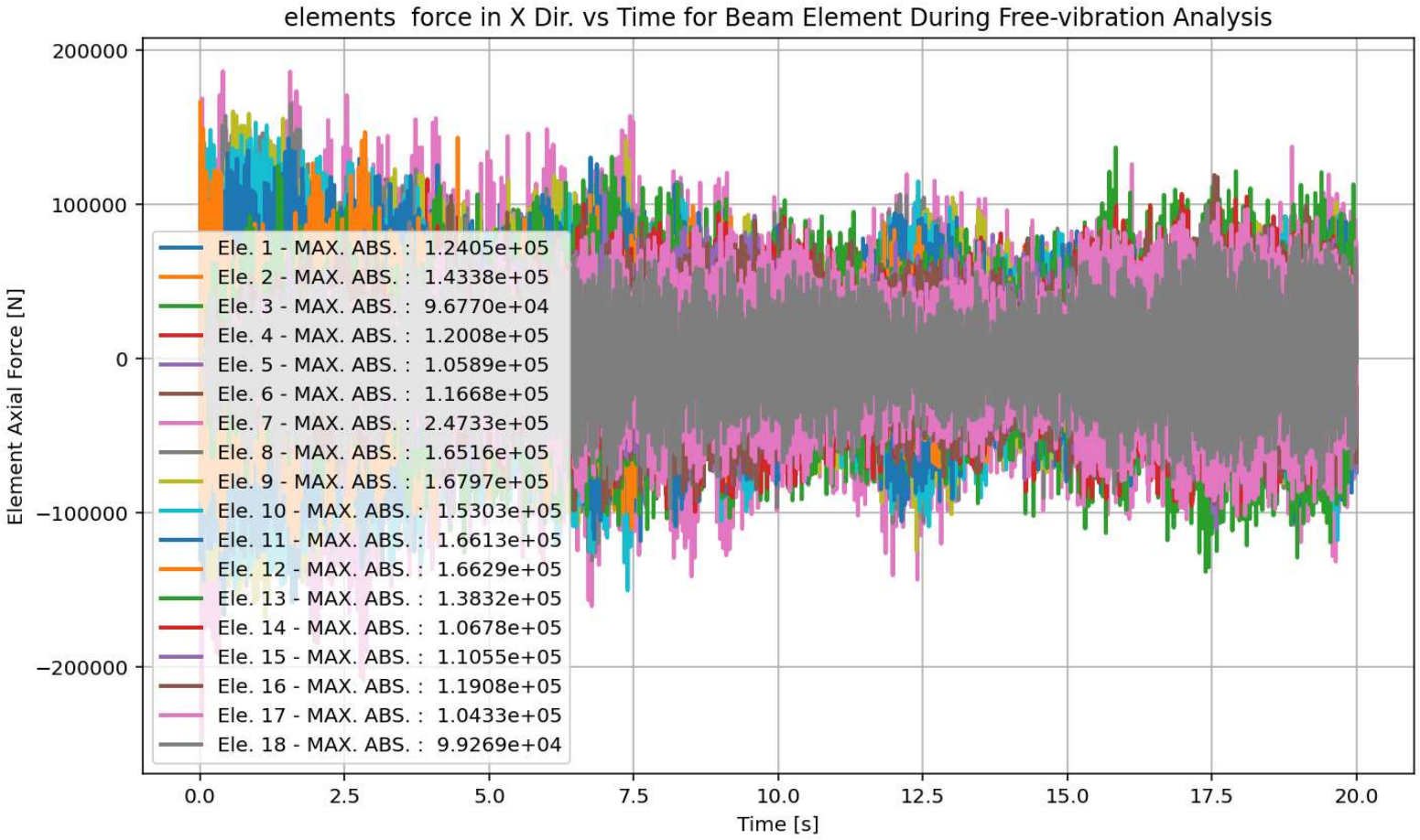


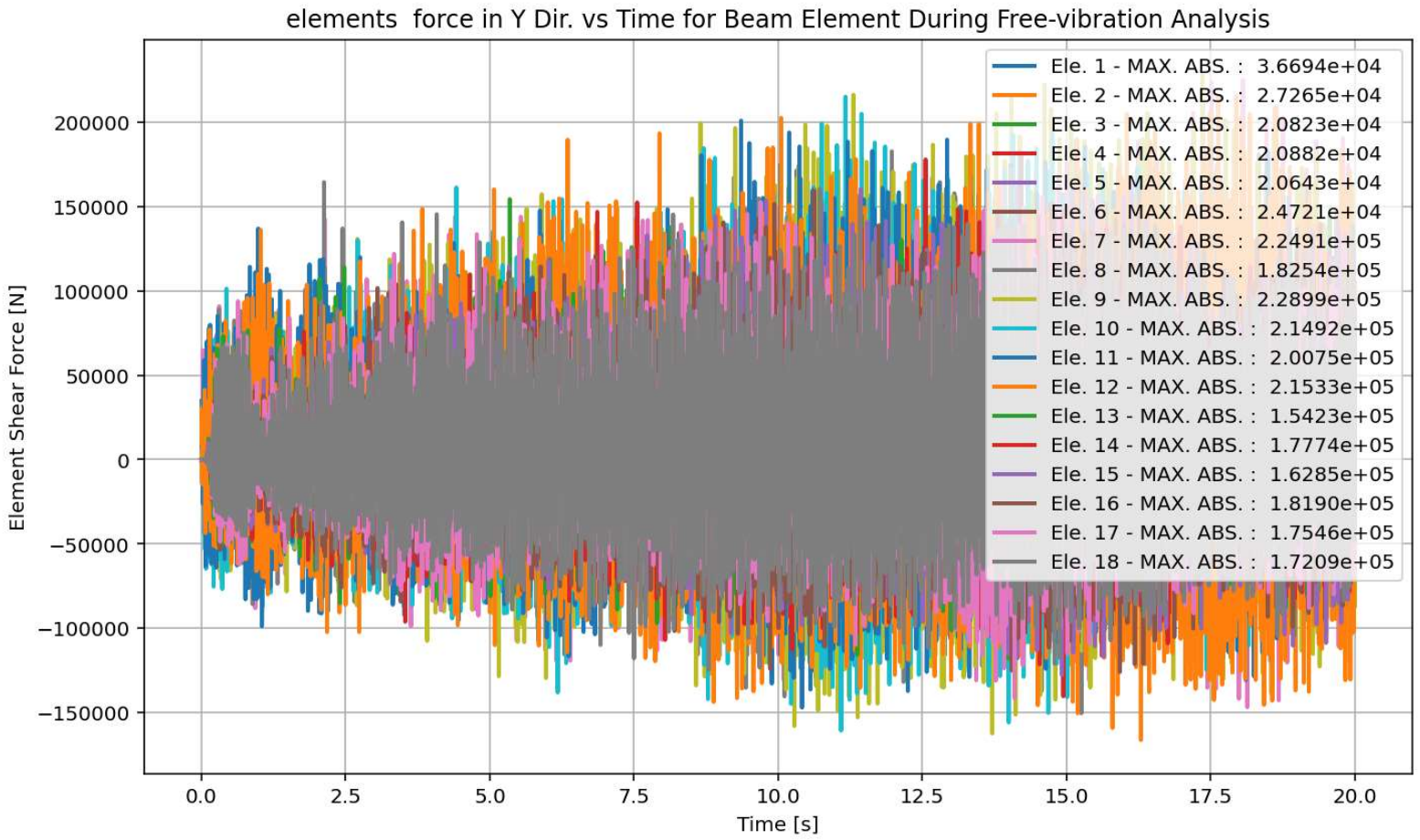
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

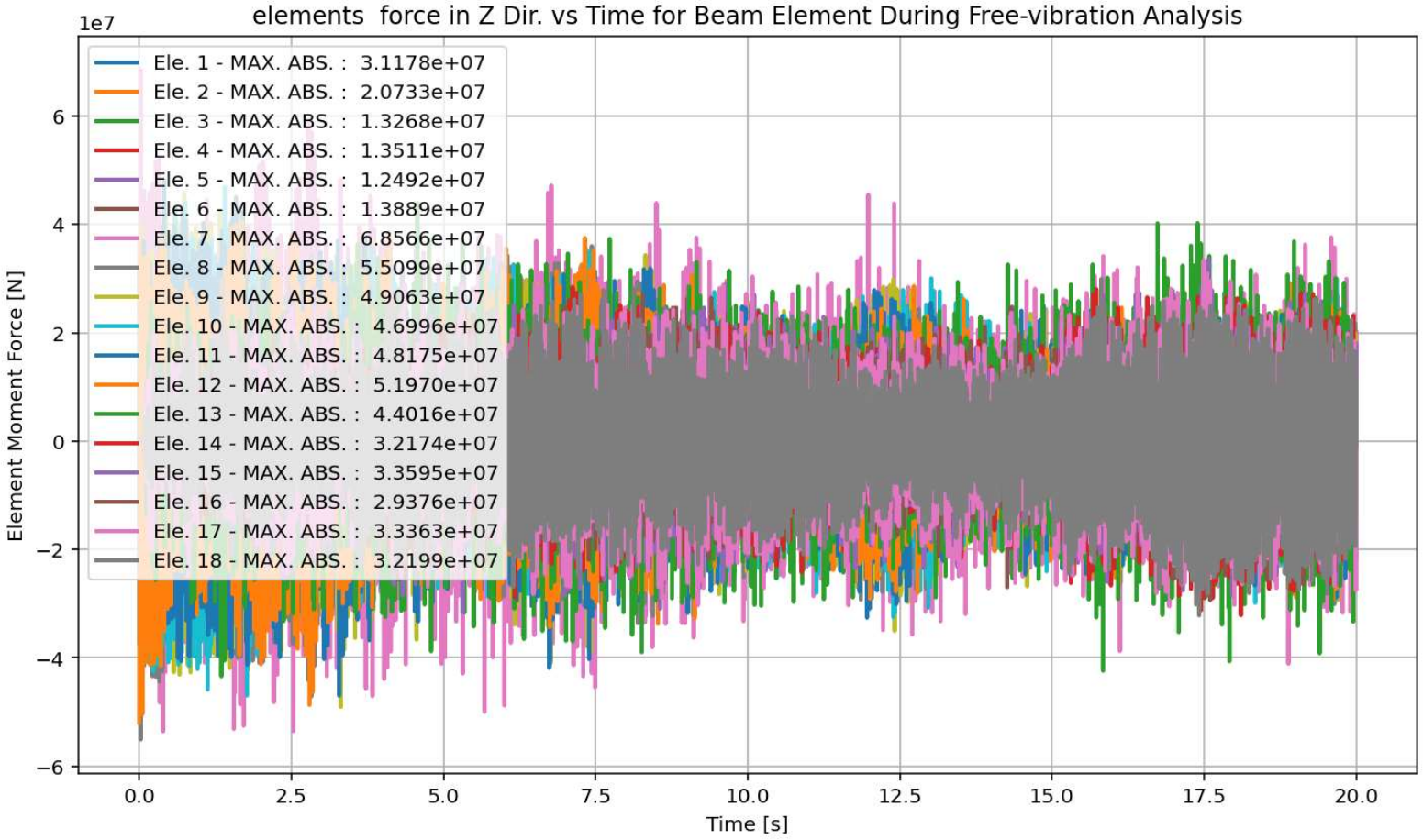
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 84, Col 34 | UTF-8 | CRLF | RW | Mem 41%

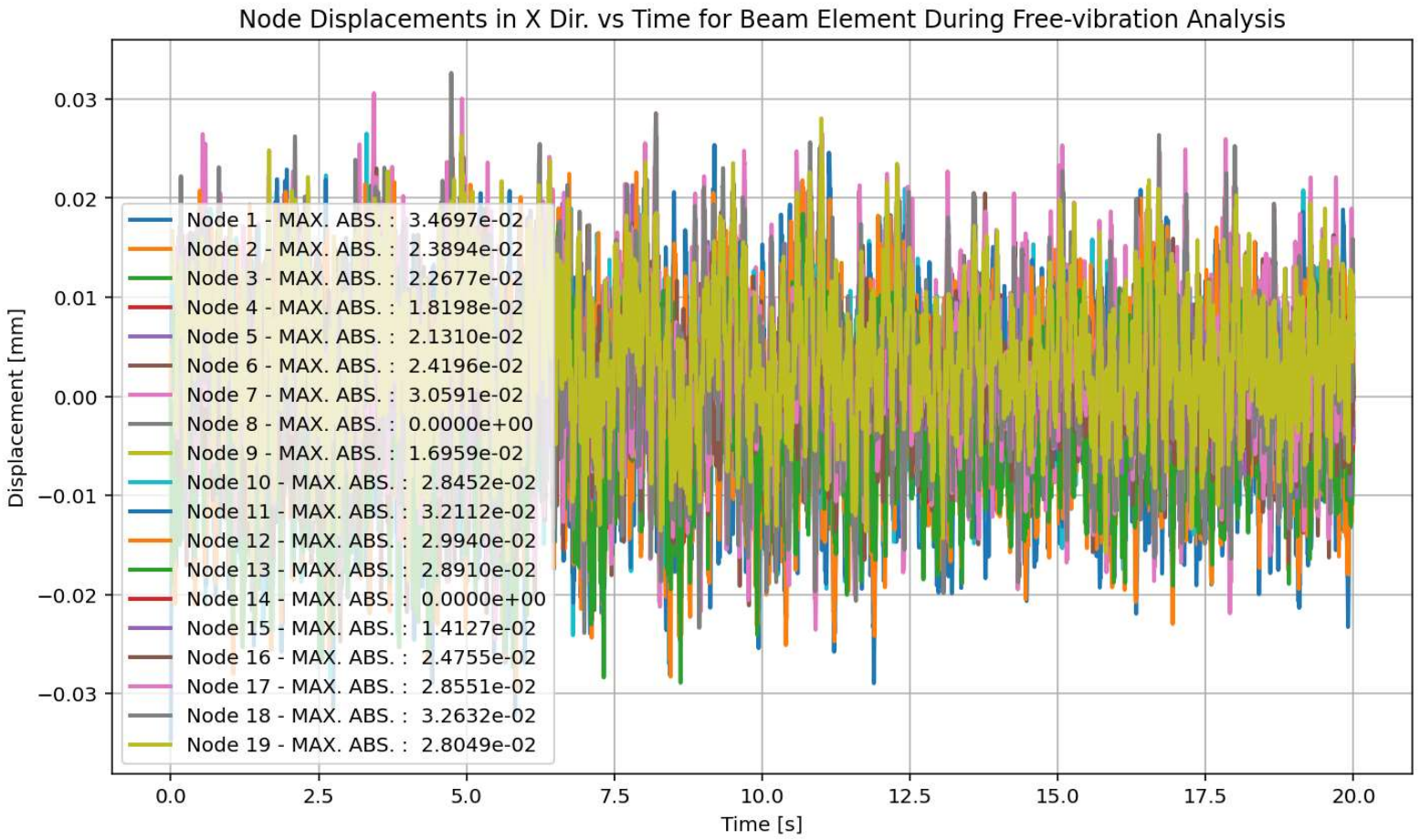

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
−0.03
−0.02
−0.01
0.00
0.01
0.02
Displacement [mm]

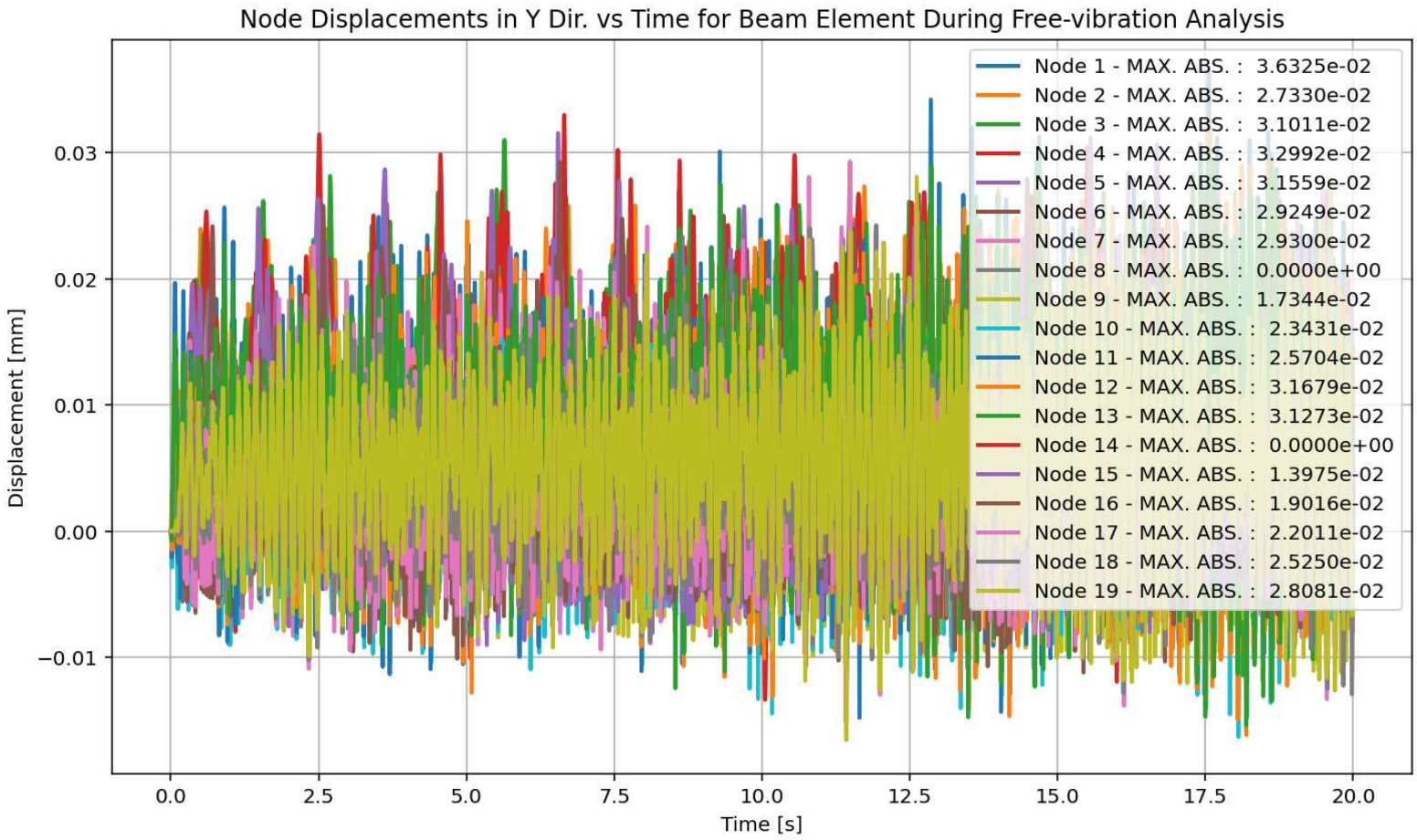

Reaction [N]
100000
0
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)
Mid Displacement [mm]
0.02
0.01
0.00
−0.01
−0.02
−0.03
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
0.010
0.005
0.000
−0.005
−0.010
Time (s)
Velocity X [mm/s]
4
2
0
−2
−4
Time (s)
Velocity Y [mm/s]
1.0
0.5
0.0
−0.5
−1.0
Time (s)
Acceleration X [mm/s²]
1000
500
0
−500
Time (s)
Acceleration Y [mm/s²]
300
200
100
0
−100
−200
−300
Time (s)

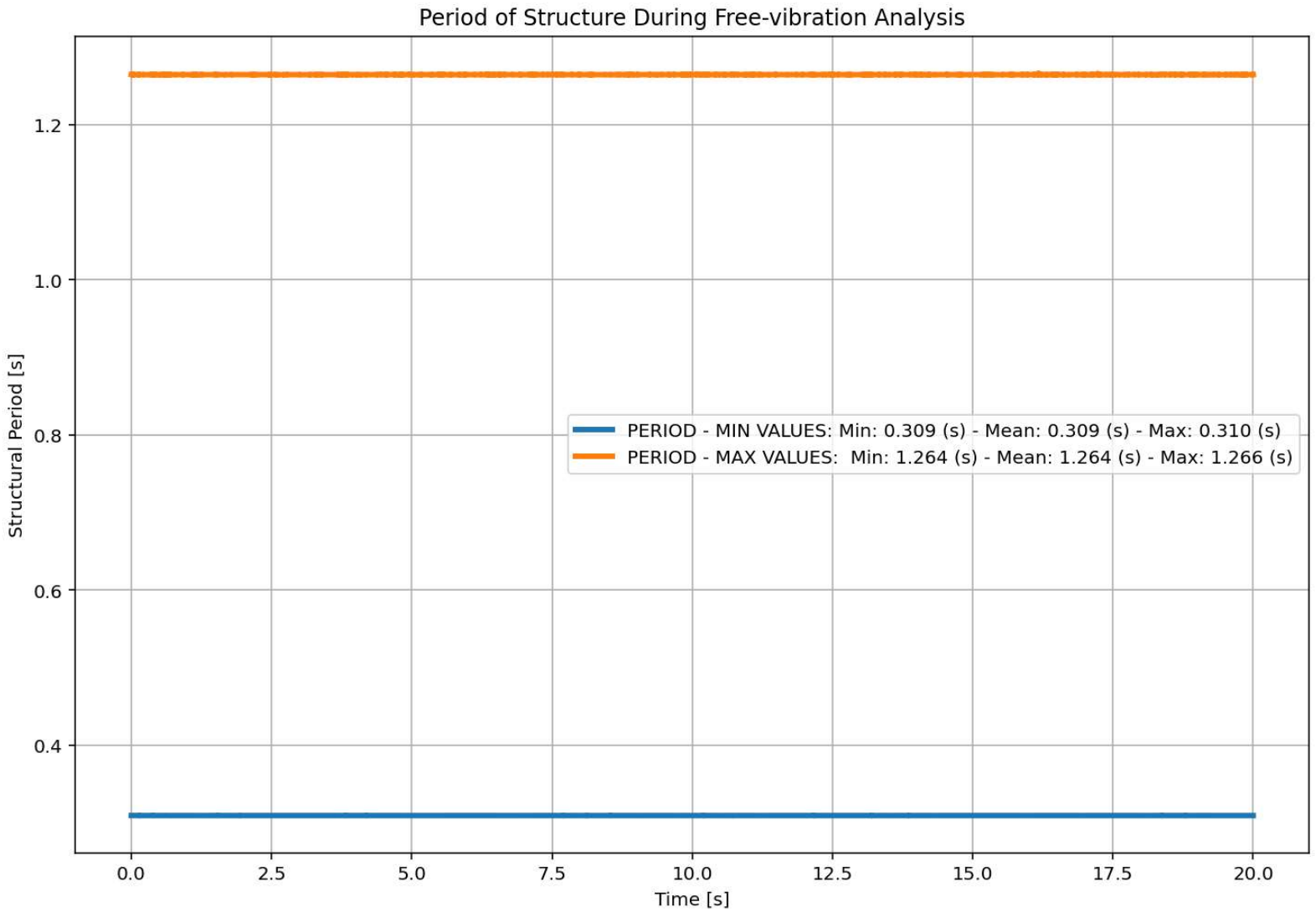

elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis
Element Axial Force [N]
Time [s]
200000
100000
0
−100000
−200000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 1.2405e+05
Ele. 2 - MAX. ABS. : 1.4338e+05
Ele. 3 - MAX. ABS. : 9.6770e+04
Ele. 4 - MAX. ABS. : 1.2008e+05
Ele. 5 - MAX. ABS. : 1.0589e+05
Ele. 6 - MAX. ABS. : 1.1668e+05
Ele. 7 - MAX. ABS. : 2.4733e+05
Ele. 8 - MAX. ABS. : 1.6516e+05
Ele. 9 - MAX. ABS. : 1.6797e+05
Ele. 10 - MAX. ABS. : 1.5303e+05
Ele. 11 - MAX. ABS. : 1.6613e+05
Ele. 12 - MAX. ABS. : 1.6629e+05
Ele. 13 - MAX. ABS. : 1.3832e+05
Ele. 14 - MAX. ABS. : 1.0678e+05
Ele. 15 - MAX. ABS. : 1.1055e+05
Ele. 16 - MAX. ABS. : 1.1908e+05
Ele. 17 - MAX. ABS. : 1.0433e+05
Ele. 18 - MAX. ABS. : 9.9269e+04

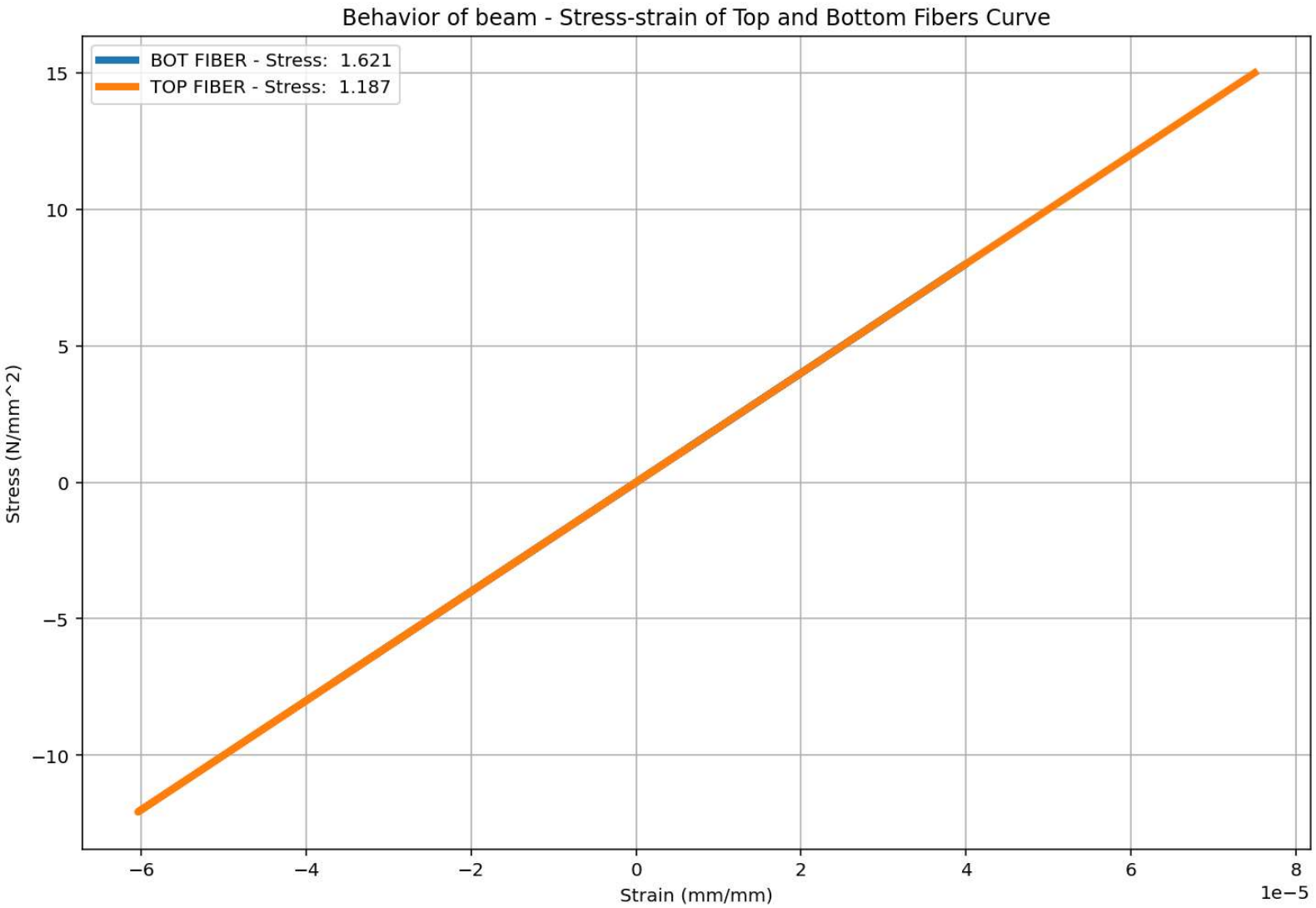

elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Element Shear Force [N]
Time [s]
200000
150000
100000
50000
0
−50000
−100000
−150000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 3.6694e+04
Ele. 2 - MAX. ABS. : 2.7265e+04
Ele. 3 - MAX. ABS. : 2.0823e+04
Ele. 4 - MAX. ABS. : 2.0882e+04
Ele. 5 - MAX. ABS. : 2.0643e+04
Ele. 6 - MAX. ABS. : 2.4721e+04
Ele. 7 - MAX. ABS. : 2.2491e+05
Ele. 8 - MAX. ABS. : 1.8254e+05
Ele. 9 - MAX. ABS. : 2.2899e+05
Ele. 10 - MAX. ABS. : 2.1492e+05
Ele. 11 - MAX. ABS. : 2.0075e+05
Ele. 12 - MAX. ABS. : 2.1533e+05
Ele. 13 - MAX. ABS. : 1.5423e+05
Ele. 14 - MAX. ABS. : 1.7774e+05
Ele. 15 - MAX. ABS. : 1.6285e+05
Ele. 16 - MAX. ABS. : 1.8190e+05
Ele. 17 - MAX. ABS. : 1.7546e+05
Ele. 18 - MAX. ABS. : 1.7209e+05

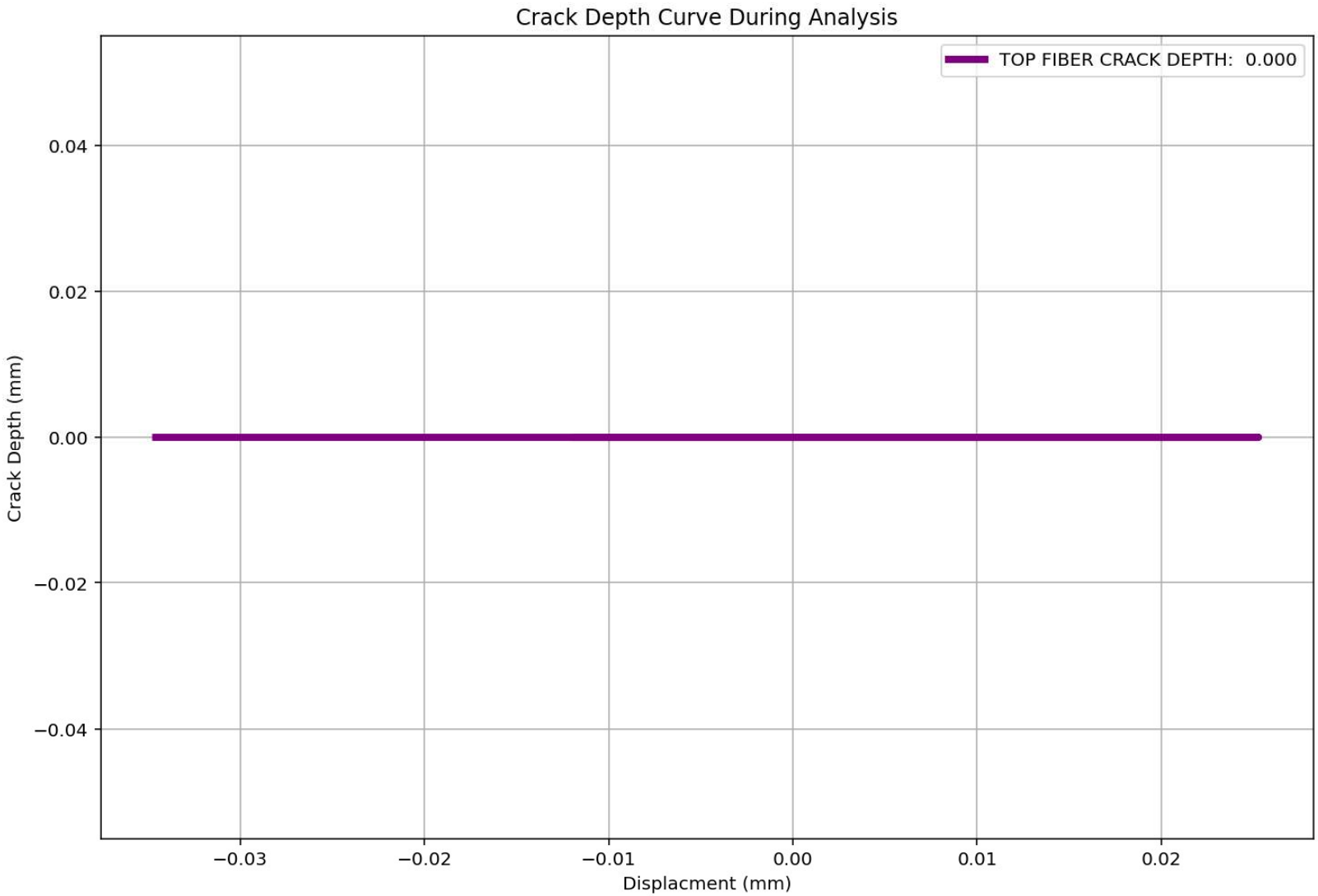

elements force in Z Dir. vs Time for Beam Element During Free-vibration Analysis
1e7
Element Moment Force [N]
Time [s]
Ele. 1 - MAX. ABS. : 3.1178e+07
Ele. 2 - MAX. ABS. : 2.0733e+07
Ele. 3 - MAX. ABS. : 1.3268e+07
Ele. 4 - MAX. ABS. : 1.3511e+07
Ele. 5 - MAX. ABS. : 1.2492e+07
Ele. 6 - MAX. ABS. : 1.3889e+07
Ele. 7 - MAX. ABS. : 6.8566e+07
Ele. 8 - MAX. ABS. : 5.5099e+07
Ele. 9 - MAX. ABS. : 4.9063e+07
Ele. 10 - MAX. ABS. : 4.6996e+07
Ele. 11 - MAX. ABS. : 4.8175e+07
Ele. 12 - MAX. ABS. : 5.1970e+07
Ele. 13 - MAX. ABS. : 4.4016e+07
Ele. 14 - MAX. ABS. : 3.2174e+07
Ele. 15 - MAX. ABS. : 3.3595e+07
Ele. 16 - MAX. ABS. : 2.9376e+07
Ele. 17 - MAX. ABS. : 3.3363e+07
Ele. 18 - MAX. ABS. : 3.2199e+07

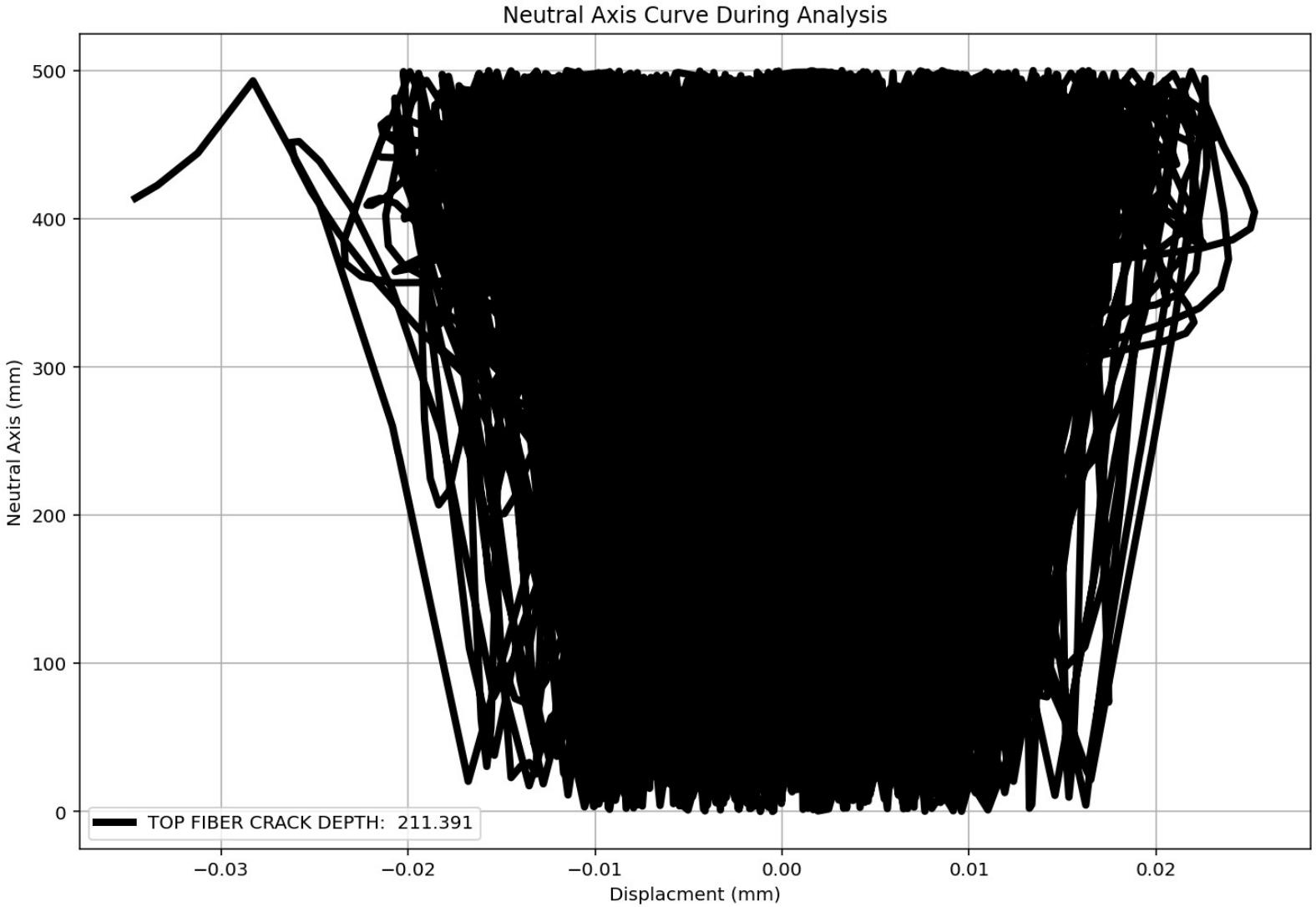

Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 3.4697e-02
Node 2 - MAX. ABS. : 2.3894e-02
Node 3 - MAX. ABS. : 2.2677e-02
Node 4 - MAX. ABS. : 1.8198e-02
Node 5 - MAX. ABS. : 2.1310e-02
Node 6 - MAX. ABS. : 2.4196e-02
Node 7 - MAX. ABS. : 3.0591e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 1.6959e-02
Node 10 - MAX. ABS. : 2.8452e-02
Node 11 - MAX. ABS. : 3.2112e-02
Node 12 - MAX. ABS. : 2.9940e-02
Node 13 - MAX. ABS. : 2.8910e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 1.4127e-02
Node 16 - MAX. ABS. : 2.4755e-02
Node 17 - MAX. ABS. : 2.8551e-02
Node 18 - MAX. ABS. : 3.2632e-02
Node 19 - MAX. ABS. : 2.8049e-02

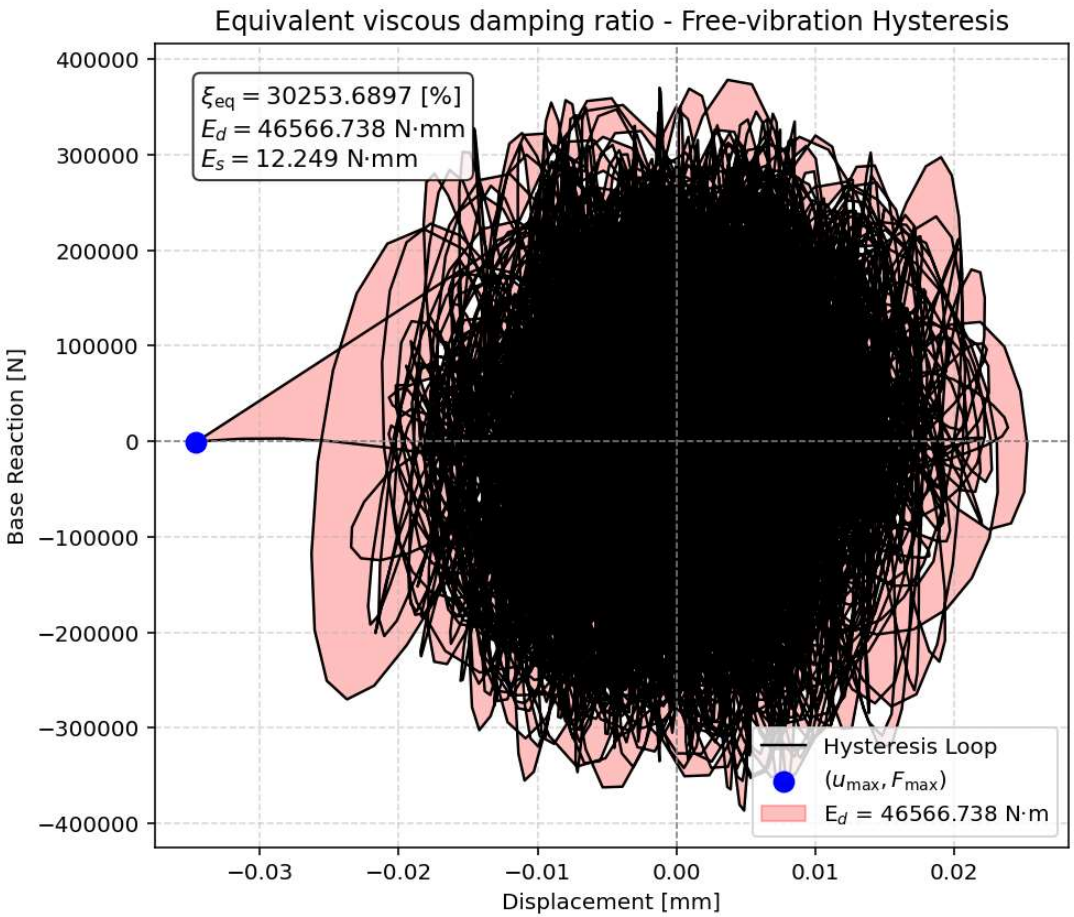

Node Displacements in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.03
0.02
0.01
0.00
−0.01
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 3.6325e-02
Node 2 - MAX. ABS. : 2.7330e-02
Node 3 - MAX. ABS. : 3.1011e-02
Node 4 - MAX. ABS. : 3.2992e-02
Node 5 - MAX. ABS. : 3.1559e-02
Node 6 - MAX. ABS. : 2.9249e-02
Node 7 - MAX. ABS. : 2.9300e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 1.7344e-02
Node 10 - MAX. ABS. : 2.3431e-02
Node 11 - MAX. ABS. : 2.5704e-02
Node 12 - MAX. ABS. : 3.1679e-02
Node 13 - MAX. ABS. : 3.1273e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 1.3975e-02
Node 16 - MAX. ABS. : 1.9016e-02
Node 17 - MAX. ABS. : 2.2011e-02
Node 18 - MAX. ABS. : 2.5250e-02
Node 19 - MAX. ABS. : 2.8081e-02

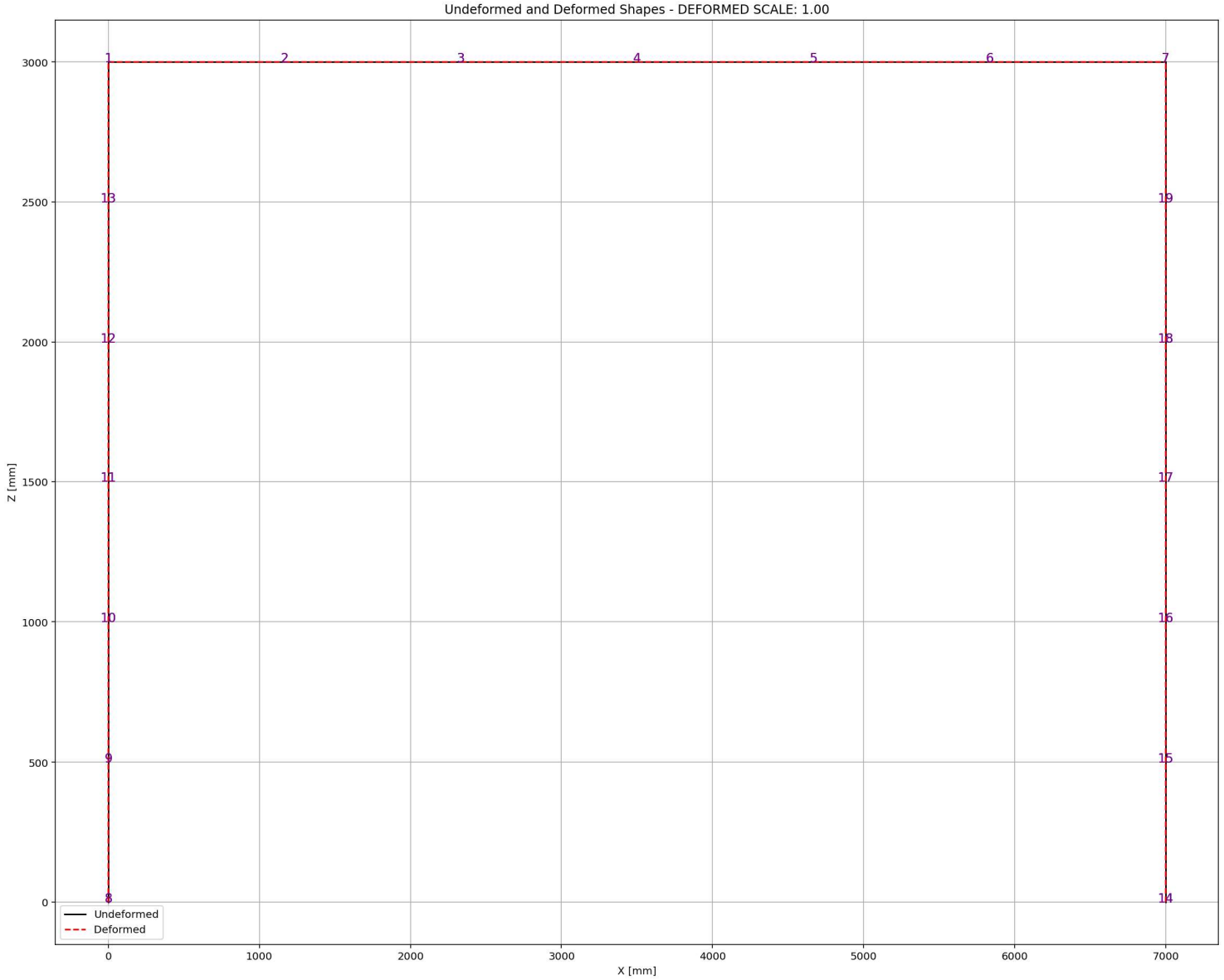

Period of Structure During Free-vibration Analysis
Structural Period [s]
1.2
1.0
0.8
0.6
0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 0.309 (s) - Mean: 0.309 (s) - Max: 0.310 (s)
PERIOD - MAX VALUES: Min: 1.264 (s) - Mean: 1.264 (s) - Max: 1.266 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 1.621
TOP FIBER - Stress: 1.187
Stress (N/mm^2)
15
10
5
0
−5
−10
−6
−4
−2
0
2
4
6
8
Strain (mm/mm)
1e−5


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−0.03
−0.02
−0.01
0.00
0.01
0.02
Displacment (mm)


Neutral Axis Curve During Analysis
Neutral Axis (mm)
500
400
300
200
100
0
−0.03
−0.02
−0.01
0.00
0.01
0.02
Displacment (mm)
TOP FIBER CRACK DEPTH: 211.391


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 30253.6897$ [%]
$E_d = 46566.738$ N·mm
$E_s = 12.249$ N·mm
Base Reaction [N]
Displacement [mm]
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
−0.03
−0.02
−0.01
0.00
0.01
0.02
Hysteresis Loop
(u_{max}, F_{max})
E_d = 46566.738 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

SEISMIC ANALYSIS

Spyder (Python 3.12)

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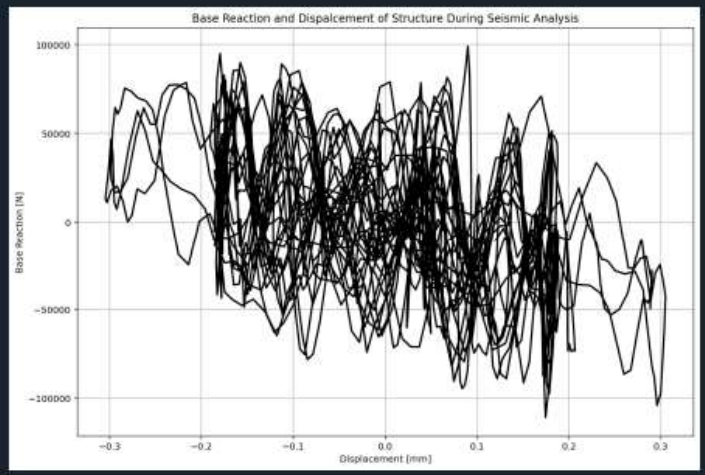
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...FRAME_8_ANA\P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py

P(t)_COMPOSITE_SECTION_FRAME_8_ANA.py | COMPOSITE_CONCRETE_SECTION_FUN_EXTRA.py

```
983     if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
984         #%% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
985         duration = 20.0                  # [s] Analysis duration
986         dt = 0.01                        # [s] Time step
987         GMfact = 9810                    # [mm/s²]standard acceleration of gravity or st
988         SSF_X = 0.05                     # Seismic Acceleration Scale Factor in X Direct
989         SSF_Y = 0.05                     # Seismic Acceleration Scale Factor in Y Direct
990         iv0_X = 0.0005                   # [mm/s] Initial velocity applied to the node
991         iv0_Y = 0.0005                   # [mm/s] Initial velocity applied to the node
992         st_iv0 = 0.0                     # [s] Initial velocity applied starting time
993         SEI = 'X'                        # Seismic Direction
994         DR = 0.03                        # Damping ratio
995
996         # Define time series for input motion (Acceleration time history)
997         if SEI == 'X':
998             SEISMIC_TAG_01 = 100
999             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
1000            # Define load patterns
1001            # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
1002            ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
1003        if SEI == 'Y':
1004            SEISMIC_TAG_02 = 200
1005            ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
1006            ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
1007        if SEI == 'XY':
1008            SEISMIC_TAG_01 = 100
1009            ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
1010            # Define load patterns
1011            # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
1012            ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
1013            SEISMIC_TAG_02 = 200
1014            ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
1015            ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
1016        print('Seismic Defined Done.')
```

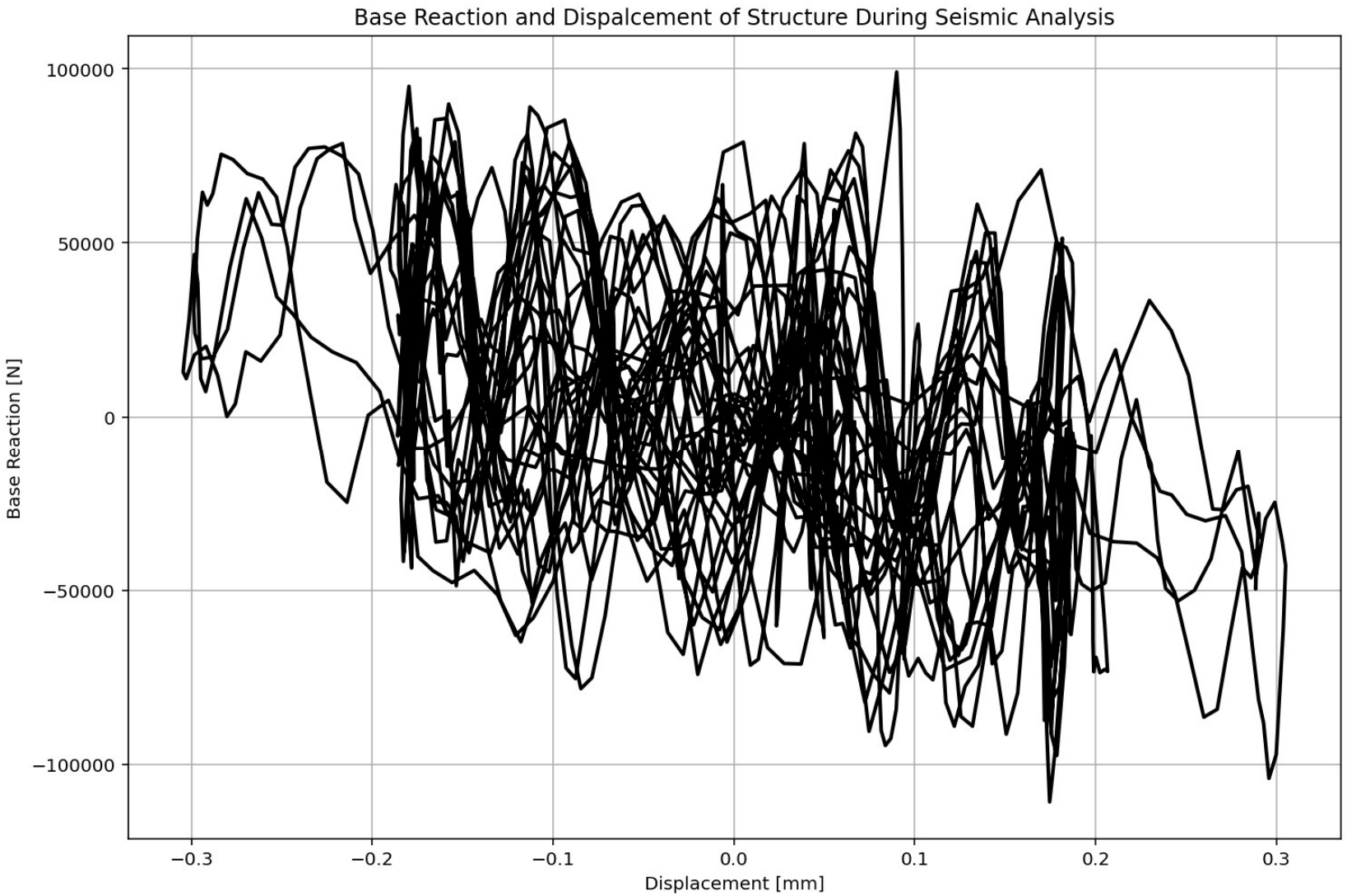
..ENERGY_DISSIPATION_CAPACITY_INDEX\10_COMPOSITE_FRAME_8_ANA

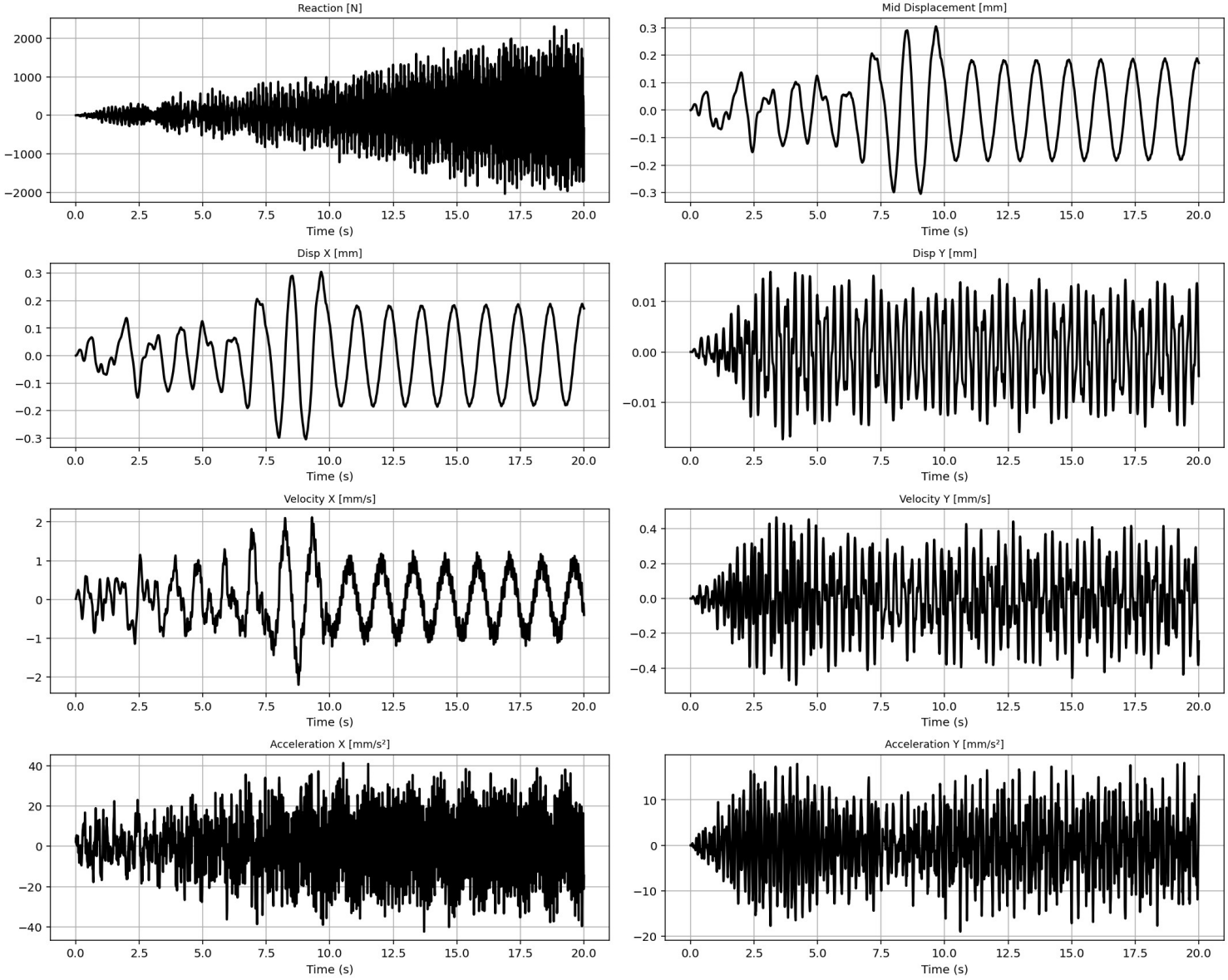
30 %

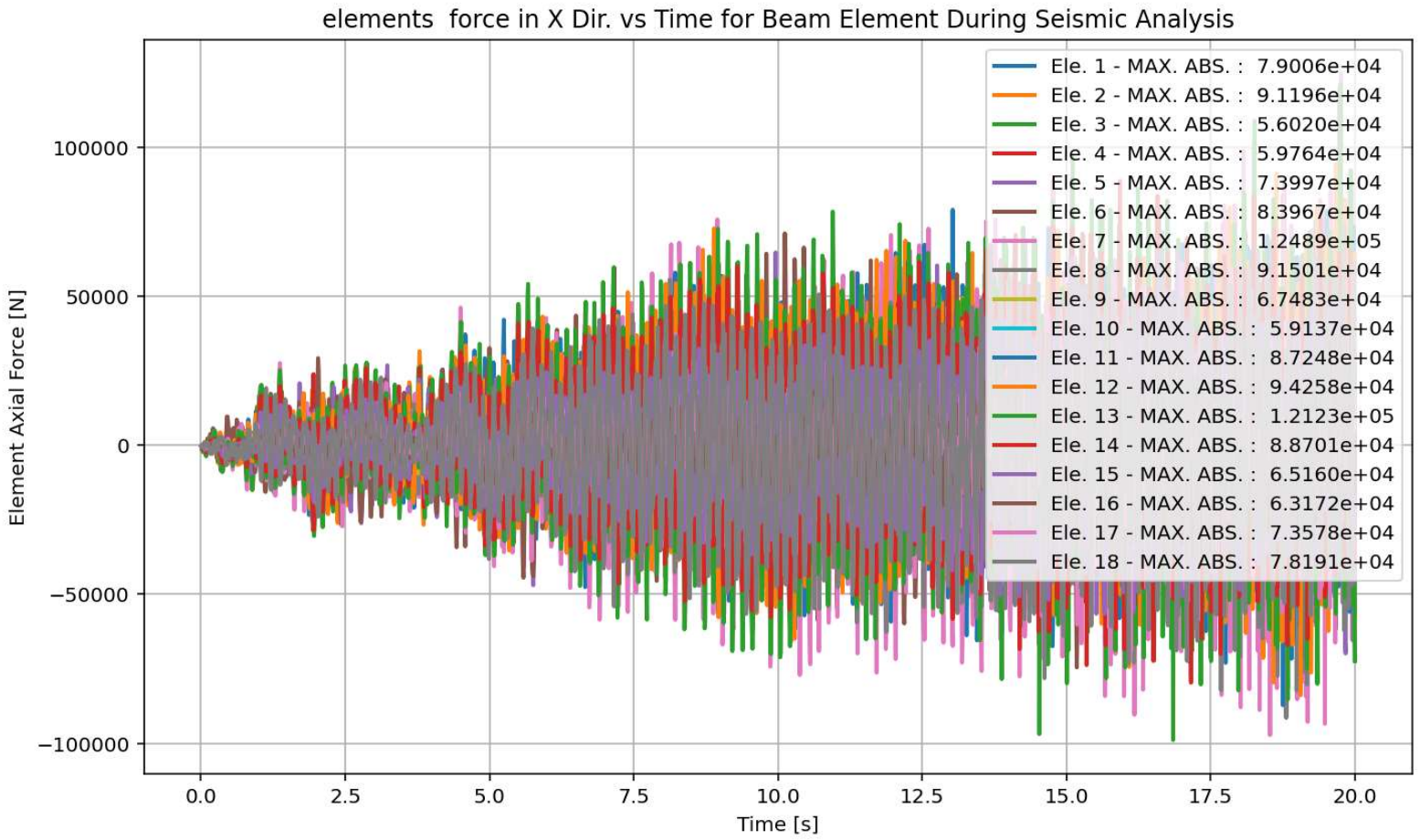


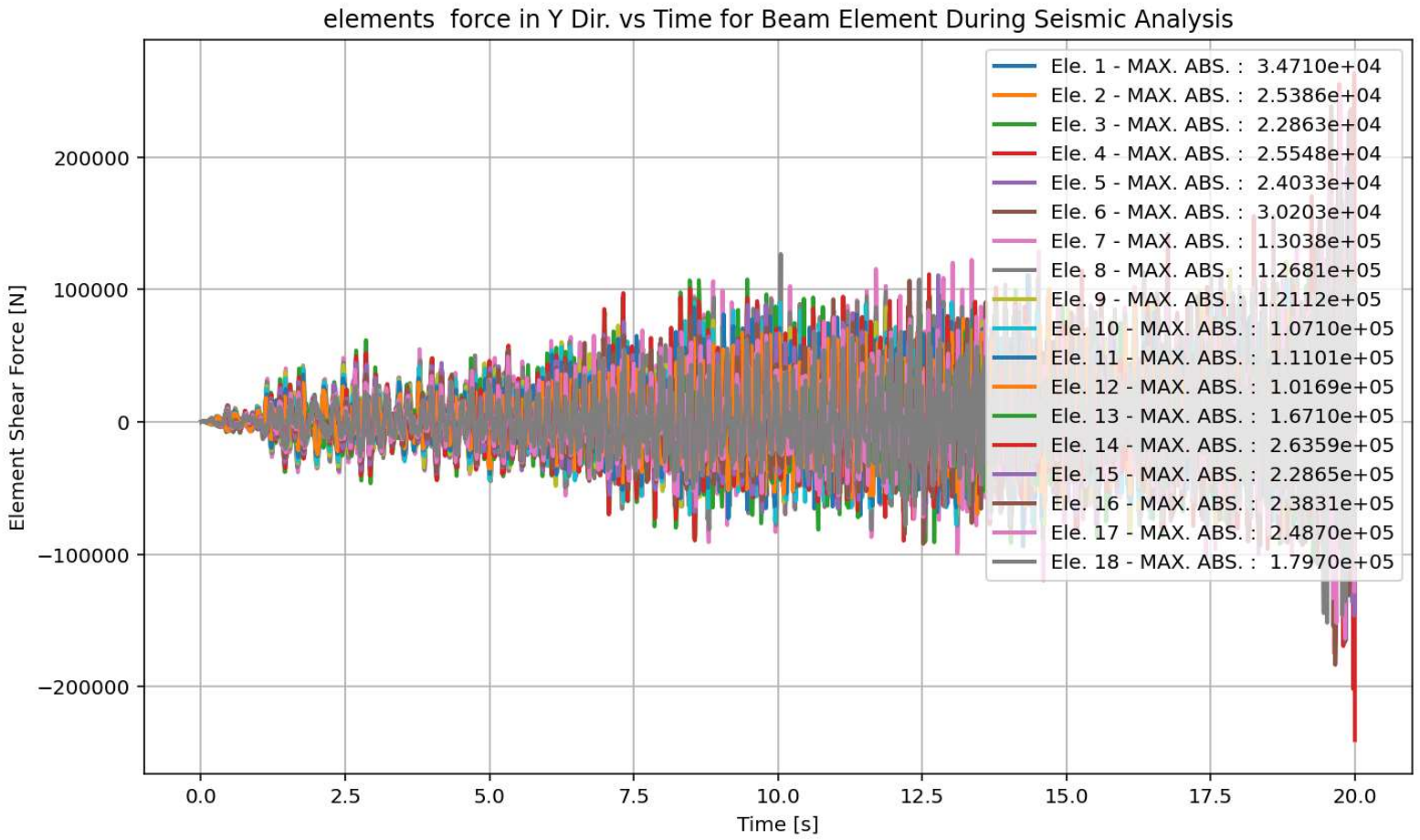
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

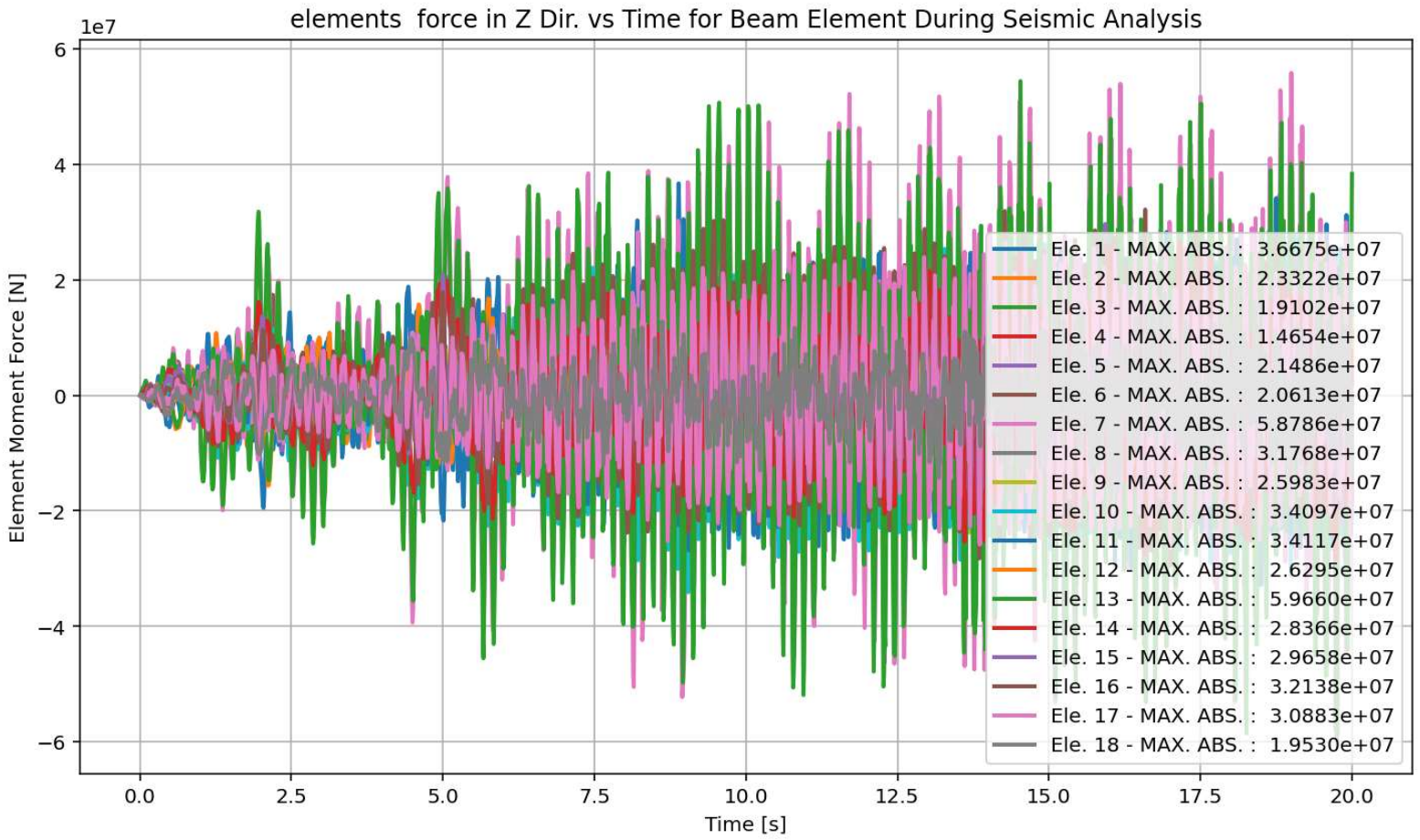
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 84, Col 34 | UTF-8 | CRLF | RW | Mem 42%

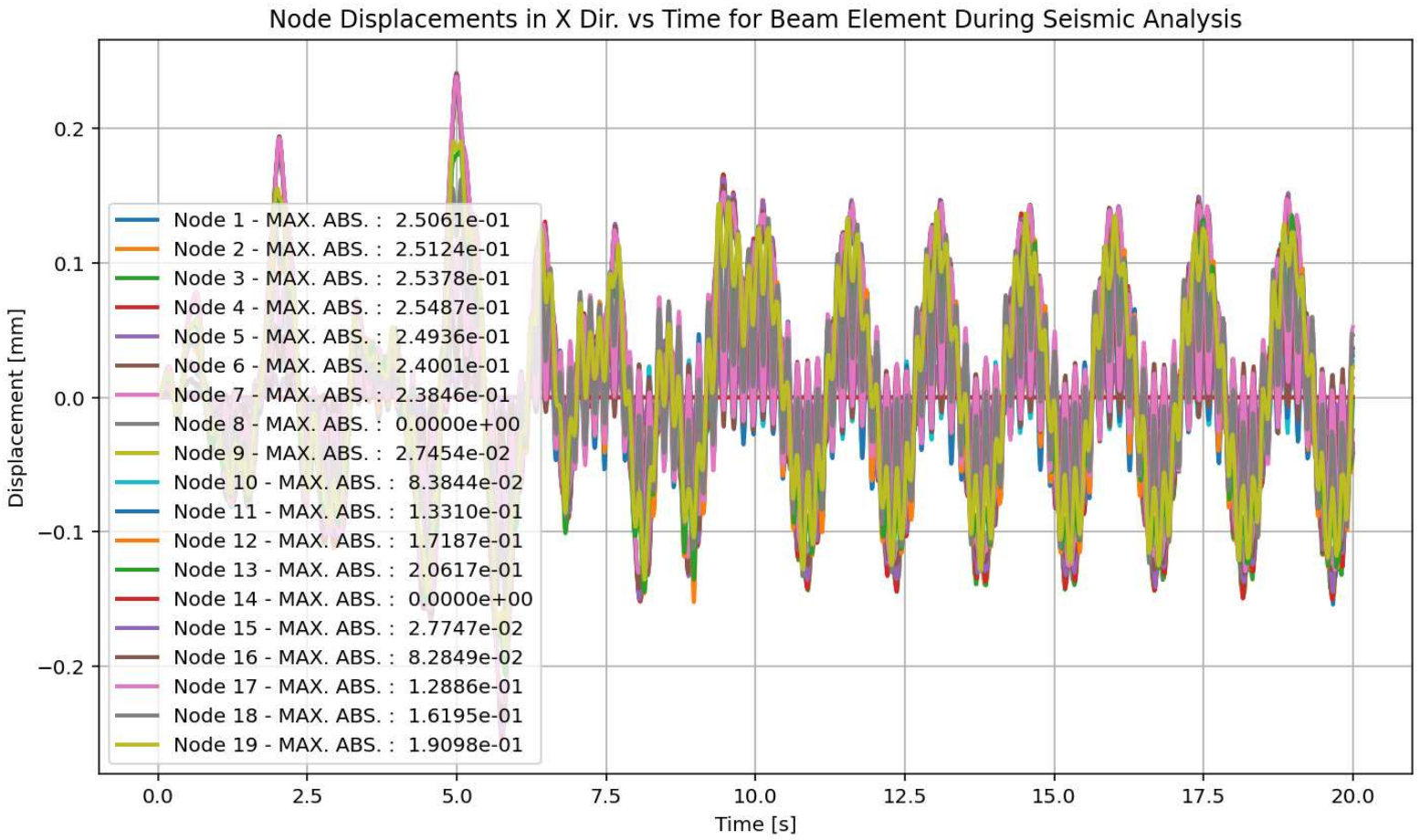

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
100000
50000
0
−50000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [mm]

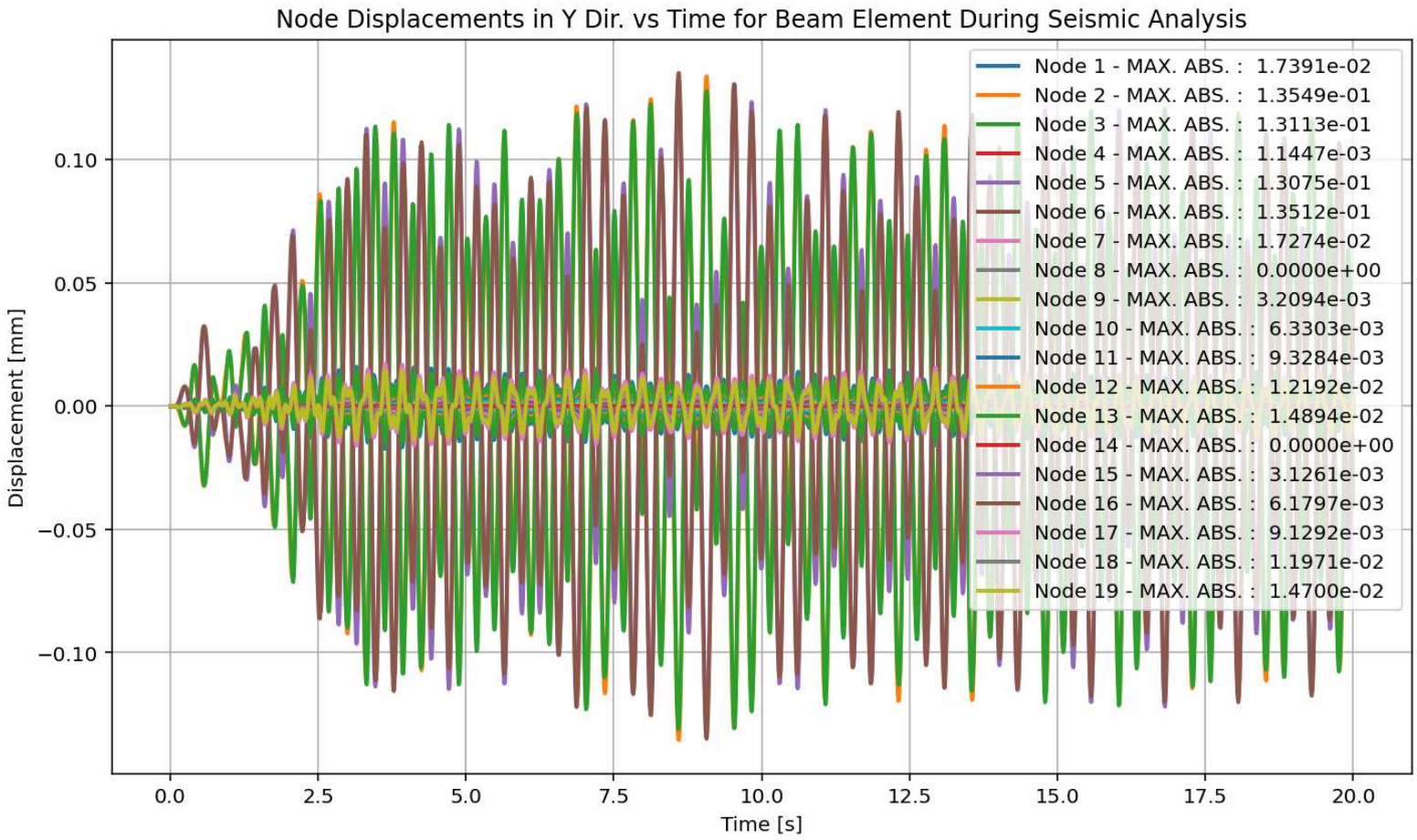

Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

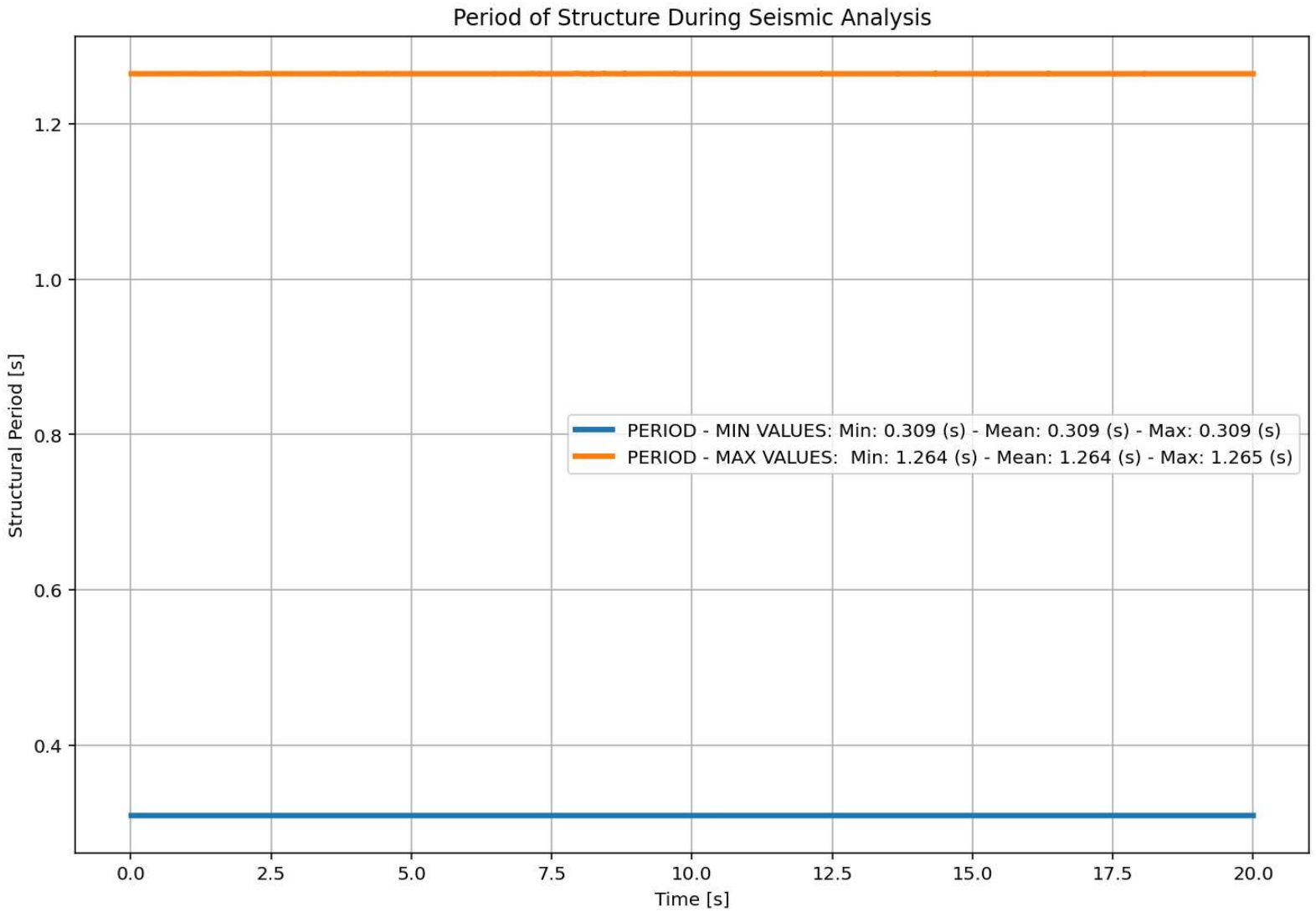

elements force in X Dir. vs Time for Beam Element During Seismic Analysis
Element Axial Force [N]
Time [s]
100000
50000
0
−50000
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 7.9006e+04
Ele. 2 - MAX. ABS. : 9.1196e+04
Ele. 3 - MAX. ABS. : 5.6020e+04
Ele. 4 - MAX. ABS. : 5.9764e+04
Ele. 5 - MAX. ABS. : 7.3997e+04
Ele. 6 - MAX. ABS. : 8.3967e+04
Ele. 7 - MAX. ABS. : 1.2489e+05
Ele. 8 - MAX. ABS. : 9.1501e+04
Ele. 9 - MAX. ABS. : 6.7483e+04
Ele. 10 - MAX. ABS. : 5.9137e+04
Ele. 11 - MAX. ABS. : 8.7248e+04
Ele. 12 - MAX. ABS. : 9.4258e+04
Ele. 13 - MAX. ABS. : 1.2123e+05
Ele. 14 - MAX. ABS. : 8.8701e+04
Ele. 15 - MAX. ABS. : 6.5160e+04
Ele. 16 - MAX. ABS. : 6.3172e+04
Ele. 17 - MAX. ABS. : 7.3578e+04
Ele. 18 - MAX. ABS. : 7.8191e+04

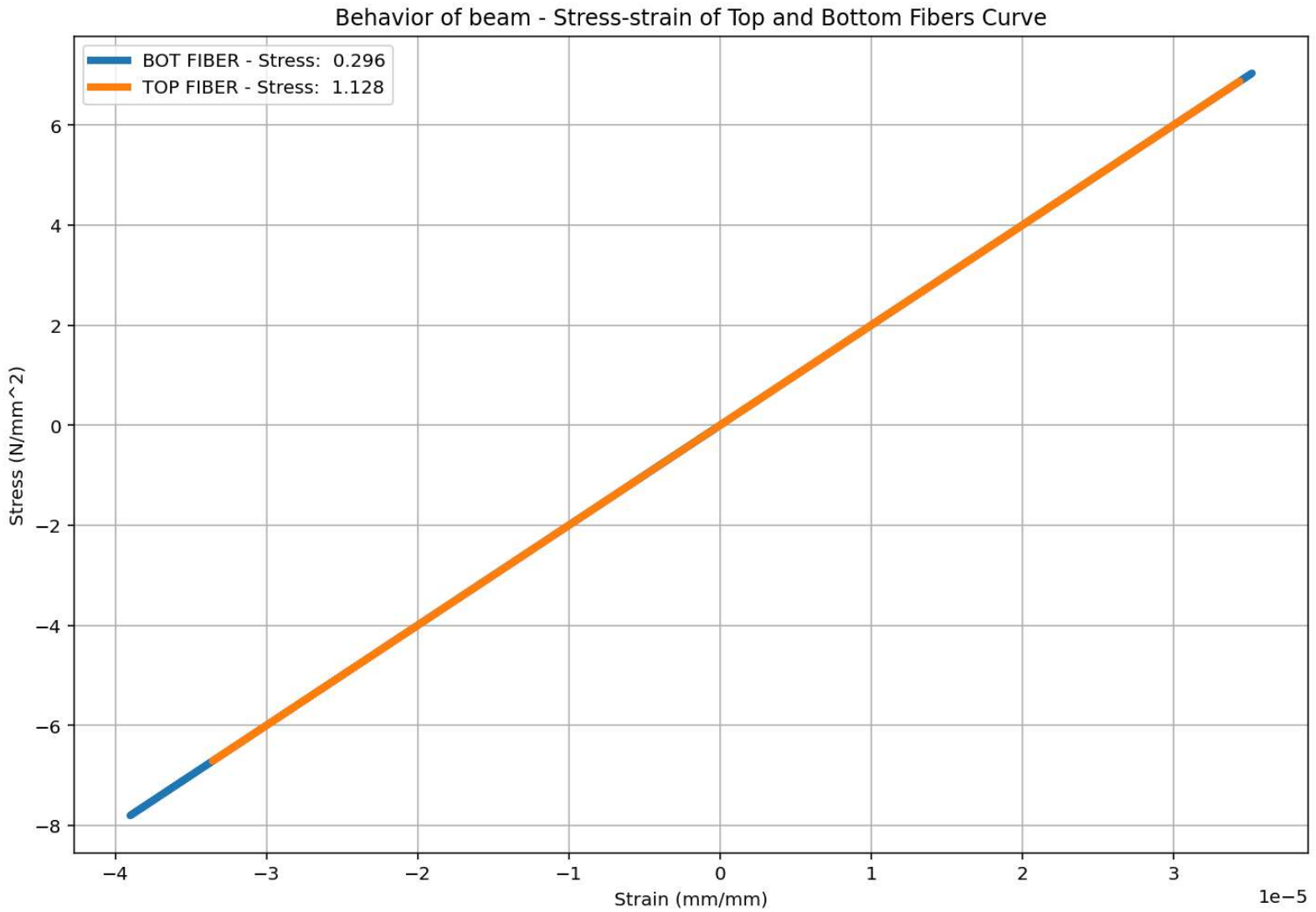

elements force in Y Dir. vs Time for Beam Element During Seismic Analysis
Element Shear Force [N]
Time [s]
200000
100000
0
−100000
−200000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 3.4710e+04
Ele. 2 - MAX. ABS. : 2.5386e+04
Ele. 3 - MAX. ABS. : 2.2863e+04
Ele. 4 - MAX. ABS. : 2.5548e+04
Ele. 5 - MAX. ABS. : 2.4033e+04
Ele. 6 - MAX. ABS. : 3.0203e+04
Ele. 7 - MAX. ABS. : 1.3038e+05
Ele. 8 - MAX. ABS. : 1.2681e+05
Ele. 9 - MAX. ABS. : 1.2112e+05
Ele. 10 - MAX. ABS. : 1.0710e+05
Ele. 11 - MAX. ABS. : 1.1101e+05
Ele. 12 - MAX. ABS. : 1.0169e+05
Ele. 13 - MAX. ABS. : 1.6710e+05
Ele. 14 - MAX. ABS. : 2.6359e+05
Ele. 15 - MAX. ABS. : 2.2865e+05
Ele. 16 - MAX. ABS. : 2.3831e+05
Ele. 17 - MAX. ABS. : 2.4870e+05
Ele. 18 - MAX. ABS. : 1.7970e+05

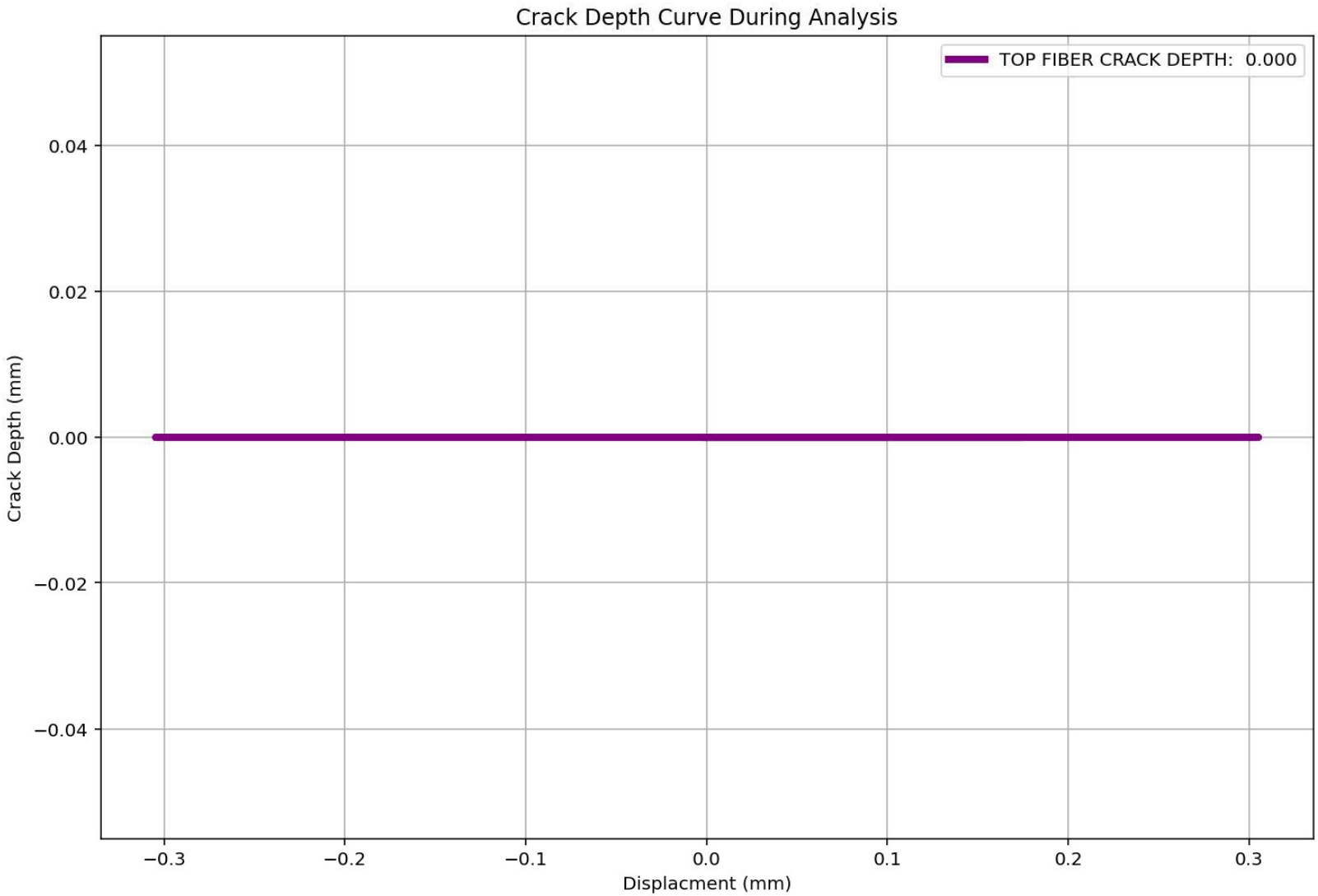

elements force in Z Dir. vs Time for Beam Element During Seismic Analysis
1e7
Element Moment Force [N]
Time [s]
Ele. 1 - MAX. ABS. : 3.6675e+07
Ele. 2 - MAX. ABS. : 2.3322e+07
Ele. 3 - MAX. ABS. : 1.9102e+07
Ele. 4 - MAX. ABS. : 1.4654e+07
Ele. 5 - MAX. ABS. : 2.1486e+07
Ele. 6 - MAX. ABS. : 2.0613e+07
Ele. 7 - MAX. ABS. : 5.8786e+07
Ele. 8 - MAX. ABS. : 3.1768e+07
Ele. 9 - MAX. ABS. : 2.5983e+07
Ele. 10 - MAX. ABS. : 3.4097e+07
Ele. 11 - MAX. ABS. : 3.4117e+07
Ele. 12 - MAX. ABS. : 2.6295e+07
Ele. 13 - MAX. ABS. : 5.9660e+07
Ele. 14 - MAX. ABS. : 2.8366e+07
Ele. 15 - MAX. ABS. : 2.9658e+07
Ele. 16 - MAX. ABS. : 3.2138e+07
Ele. 17 - MAX. ABS. : 3.0883e+07
Ele. 18 - MAX. ABS. : 1.9530e+07

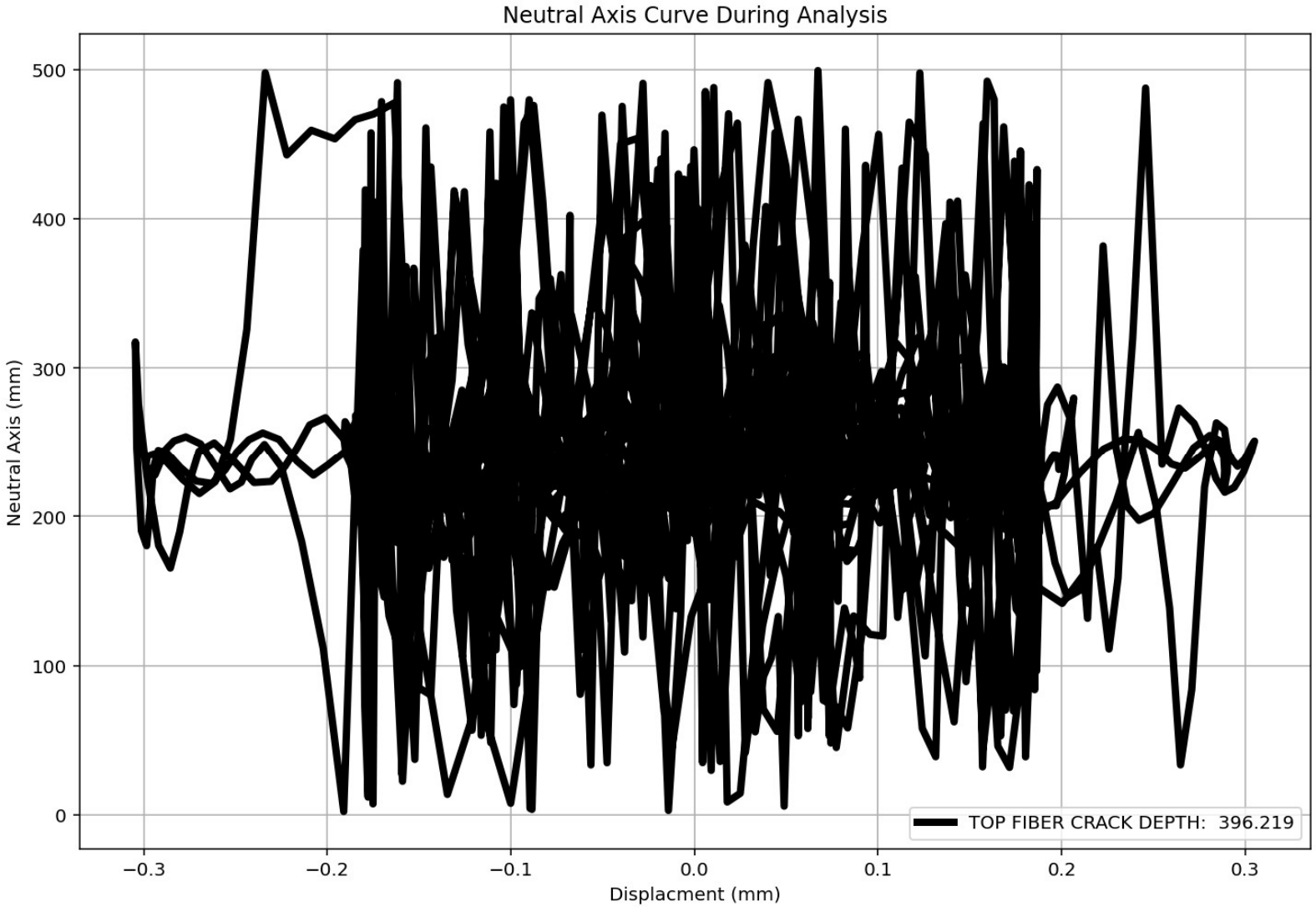

Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
0.2
0.1
0.0
−0.1
−0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 2.5061e-01
Node 2 - MAX. ABS. : 2.5124e-01
Node 3 - MAX. ABS. : 2.5378e-01
Node 4 - MAX. ABS. : 2.5487e-01
Node 5 - MAX. ABS. : 2.4936e-01
Node 6 - MAX. ABS. : 2.4001e-01
Node 7 - MAX. ABS. : 2.3846e-01
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.7454e-02
Node 10 - MAX. ABS. : 8.3844e-02
Node 11 - MAX. ABS. : 1.3310e-01
Node 12 - MAX. ABS. : 1.7187e-01
Node 13 - MAX. ABS. : 2.0617e-01
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 2.7747e-02
Node 16 - MAX. ABS. : 8.2849e-02
Node 17 - MAX. ABS. : 1.2886e-01
Node 18 - MAX. ABS. : 1.6195e-01
Node 19 - MAX. ABS. : 1.9098e-01

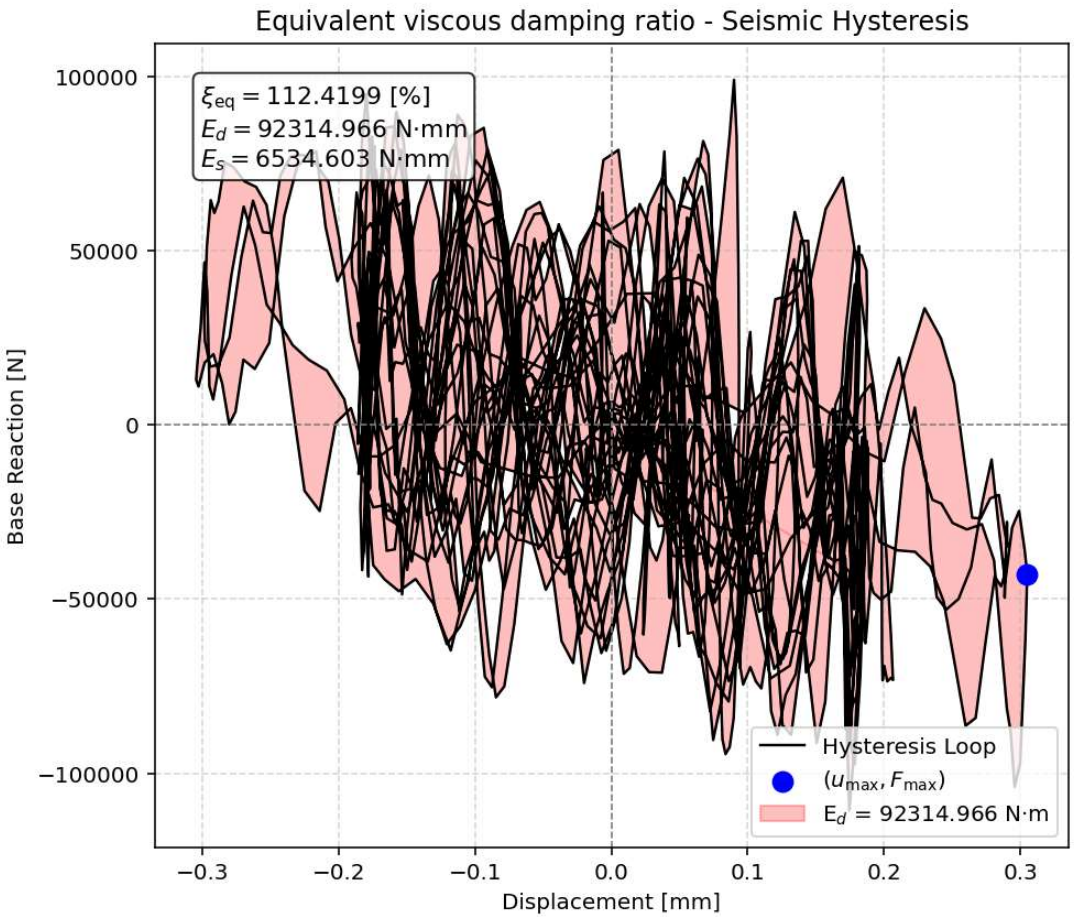

Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 1.7391e-02
Node 2 - MAX. ABS. : 1.3549e-01
Node 3 - MAX. ABS. : 1.3113e-01
Node 4 - MAX. ABS. : 1.1447e-03
Node 5 - MAX. ABS. : 1.3075e-01
Node 6 - MAX. ABS. : 1.3512e-01
Node 7 - MAX. ABS. : 1.7274e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 3.2094e-03
Node 10 - MAX. ABS. : 6.3303e-03
Node 11 - MAX. ABS. : 9.3284e-03
Node 12 - MAX. ABS. : 1.2192e-02
Node 13 - MAX. ABS. : 1.4894e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 3.1261e-03
Node 16 - MAX. ABS. : 6.1797e-03
Node 17 - MAX. ABS. : 9.1292e-03
Node 18 - MAX. ABS. : 1.1971e-02
Node 19 - MAX. ABS. : 1.4700e-02

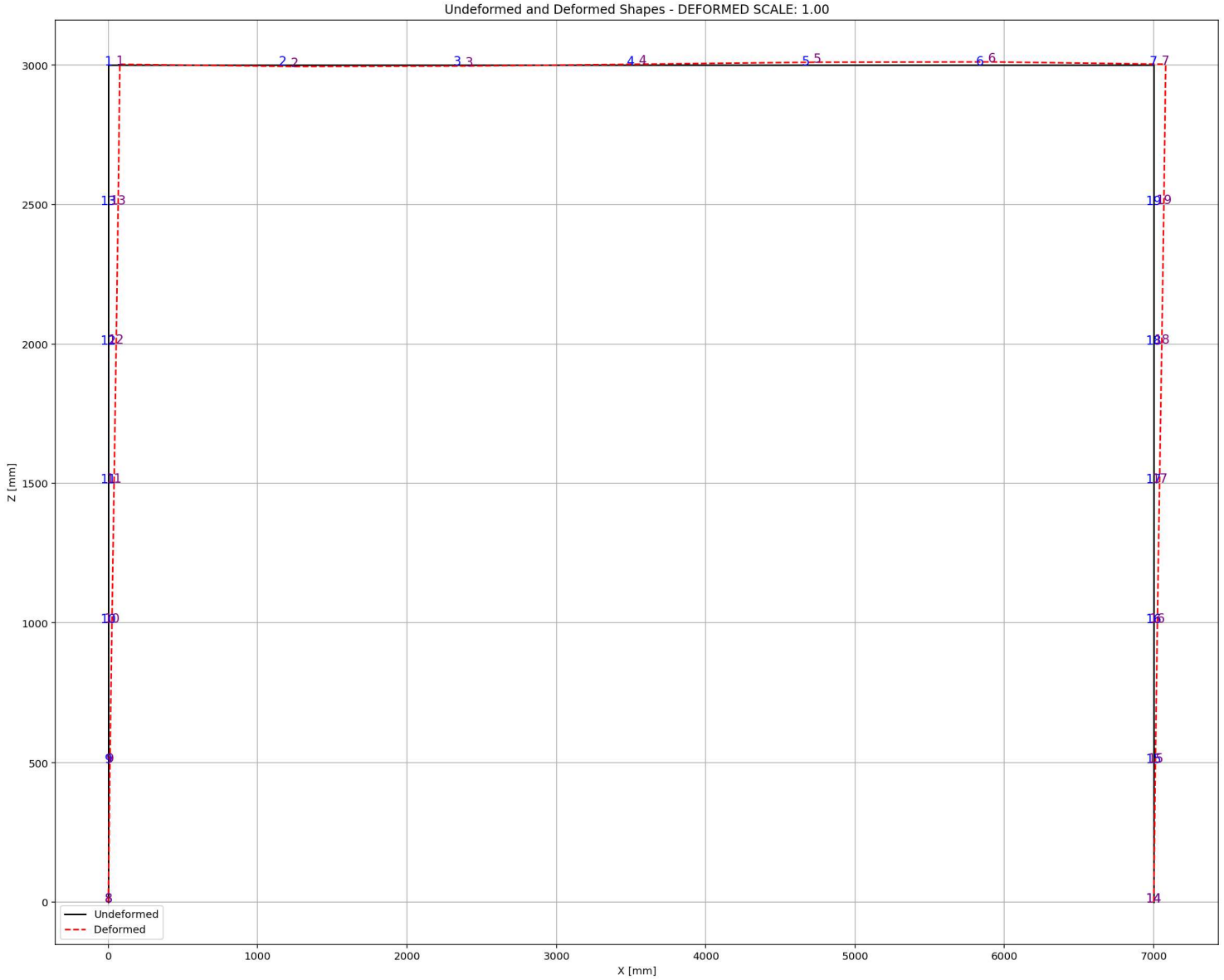

Period of Structure During Seismic Analysis
Structural Period [s]
1.2
1.0
0.8
0.6
0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 0.309 (s) - Mean: 0.309 (s) - Max: 0.309 (s)
PERIOD - MAX VALUES: Min: 1.264 (s) - Mean: 1.264 (s) - Max: 1.265 (s)

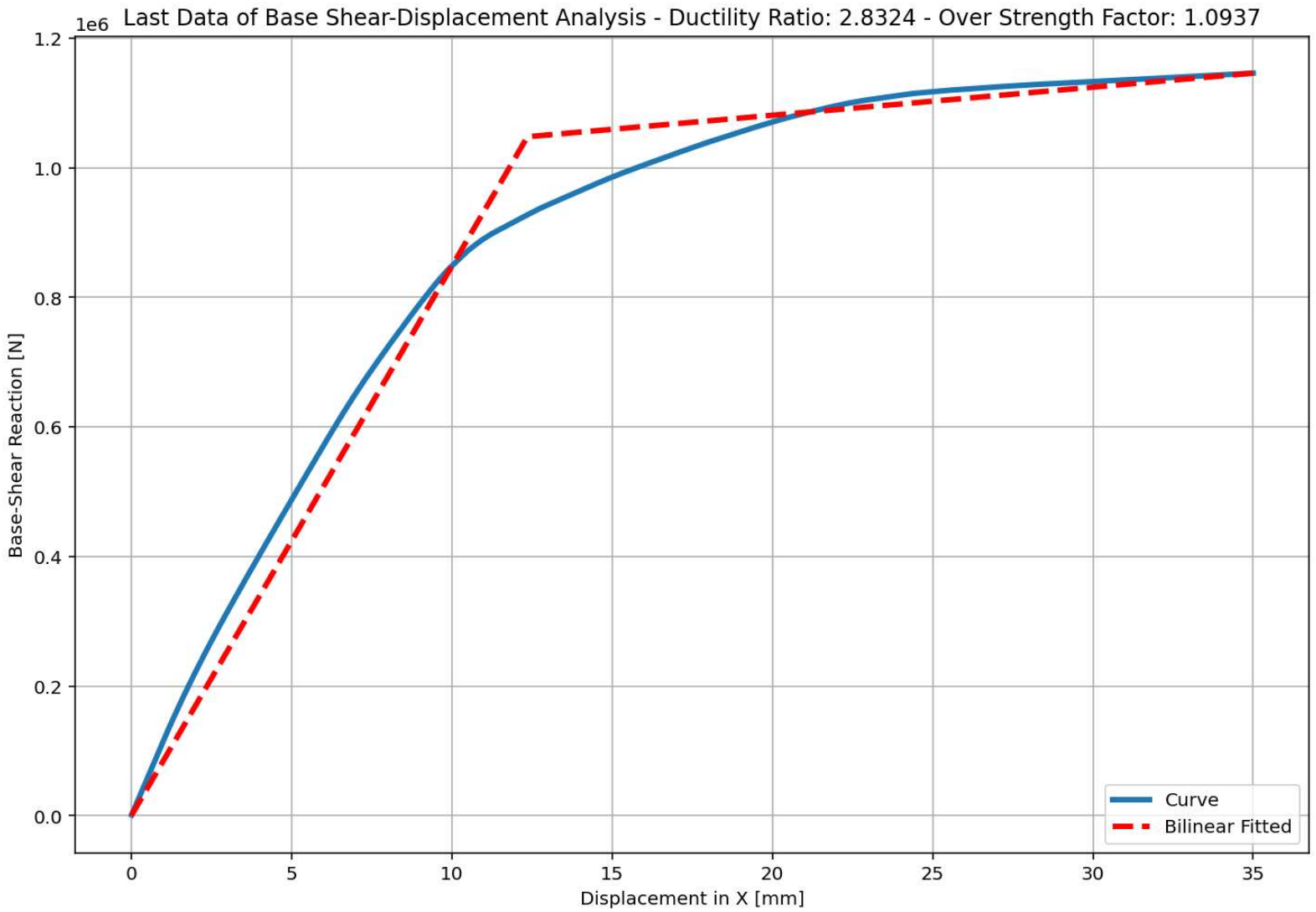

Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.296
TOP FIBER - Stress: 1.128
Stress (N/mm^2)
6
4
2
0
−2
−4
−6
−8
−4
−3
−2
−1
0
1
2
3
Strain (mm/mm)
1e−5


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacment (mm)

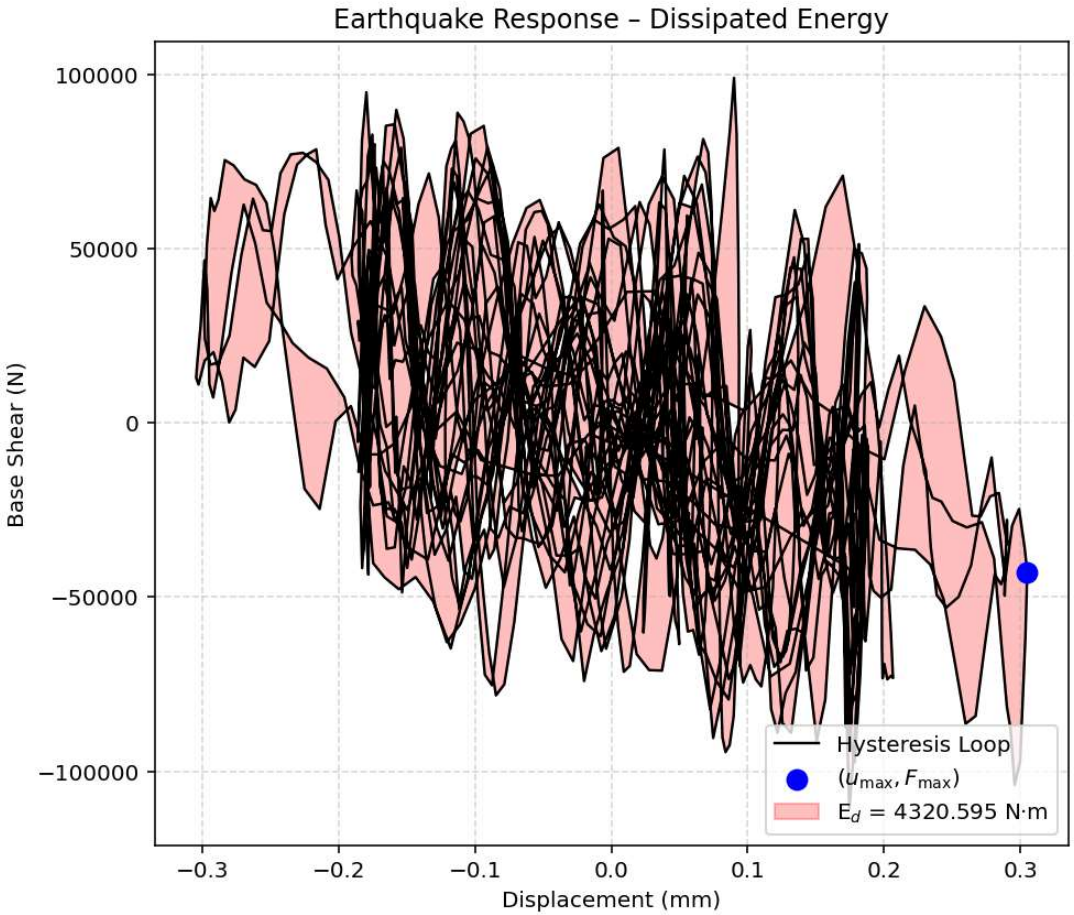

Neutral Axis Curve During Analysis
Neutral Axis (mm)
500
400
300
200
100
0
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacment (mm)
TOP FIBER CRACK DEPTH: 396.219

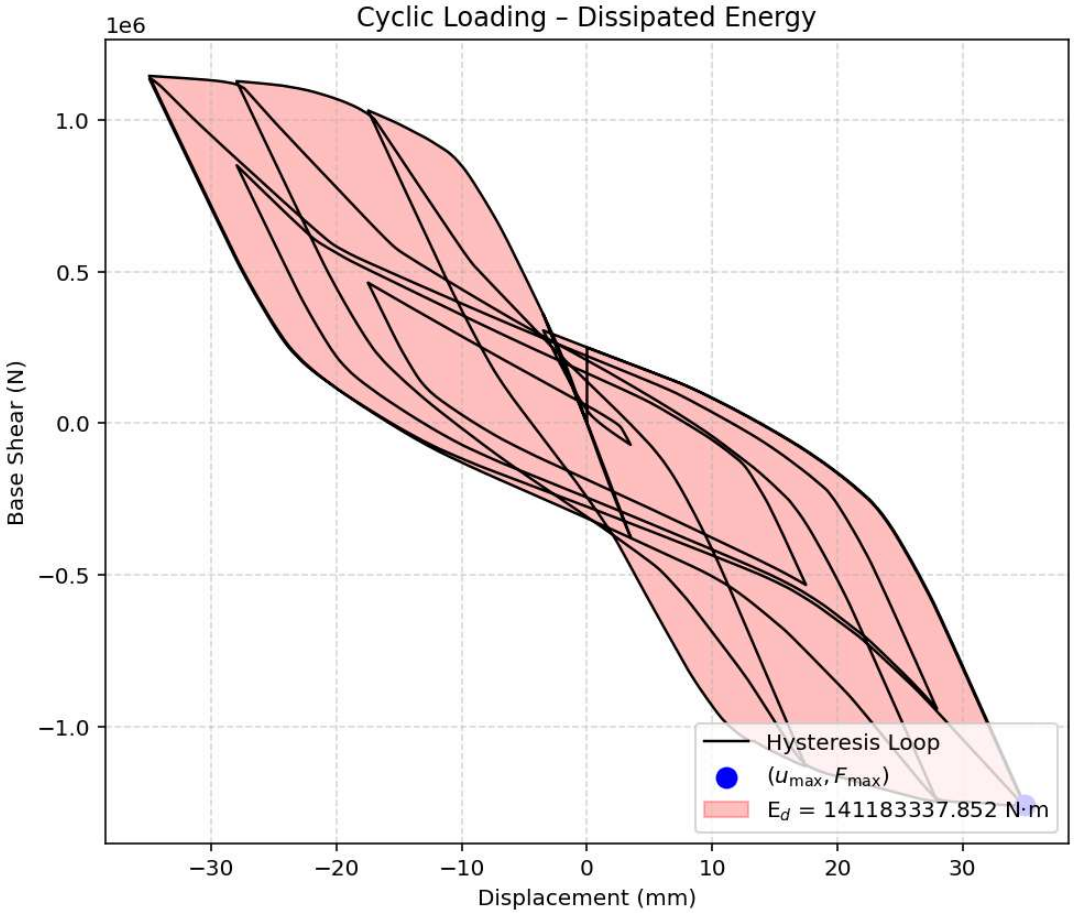

Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 112.4199$ [%]
$E_d = 92314.966$ N·mm
$E_s = 6534.603$ N·mm
Base Reaction [N]
Displacement [mm]
100000
50000
0
−50000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 92314.966 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
Undeformed
Deformed


Last Data of Base Shear-Displacement Analysis - Ductility Ratio: 2.8324 - Over Strength Factor: 1.0937
1e6
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Base-Shear Reaction [N]
0
5
10
15
20
25
30
35
Displacement in X [mm]
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.23570961e+01 1.04798136e+06]
 [3.50000000e+01 1.14617861e+06]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :     84808.06
 Structure Plastic Stiffness :     32747.96
 Structure Tangent Stiffness :     4336.78
 Structure Ductility Ratio :      2.83
 Structure Over Strength Factor:   1.09
+-------------------------------------------------+
Over Strength Coefficient (Ω0):     1.0937
Displacement Ductility Ratio (μ):    2.8324
Ductility Coefficient (Rμ):         2.8324
Structural Behavior Coefficient (R): 3.0978
Structural Ductility Damage Index in X Direction: -53.2265 (%)
```


Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (mm)
100000
50000
0
−50000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 4320.595 N·m


Cyclic Loading – Dissipated Energy
1e6
1.0
0.5
0.0
−0.5
−1.0
Base Shear (N)
−30
−20
−10
0
10
20
30
Displacement (mm)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 141183337.852 N·m

Dissipated Energy from Earthquake= 4320.60 N·mm
Dissipated Energy from Cyclic Displacement= 141183337.85 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.003 [%]
ZONE 1: VERY MINOR DAMAGE